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1998

SPECIAL FEATURES
Education, training and labour market reform

SWEDEN

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1997-1998

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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Table of contents

Assessment and recommendations 1

I. Recent trends and prospects 17

Economic activity remains moderate 19
The labour market and inflation 29
The short-term outlook 34

II. Macroeconomic policies 38

Monetary policy 38
Fiscal policy 51

III. Implementing structural reform: a review of progress 65

Introduction 65
Developments in structural reform 67
Overview and scope for further action 78

IV. Education and training 83

Introduction 83
Formal education: developing a new framework 84
Economic impact of the education system 101
Life-long learning: a pioneering effort 107
Assessment and scope for further action 119

V. Entrepreneurship and job creation 126

Entrepreneurship and enterprise growth 126
Start-up of enterprises: incentives and impediments 133
Obstacles to expansion 138
Policies to promote entrepreneurship 147
Assessment and recommendations 151

Notes 154

Annexes

I. Supporting material to Chapter II 161
II. Calendar of main economic events 164

Statistical annex and structural indicators 169

Boxes

1. Tax structure and labour-market outcomes 71
2. Implementing the OECD Jobs Strategy – an overview of progress 81
3. The educational framework: initial education 89
4. The educational framework: adult education and training 110
5. Educational reform in the 1990s 120
6. Recommendations to improve effectiveness and efficiency in the education sector 124
7. A recent example of Swedish entrepreneurship: IKEA 128
8. Government programmes for assisting new, small and medium-sized enterprises 148
9. The Swedish government's programme for growth and employment 150

Tables

Text

1. Main demand components 19
2. Export performance 25
3. Commodity composition of goods exports and imports 28
4. Balance of payments 29
5. Wages and salaries in the private sector 33
6. Economic policy assumptions and the external environment 35
7. Short-term outlook 36
8. Budget plans and outcomes 50
9. Contributions to budget consolidation 53
10. The central government borrowing requirement 54
11. Expenditure ceilings 1997-2000 60
12. Public sector revenues and expenditures in selected OECD countries 62
13. Education expenditure, students and teachers 93

14. Monthly net income of youths in education, at work and on social benefits 94
15. Student-to-teacher ratios and teaching hours by level of education 95
16. The achievement of Swedish pupils in an international comparison 99
17. Percentage of students leaving secondary education without a certificate 100
18. Productivity levels in OECD countries 102
19. Output from and input into the education system 103
20. The effect on hourly wages of post-compulsory schooling 105
21. Effects of taxes and grants on the return from undertaking university education 106
22. Traditional and alternative education avenues 108
23. The effects of formal adult education (*Komvux*) 115
24. Enterprise-organised training in the private sector by enterprise size and length of training 118
25. Share of employment in selected sectors, by enterprise size 127
26. Company characteristics – IKEA 128
27. New enterprises and bankruptcies by sector 130
28. Registration requirements and costs 137
29. Bankruptcy proceedings 138
30. Effective marginal tax rates 143
31. Characteristics of the distribution sector in some European countries 146

Annexes

A1. Inflation targeting regimes in OECD countries 162

Statistical annex and structural indicators

A. Selected background statistics 170
B. Supply and use of resources (current prices) 171
C. Supply and use of resources (1991 prices) 172
D. General government revenue and expenditure 173
E. Balance of payments 174
F. Labour 175
G. Production by sector and employment 176
H. Foreign trade, total and by area 177
I. Prices and wages 178
J. Money and credit 179
K. Production structure and performance indicators 181
L. Labour market indicators 184
M. The public sector 185

Figures

Text

1. Macroeconomic performance 18
2. Private consumption 20
3. Price index for houses and Stockholm stock exchange index 21
4. Gross fixed investment by sector 22
5. Relative prices and trade performance 24
6. Resource use in the manufacturing sector 26
7. Resource use in the service sector 27
8. Hours worked and productivity growth 30
9. Unemployment, labour market programmes and labour supply 32
10. Short-term interest rates and the exchange rate 40
11. Inflation expectations 42
12. Bond yields 43
13. Indicators of monetary conditions 46
14. The Taylor rule 47
15. Components of relative unit labour cost 49
16. General government budget balances 55
17. General government debt and the primary balance 56
18. General government expenditure and tax ratios 58
19. Taxation at different income levels 68
20. Effective tax rates at different income levels 69
21. State of implementation of the EU directives on the internal market 75
22. Structure of the labour force 79
23. Educational spending 85
24. Population and labour force by highest educational attainment 86
25. The Swedish education system 87
26. Enrolment trends in universities 92
27. Enrolment and time use in education 96
28. Literacy levels of adults 98
29. Marginal tax rates 106
30. Features of labour market training 112
31. Fast-growing companies in selected sectors 131
32. Tax wedges and the service sector 136
33. Impediments to enterprise growth 139
34. Venture capital 141

BASIC STATISTICS OF SWEDEN

THE LAND

Land area, (1 000 sq. km)	411
Lakes (1 000 sq. km)	39
Arable area (1 000 sq. km)	30
Woodland (1 000 sq. km)	228

Inhabitants in major cities, including suburbs (31 December 1995) thousands:	
Stockholm	1 570
Göteborg	767
Malmö	503

THE PEOPLE

Population (31 December 1996), thousands	8 924
Number of inhabitants per sq. km	20
Net natural increase (average 1992-96), thousands	15.8

Net natural increase per 1 000 inhabitants (average 1992-96)	1.8
Net migration (average 1992-96), thousands	40.2

PRODUCTION

Gross domestic product in 1996 (Kr billion)	1 688.2
GDP per head, US$	28 206

Gross fixed capital formation in 1996:	
Per cent of GDP	14.8
Per head, US$	4 170
Employment, 1996:	
Total civilian, thousands	3 963
Per cent of total:	
Agriculture, forestry, fishing	2.9
Industry	26.1
Other	71.0

THE GOVERNMENT

Per cent of GDP in 1996:	
Public consumption	26.2
General government current revenue	61.6
Public gross fixed capital formation	2.0

Composition of Parliament (number of seats):	
Social Democrats	161
Conservatives	80
Centre	27
Liberals	26
Left Party	22
Environmental party	18
Christian Democrats	15
	349

Last general election: September 1994
Next general election: September 1998

FOREIGN TRADE

Exports of goods and services, 1996 as per cent of GDP	40.0
Main exports in 1996 (per cent of total merchandise exports):	
Wood, pulp and paper	15.0
Metal and metal products	10.1
Machinery and electronic products	33.8
Transport equipment	14.1

Imports of goods and services, 1996 as per cent of GDP	33.3
Main imports in 1996 (per cent of total merchandise imports):	
Food, beverages, tobacco	6.9
Fuel and energy	8.6
Machinery and electronic products	31.5
Transport equipment	9.9

THE CURRENCY

Monetary unit: Krona

Currency units per US$, average of daily figures:	
Year 1996	6.71
December 1997	7.79

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Sweden by the Economic and Development Review Committee on 15th December 1997.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 15th January 1998.

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The previous Survey of Sweden was issued in December 1996.

Assessment and recommendations

Overview of current policy issues

The pace of economic growth has quickened in the period since the Swedish economy was last reviewed in October 1996, and the upturn is projected to strengthen into 1998 and 1999. The prerequisites for renewed growth have improved significantly as balance is being restored to public finances and as the credibility of the monetary policy framework has strengthened, serving to ease financial-market pressures and to bring about lower interest rates. Against this background, the *Survey* begins with an analysis of the forces behind the expansion and the contributions made by monetary and fiscal policy, while also discussing the medium-term policy implications of the Swedish decision not to be a member of the Economic and Monetary Union at its inception. The *Survey* then examines progress in structural reform from the perspective of the recommendations made in the *OECD Jobs Strategy*. Labour market imbalances have persisted despite improved output prospects and they remain both the main risk to the sustainability of the upturn and an obstacle to a permanent improvement in economic performance. Strengthening the competencies of the workforce has taken on crucial importance in the government's strategy of reconciling high overall employment with low wage differentials, and a separate chapter reviews efforts to improve education performance and resource utilisation in formal and adult education. Regulatory and incentive structures which allow entrepreneurs to exploit business opportunities are also integral to a

successful jobs strategy and factors influencing the entrepreneurial climate in Sweden are assessed in the final chapter.

A broader-based pick-up in activity should be sustained into 1998-99

Following the pause in late 1995 and early 1996, GDP growth increased to an estimated 1¾ per cent in 1997. Exports have been the mainstay of the expansion, but both private consumption and business investment have benefited from falling interest rates and capital gains following from higher asset prices, so that domestic demand growth is now playing an equally important role. With the 1994-98 fiscal consolidation programme coming to an end, the growth in disposable incomes and private consumption is set to pick up, the contribution from net exports diminishing as imports respond strongly to higher domestic demand. GDP growth may be around 2½ per cent in both 1998 and 1999. Productivity gains have been the main engine of growth, but the fall in employment appears to have bottomed out in mid-1997, and moderate employment gains are expected over the next two years. Unemployment should fall below 8 per cent in 1998 and approach 7 per cent in 1999, excluding the internationally large share of the working-age population in active labour-market policy programmes.

Wage- and price inflation have fallen, but wage growth will hold back employment expansion

Wage growth has slowed, following the profile set out in the three-year settlement agreed in the 1995 bargaining round. Against the background of heavy labour-shedding, those agreements which came up for renewal in 1997 (25 per cent of the labour market) resulted in moderate, one-year, increments to wages. The 1998 wage bargaining round will cover all agreements in the labour market and wage moderation should be sustained by the fall in inflation expectations: though price inflation picked up during 1997 as indirect tax increases were increased and the direct

effects of falling interest rates and krona appreciation unwound, it should nevertheless remain around 2 per cent. Conciliation mechanisms recently established by the organisation of employers and employees in manufacturing and retail trade should help to contain bargained wage increases and allow exporters to maintain competitiveness. However, given the rigidity of wage relativities within the economy, it is unlikely that wages in the economy at large will be consistent with a marked expansion of employment. Indeed, with the wage round taking place in an environment of increasing overall labour demand and shortages of skilled labour in some sectors, the danger of too-high overall nominal wage increases is the chief risk to the projections. Such a development would require a monetary response, forcing producers to reduce costs through further labour shedding.

Having achieved price stability, monetary policy is supporting the expansion

Monetary policy has played a crucial role in the disinflation process and is helping to sustain the upturn. The inflation-targeting framework, which sets a target for headline inflation of 2 per cent with a fluctuation margin of 1 per cent on either side, has gained increased credibility over the past two years as inflation expectations in both labour and financial markets have been contained within the tolerance range of 1 to 3 per cent. Tight fiscal policy, receding price expectations and an appreciation of the Swedish krona created scope for a significant lowering of short-term rates during 1996. Short- and long-term interest differentials against German rates have narrowed considerably and this has translated into lower lending rates of mortgage institutes and banks. With the 2 per cent inflation objective now embedded in the plans of all parties in the labour market, monetary conditions should remain favourable to growth. A further reduction in the risk premium in Swedish interest rates should occur as sound public finances and low inflation becomes more deeply embedded. In this respect, the

recent comprehensive revision of the legal framework for monetary policy will help underpin the inflation objective by establishing the goal of price stability in law and the central bank's independence. Inflation targeting implies the precedence of domestic stabilisation objectives over a relatively stable exchange rate so that as the output gap narrows, rates may need to rise to restrain domestic demand. Since the present Swedish cycle now appears to be synchronised with that of the ERM core, the rise in short-term interest rates will tend to coincide with higher external interest rates.

Monetary policy requirements are similar outside as inside the EMU

While Sweden appears set to meet the membership requirements for Economic and Monetary Union, weak public support for introducing the Euro means that the country will not participate in the monetary union at its inception. This means that Swedish monetary policy will continue to be anchored by a national inflation target. The policy requirements stemming from the objective of price stability will, however, remain the same, whether or not Sweden is bound into an irrevocably fixed exchange rate. Indeed, if the inflation objective is to be fully credible, it has to be the overriding objective of the central bank. It may be argued that non-participation in the monetary union enhances monetary policy discretion and avoids the risk that an EMU-wide monetary stance could in some conditions become inappropriate for Sweden's economic circumstances. In principle inflation targeting allows for cyclical asymmetries to show up in interest-rate differentials and exchange-rate fluctuations. But the inflation target also implies that Swedish inflation will develop in parallel with its neighbours, so that exchange-rate fluctuations would have to broadly cancel out over time and the option to cushion increasing relative unit costs via the exchange rate is no longer available. It should also be noted, in this respect, that in the past

monetary accommodation of this type and subsequent policy correction have been more important than fundamental differences of economic structure in causing the Swedish economy to be out of phase with other European countries. Instead, the challenge is to make wage behaviour more flexible, which would also allow a speedier adaptation to asymmetric shocks should Sweden exercise the option of entering EMU at a later date. Sweden's financial-market structures appear to be well-adapted to prevent any problems which might stem from the operation of a common monetary policy.

Fiscal balance has been restored and medium-term goals are ambitious

Fiscal consolidation has been a fundamental ingredient in recreating confidence in the Swedish economy and the continuing commitment to sound public finances has been written into the Swedish Convergence programme to the European Union covering the years 1995-98. The programme, adopted in 1994-95 with a minor reinforcement in 1996, included both tax increases and expenditure cuts to strengthen general government finances by some 8 per cent of GDP. In addition to this programme of fiscal consolidation, the local government expenditure share of GDP will have fallen by close to 2 percentage points by 1998. From a general government budget deficit of above 10 per cent of GDP in 1994, public finances are scheduled to be back in balance in 1998. Assisted by an appreciation of the currency, privatisation proceeds and liquidation of assets to reduce debt, the general government gross debt ratio has been declining since 1994 and the borrowing requirement of central government has fallen close to zero. Recognising the challenges posed by the cyclical sensitivity of public finances and the contingent liabilities building up as a result of the ageing of the population, fiscal retrenchment is to continue towards an objective of an average 2 per cent budget surplus over the economic cycle.

... but the sensitivity of the budget to the cycle makes them vulnerable

This fiscal objective is more ambitious than the goals adopted by any of the future EMU participants, and the very high cyclical sensitivity makes its realisation dependent on maintaining substantial surpluses at the peak of the economic cycle. This requirement poses a particular challenge in terms of the need for continuous fiscal discipline in order to avoid the danger that periods of high surplus are used to increase public spending, necessitating offsetting (and pro-cyclical) fiscal action during the trough years of the cycle. Shifts of this kind, which can often result in a ratcheting up of tax and transfer-benefit rates are highly detrimental to growth and resource allocation. Having brought public finances back into balance, thereby restoring confidence in the overall policy framework, the challenge is to combine sound public finances with stability in the framework for private-sector decision making. To this end, the Swedish medium-term fiscal strategy is to allow the deficit for individual years to vary with the cycle, subject to the overall objective of a 2 per cent surplus over the cycle as a whole. Such a strategy should help to avoid the danger of pro-cyclical budgeting noted above. However, the framework remains sensitive to the underlying assessment of the potential growth rate of the economy and the evolution of output gaps. If by 2001, the economy may be close to the peak of the cycle, as is implicit in the OECD projections, further fiscal measures may be needed to achieve the overall surplus objective.

Greater priority needs to be given to cutting taxes and reducing transfer spending

Substantial challenges remain in controlling public expenditures and reducing taxation. A combination of tax increases and expenditure cuts was seen as necessary to gain widespread acceptance of the fiscal consolidation, so that Sweden continues to have a tax level matched only by Denmark among OECD countries and which imposes serious disincentives to savings, labour supply and human capital investment. Financial saving in excess of targets could

have been used for debt reduction or tax cuts rather than higher spending. As it is, the Swedish emphasis on welfare spending reflects deeply held egalitarian convictions in Swedish society. But the emphasis on social aims over incentives has to seen against the background that as large a share of benefits relates to redistribution over the life-cycle of the individual as to inter-personal redistribution, even when the old-age pension is excluded. Furthermore, the share of the population relying on transfers as their main source of income has persistently grown (from 16 to 31 per cent over a period of less than 30 years), a trend which will be exacerbated as the population ages unless corrective action is taken. Long-term budget sustainability requires *i)* a continued decrease in public expenditures as a share of GDP as a means of reaching the 2 per cent surplus target and allowing more scope for reducing taxation; *ii)* reducing tax expenditures, which presently amount to the equivalent of 8 per cent of GDP, in order to broaden the tax base and lower marginal tax rates and their associated welfare loss; and *iii)* supplementing the planned pension reform, which is set up to be robust *vis-à-vis* adverse macroeconomic and demographic developments with measures that reduce the government's contingent liabilities in social insurance. Such measures would rely, to a larger extent than is currently the case, on individual savings playing a role in providing for retirement.

The employment programme is less broadly-based than recommended in the Jobs Study follow-up

The government's employment programme reflects a continuing strong attachment to income-distribution objectives embedded in the traditional Swedish model. According to this approach, there exists a trade-off between equity objectives and increased work incentives, insofar as the latter would imply greater wage inequalities, particularly at the lower end of the pay spectrum. Where negotiated pay is too high to match marginal productivity, the emphasis of the model is on raising labour efficiency via investment in

human capital. The employment programme thus relies on efforts in adult and tertiary education to upgrade the qualifications of the workforce and hence validate the existing strong compression of wages. An emphasis on human capital enhancement is consistent with the recommendations of the *OECD Jobs Strategy*, but the strategy also emphasises the possibilities for improving incentive structures while also enhancing equity objectives. In particular, unemployment risk is by far the most important determinant of life-time income inequality. And relatively large income differentials between new entrants and established members of the work-force may be consistent with life-time equity objectives where there is progression through the income scale. At present, access to ordinary jobs is difficult for low-skilled entrants to the workforce, including a large segment of the youth cohorts, acting to prevent them from acquiring on-the-job training and moving on to better-paid jobs. The 1996 *OECD Economic Survey of Sweden* thus offered a broad set of recommendations designed to increase wage and labour cost flexibility, re-formulate unemployment benefits and related transfer schemes and enhance active labour market policies, as well as to raise human capital formation.

... and needs also to reform incentive structures in the labour market

The reform programme incorporated in the 1996 Employment Bill embraced quite broad-ranging proposals for improving labour market incentives to employers and employees. Most of these proposals have now been implemented, but the decision to impose a total time limit for unemployment benefits was reversed before entering into force and has subsequently been abandoned. The legal framework for wage bargaining will be basically unchanged when the social partners enter the 1998 round, further efforts in this field awaiting committee recommendations. Pressures to modify wage structures have generally been impeded by recent social-security and labour-market

initiatives, encompassing not only higher replacement rates but also a temporary public sector-job programme for older workers and an offer available to the end of 1997 for older long-term unemployed to take early retirement at UI benefit level. Modifications to existing employment protection legislation, *inter alia* with respect to fixed-term contracts, give enterprises an additional measure of flexibility, but the last-in-first-out rule remains the basic principle of the law and the added scope for concluding agreements at local level has been resisted at industrial union level. A framework conducive to long-term growth and employment creation would need to implement the full reform programme set out in the Employment Bill.

The performance of the education system is crucial for the employment strategy

Against this background, the effort to improve the performance of the Swedish education system takes on a crucial importance for the success of the government's employment strategy. Education has always been an integral feature of the Swedish economic and social model, combining the aims of equal opportunities and employment generation through rapid human capital formation. The total public and private resources devoted to education in Sweden are among the highest in the OECD area. Functional literacy levels are high by international comparison. However, the relative decline in Swedish economic performance has focused attention on the failure of the system up to the early 1990s to produce highly-qualified labour which can adapt to new requirements and changing consumer preferences. The expansion in higher education is meant to address this problem. The focus on education has been underpinned by a recognition that large differences in unemployment risk can only be avoided if the competencies of low-skilled workers are upgraded to match the strong compression of wage differentials. The formulation of effective policies in these respects has embraced both the formal education system and the scope and design of adult education. Building

on an enrolment rate in upper secondary education of close to 100 per cent, the long-term expectation is to have 50 per cent of youths enter tertiary education. Adult education capacity is also being extended, in an effort to reach the estimated 1 million persons with only primary education or outdated formal competences and provide them with competences at upper secondary level.

Increased outlays on formal education need to be accompanied by greater efficiency

If ambitions in the educational sphere are to be achieved, and additional outlays used to best effect, some specific weaknesses of the formal education system need to be addressed. Sweden has a relatively well-educated population, but the output of the formal education system has, nevertheless, been unsatisfactory in terms of years of schooling received, the slow transition from secondary to tertiary education and the long spells spent in tertiary education. Moreover the performance of the system is in some respects inadequate as regards the achievement of Swedish pupils in international aptitude tests, while there are deeply entrenched differences in skill acquisition among members of a given age-cohort. Following a wide-ranging decentralisation in 1991, the responsibility for dealing with inefficiencies at the primary and secondary levels has partly fallen on municipalities, which now allocate resources to primary and secondary schools. On the other hand, the slow transition of pupils through the system should be seen in light of national standards which for a long time embraced a wide range of (non-educational) criteria for advancement and set little store by nationwide pedagogical evaluations. Moreover, a generous students' grants system has made long courses at tertiary level economically feasible. The *Survey* makes a number of recommendations to correct these weaknesses:

- *To ensure the acquisition of basic skills at compulsory level*, the time both of teachers and pupils could be utilised more fully. Teaching hours are low. Recent agreements

between municipalities and teachers' organisations on wages and working conditions should lead to a more extensive use of school facilities and pupils' time during working hours, while rewards for educational innovation should pave the way for performance-related pay;

- *To stimulate educational innovation*, the decentralisation of primary and secondary schooling should be completed. This implies a more explicit use of educational vouchers at municipal level, more explicit independence for school administrations combined with more parental influence, and a continuation of the trend towards decentralised wage setting, all of which options are available to the municipalities and their employee organisations within the existing legal framework;
- *To ensure more uniform educational standards*, nationwide testing should be extended and possibilities of progressing with incomplete marks reduced as will be the case in upper-secondary education from 1998;
- *To reduce the time spent in both upper secondary and tertiary education*, the structure of student grants and loans should be reviewed. A more generous education grant has been seen as a way of maintaining private incentives to education, but such a strategy entails heavy dead-weight costs as grants have to be financed out of general taxes, and work only through a general incentive to study and do not provide guidance as to which studies to pursue.

... while the returns to vocational education may be lower than anticipated

With respect to vocational education, the shortcomings of the formal education system prompted a series of reforms to upper secondary education over the 1992-96 period. All programmes have been standardised at three years, and minimum requirements have been set for enterprise-based training. The abandonment of traditional apprenticeship training decades ago is also about to be reversed, with a modernised system being set up and extended to tertiary

level. Public efforts to increase the provision of formal education for above-age students will rely on supplementing these freely-provided education services with more generous grants and borrowing facilities. The overall effect may be to alleviate the consequences of not seeking an earlier qualifying education. But insofar as these efforts are not supplemented by increased reliance on price signals from the labour market to guide the allocation of labour and educational choice, there is a clear risk that the overall returns may be lower than currently expected. Efforts in this sphere is not to be viewed as a long-term substitute for maintaining and improving the output of the formal education system for standard-age pupils and modifying labour market structures to give incentives for youths to seek out education in demand by enterprises.

The adult education effort needs to be more operationally relevant

In the sphere of adult education and training there is a division of responsibilities that leaves the public sector as the main provider of training for the unemployed and enterprises as the provider of training services for their employees. This should be the most efficient way of organising this market, relying on private incentives for the broad segment of the workforce while focusing public resources on those affected by labour market imperfections. This division of responsibility has been strengthened over the 1990s. However, enterprise-based training is heavily skewed in favour of the highly-qualified and would, more importantly, seem to a large extent to offer training as a supplement to wages paid. The allocation of training resources within enterprises would benefit from individual wage contracts that reward long-term employer-employee commitments.

Product markets are competitive, but deregulation should be completed, and barriers to expansion should be reduced

One of the conclusions of the *OECD Jobs Study* was that the countries best equipped to exploit future job-creation opportunities will be characterised by competitive product markets and dynamic entrepreneurship. In the Swedish case, the *Survey* concludes that the economy is relatively free of competitive barriers, but that the process of completing regulatory reform is now progressing rather slowly in some areas. While the regulatory framework for the public utilities has been extensively liberalised in sectors such as air-transport, postal and telecommunications sectors the incumbent still exerts significant market power. Competition is still also weak in a number of other domestic-oriented sectors, such as distribution, construction, food-processing and agriculture, owing to a variety of government interventions (*inter alia* zoning regulations and support schemes). Perhaps of greatest significance for future growth potential is the fact that the efficiency of the large-enterprise sector, with its dynamic multi-nationals, contrasts with the apparent inability of small enterprises to expand into and beyond medium size. Policy efforts have tended to rely on targeted tax relief for new and small business, as well as on building institutions to stimulate diffusion of knowledge and technology. However, the difficulties faced by small enterprises with respect both to start up and expansion may be attributed to weaknesses in their legislative and economic environment:

- High marginal taxes and incomplete loss offsets may be seen as a deterrent to entrepreneurship, and several features introduced to support smaller enterprises act to produce high effective marginal tax rates as they expand;
- Labour-market regulations also act to impose costs on small enterprises, insofar as they have less capacity to adapt to the extensive employment protection legislation and may be particularly vulnerable to secondary action allowed under industrial relations legislation. No separate provisions for small and medium-sized enterprises have been introduced.

... calling for initiatives to nurture a more entrepreneurial climate

To deal with these impediments to expansion, new initiatives would be needed to reduce the constraints imposed by job-protection legislation, if necessary by granting statutory exemptions to small enterprises and to further promote a ‘‘neutral’’ system of taxation, which does not penalise companies for expanding. Enforcement of competition laws should include fair and equal competition with government entities providing market-produced goods and services. The introduction of further support schemes for selected parts of the business sector does not seem advisable. A case might be made for increased local authority involvement in nurturing an entrepreneurial climate. In some cases, regional authorities and municipalities might be better placed than the central government to diagnose and solve the problems. However, the most important role for public authorities remains the provision of a stable macroeconomic situation, a neutral system of taxation, a well-functioning labour market and a competitive business sector environment.

Summing up

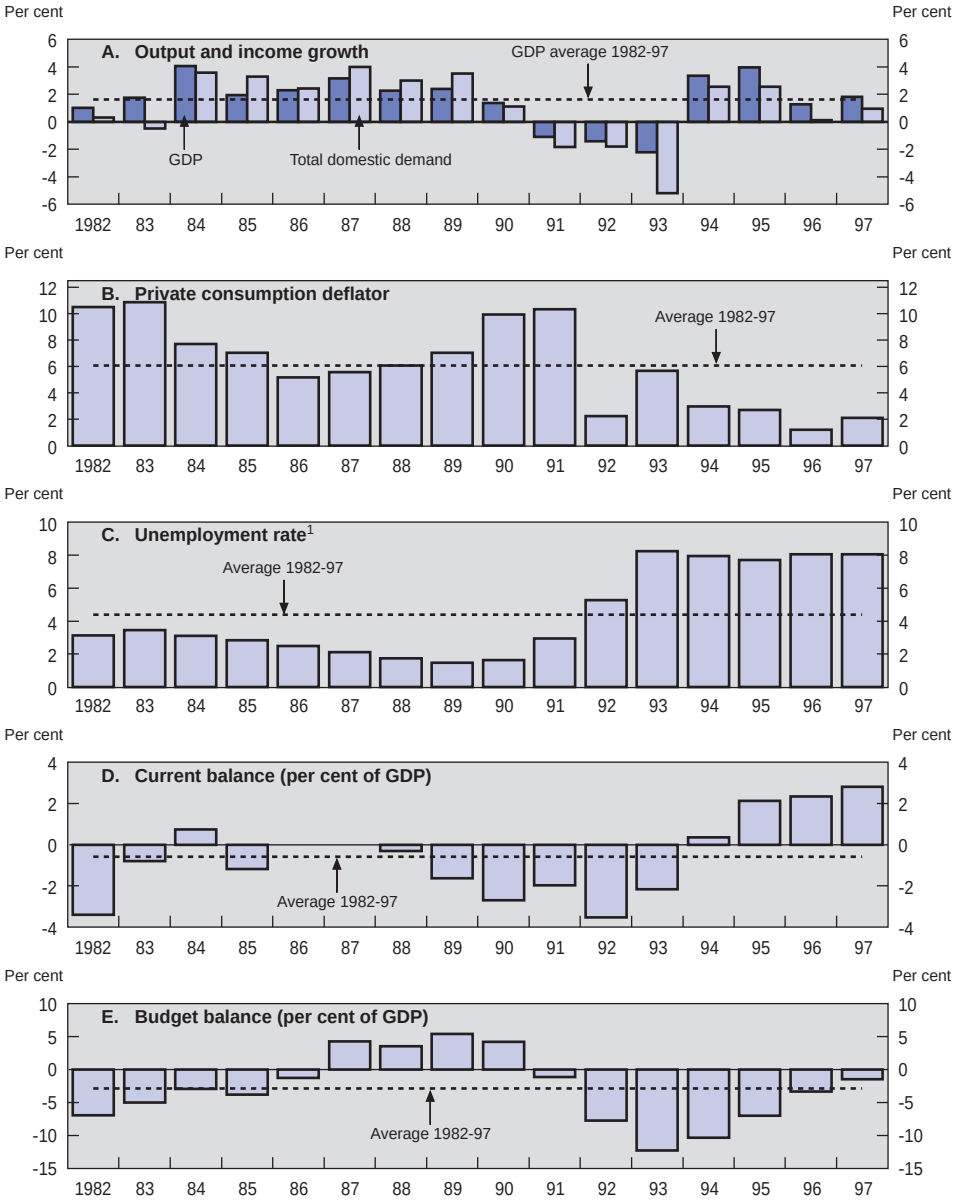
To sum up, the improvement of macroeconomic fundamentals has proceeded at a rapid pace over the past few years and is remarkable in an international context. Fiscal discipline has been restored, helping to underpin the inflation targeting framework of monetary policy, which has gained broad acceptance and credibility in both labour and financial markets. The benefit to the economy is the gradual upturn in domestic demand, fuelled by low interest rates, which should make for above-potential economic growth in the next two years. Progress has also been made in structural reform, but more slowly and less completely, and it is an open question whether this will be sufficient to raise the long-term performance of the economy, after a number of decades when Swedish growth has been lower than the average for the OECD. Product markets are flexible, but labour market structures are more rigid; they weaken employment prospects for youths and the low-skilled,

necessitating high public outlays for education, labour market programmes and income-replacing transfers. In this context, a stronger emphasis on improving incentives to job-seekers would serve to improve labour market outcomes, particularly if combined with polices to raise the return on the substantial resources committed to education and to increasing the employment potential of the small and medium-sized enterprises. This would provide the basis for a fuller and more balanced use of the country's resources while ensuring that future social welfare commitments are fulfilled.

I. Recent trends and prospects

The economy is now in the fifth year of an expansion which has relied on exports and business fixed investment for most of its momentum (Figure 1). Reflecting this dependence, the strength of activity has fluctuated with shifts in Swedish competitiveness and capacity constraints in the tradeable sector; it accelerated strongly until the latter part of 1995, then slowed and has subsequently quickened again since mid-1996 (Table 1). Business sector investment is still the most buoyant demand component, and export growth has kept up well in the face of the real currency appreciation since mid-1995, but the upswing is now being underpinned by an acceleration in private consumption growth. Public consumption and investment are continuing to decline, reflecting the ongoing process of fiscal consolidation. Housing investment, having rebounded in 1996 as the result of temporary investment subsidies, collapsed upon their withdrawal at the end of 1996. Unemployment is starting to fall, but this reflects higher enrolment in adult and tertiary education, while employment creation has remained weak, although it should gradually pick up as the upswing strengthens. Monetary conditions are favourable and the termination of the fiscal consolidation programme will feed into higher growth in household disposable income and consumption. Underlying inflation is low and should remain close to 2 per cent. However, the expansion will continue to be based on strong productivity growth in manufacturing, while strong wage-wage links could lead to average wage increases in excess of productivity gains in other sectors, restricting the growth of labour demand.

Figure 1. **MACROECONOMIC PERFORMANCE**



1. Labour force survey.
Source: OECD.

Table 1. **Main demand components**

Seasonally adjusted growth at annual rate, per cent

	1995	1996	1995		1996		1997
			I	II	I	II	I
Private consumption	0.8	1.3	0.4	0.7	0.9	2.7	1.5
Public consumption	–0.9	–0.2	0.5	–2.3	–1.4	4.6	–7.4
Gross capital formation	12.4	3.7	18.2	12.6	2.3	–1.6	–3.7
Housing	–23.5	13.1	–27.8	3.7	28.7	–4.7	–27.4
Business	27.7	4.7	41.7	14.9	2.0	0.9	6.1
Public	–5.7	–8.0	–7.5	–10.9	–9.2	–2.5	–5.9
Change in stocks [1]	0.5	–1.1	–0.1	–0.4	–0.9	–2.2	2.0
Exports of goods and services	12.9	6.1	16.0	4.7	5.1	9.5	11.2
Total demand	5.4	1.9	6.5	2.3	1.1	2.9	3.6
Imports of goods and services	10.2	3.7	11.7	4.9	1.4	7.3	13.7
GDP	3.9	1.3	5.5	2.2	0.0	2.8	0.6

1. Contribution to GDP.

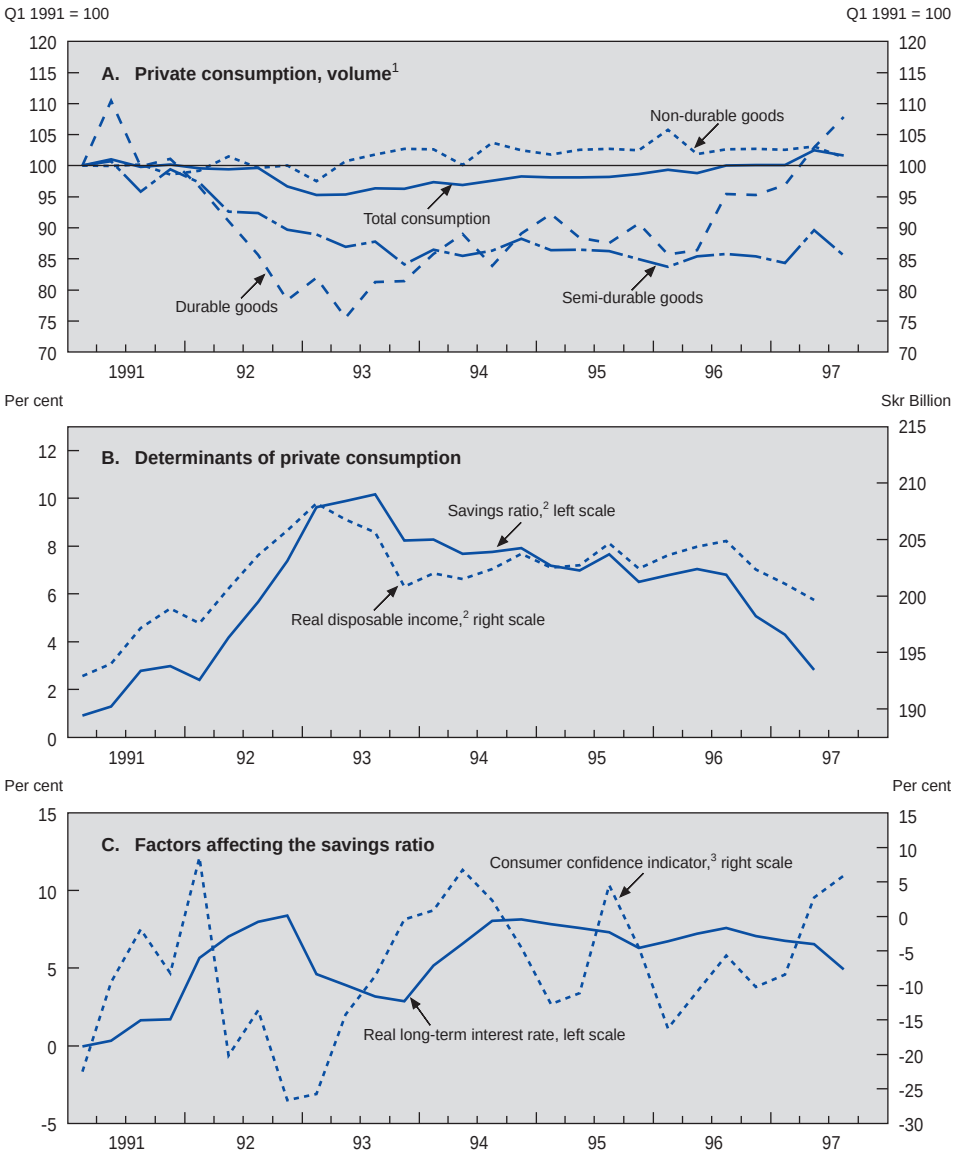
Source: Statistiska Centralbyrån (SCB); OECD.

Economic activity remains moderate

Private domestic demand continues to expand

The pick-up in private consumption has mainly affected durable goods (Figure 2, panel A), reflecting pent-up replacement demand as a result of the low spending on such items in the 1990s. New registrations of cars, for example, dropped by 60 per cent between 1989 and 1993, but are currently recovering; in 1997 they were close to one-third higher compared with 1996. Spending on semi-durable goods is also growing, indicating that the period of consumer retrenchment has ended. In all, the growth in private consumption has been sufficient to bring it back to the level prevailing in the beginning of the 1990s. Consumer spending, however, has not been financed by increased disposable income, but by a decline in the savings ratio (Figure 2, panel B). Consumer confidence has risen with improved expectations both for the outlook for Swedish economy and for households' own economic situation as the fiscal consolidation process is completed (Figure 2, panel C). The recovery of the housing market has also quickened (Figure 3) with both higher prices and turnover. Higher house prices generate positive wealth effects on consumer spending, while higher turnover

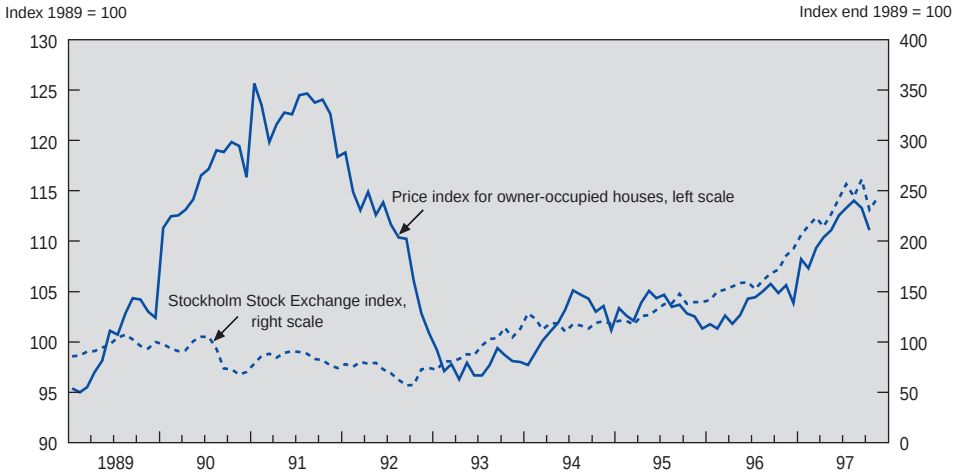
Figure 2. **PRIVATE CONSUMPTION**



1. Seasonally adjusted.
2. Seasonally adjusted, 4 quarter moving average.
3. Breaks in the series in end-1992 and in end-1995.

Source: OECD; Statistiska Centralbyrån (SCB).

Figure 3. **PRICE INDEX FOR HOUSES AND STOCKHOLM STOCK EXCHANGE INDEX**

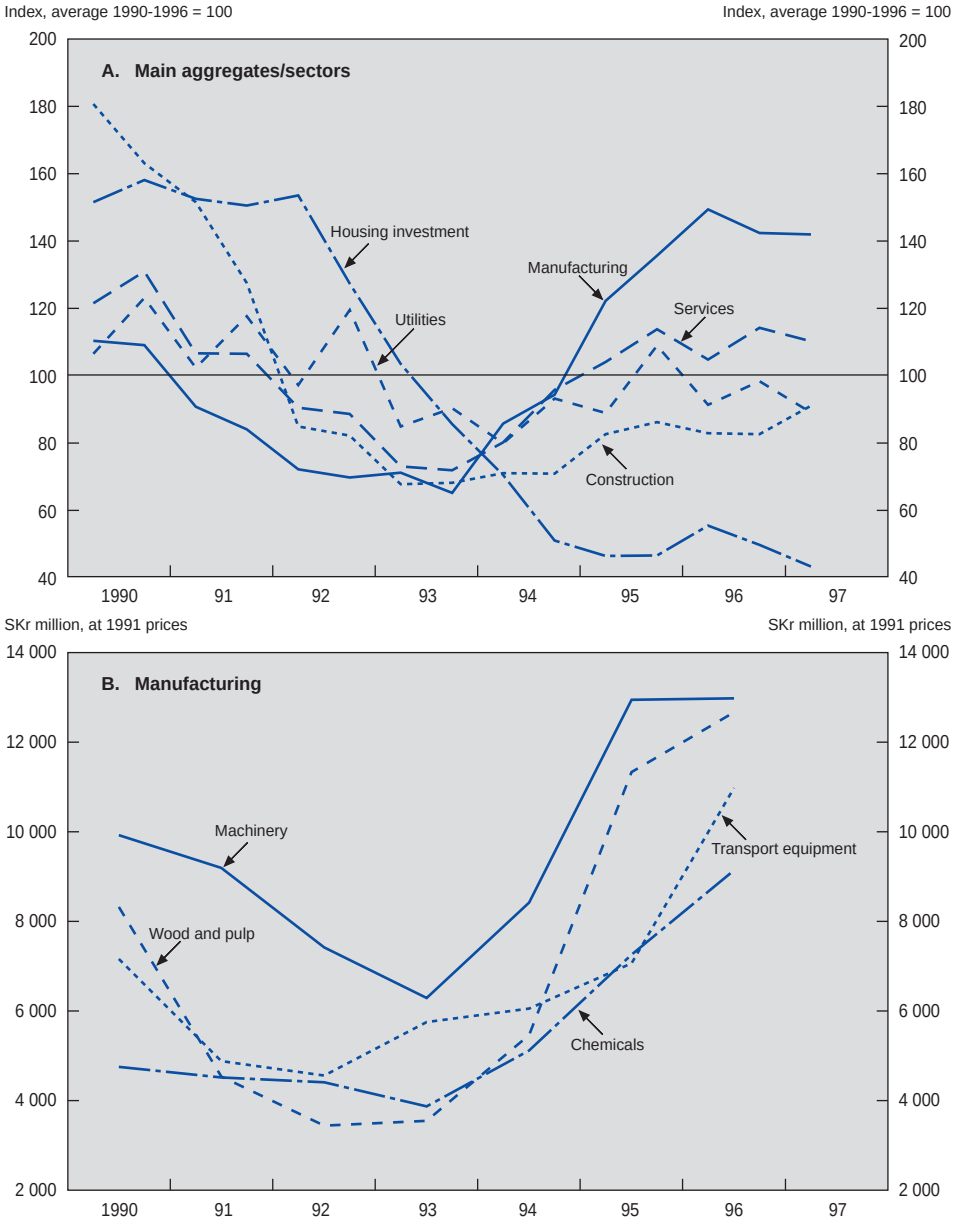


Source: SCB.

leads to additional spending on other durable goods, further underpinning private consumption growth. To this should be added the effects of the 24 per cent increase in share prices on the Stockholm exchange index during 1997, although the short-term relationship between share prices and private consumption has historically been rather weak.[1]

The recent recovery in house prices has not been sufficient to make up for the faster development in construction costs since 1990. The upswing in housing investment in 1996 can be linked to a temporary subsidy for the construction of new houses and housing improvements. The scheme was terminated in December 1996, thereby leading to a substantial fall in housing investment in 1997. A self-sustained recovery in housing investment would appear to depend on a further and more broadly-based[2] rise in house prices. Business investment, on the other hand, remains buoyant after the temporary weakening in early 1997 (Figure 4, panel A). The investment-to-GDP ratio increased to an estimated 12 per cent in 1997 from 9 per cent in 1994, and is close to the level prevailing at the end of the 1980s. The strength in investment should be seen against the

Figure 4. **GROSS FIXED INVESTMENT BY SECTOR**



Source: SCB.

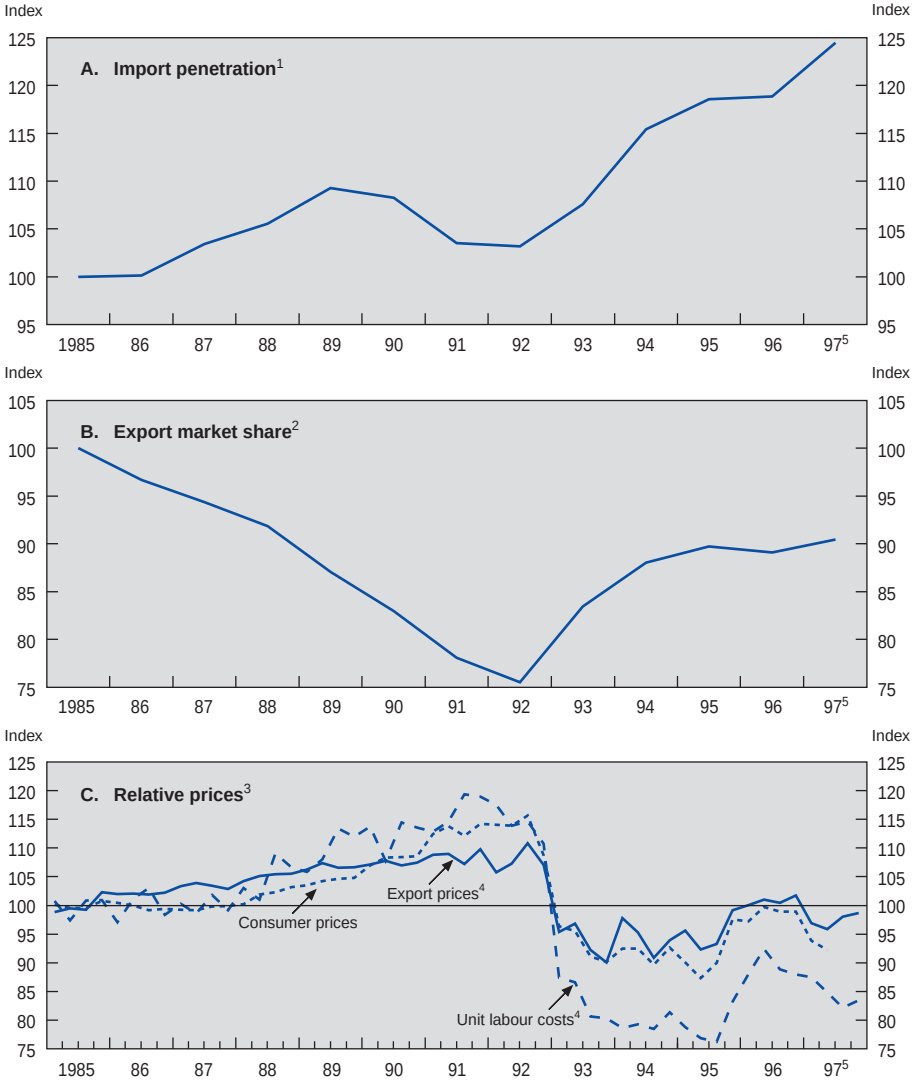
background of a somewhat deteriorating but still healthy profit situation in the business sector, a 2 percentage points drop in long term interest rates since early 1996, and strong demand for Swedish export goods. Scale-based industries with a reliance on price competition, such as wood and pulp, chemicals and transport, have seen particularly strong investment growth (Figure 4, panel B). In addition, service sector investment is recovering as business services have been boosted by the expansion in the manufacturing sector and as household services have benefited from the recovery in domestic demand.

Trade performance and the external sector

The pace of export growth has quickened since the slowdown in late 1995 and early 1996. However, the strong gains in export market shares – an increase of a fifth over the period 1993-95 – have come to an end (Figure 5, panel A). Unit labour costs have increased following the 11 per cent effective krona appreciation in 1996 and higher wage inflation (Figure 5, panel B). Nevertheless, relative Swedish export prices have been falling since late 1996 and a small increase in market shares may have been recorded for 1997 (Table 2, panel A). With respect to the destination of Swedish exports, exports to non-OECD countries – the fastest growing markets internationally up to mid-1997 – have been growing almost three times as fast as the market growth in those countries since 1993. Within Europe, Swedish goods have been gaining market shares in the Nordic countries, while the trade performance in other European countries, such as in the United Kingdom and Germany, has been less impressive in recent years. Germany, in particular, has become much less important as a destination for Swedish exports during the 1990s (Table 2, panel B). It should be noted that one single electronics company, Ericsson, has seen export growth of such a magnitude that it explains almost half of the total volume growth of exports. Swedish exports have also benefited from a favourable geographical composition, even though the recent turbulence in South-East Asia could pose a potential problem for Swedish exports. With export growth so far being driven by a limited range of products, Swedish exports could be sensitive to potential abrupt changes in demand patterns. However, during 1997 exports from the traditional industries (engineering, pulp and paper) strengthened, broadening the base for a continued expansion.

Figure 5. **RELATIVE PRICES AND TRADE PERFORMANCE**

Index 1985 = 100



1. Import of goods relative to import-weighted final demand (using 1985 input-output coefficients).
2. Manufactured goods at fixed 1991-prices.
3. Sweden relative to an export-weighted average of 25 OECD countries; measured in a common currency.
4. Manufacturing industry.
5. Estimates.

Source: OECD.

Table 2. **Export performance**

A. Export growth and aggregate market shares

	Seasonally-adjusted growth at annual rate, per cent						
	1994	1995	1996	1996		1997	
				I	II	I	II
Manufactured products							
Relative export prices	0.9	0.7	6.0	8.1	2.1	–10.0	5.1
Export volume growth	18.4	12.6	6.1	7.6	5.2	9.9	15.9
Market growth	12.3	10.5	6.8	7.7	6.3	9.9	8.2
Market share	5.5	1.9	–0.7	–0.1	–1.0	0.0	7.1
Total goods							
Export volume growth	16.3	11.4	6.0	7.2	6.1	10.1	14.5
Market growth	11.5	9.5	6.6	7.6	6.3	9.3	8.0
Market share	4.5	1.9	–0.6	–0.3	–0.2	0.8	6.0

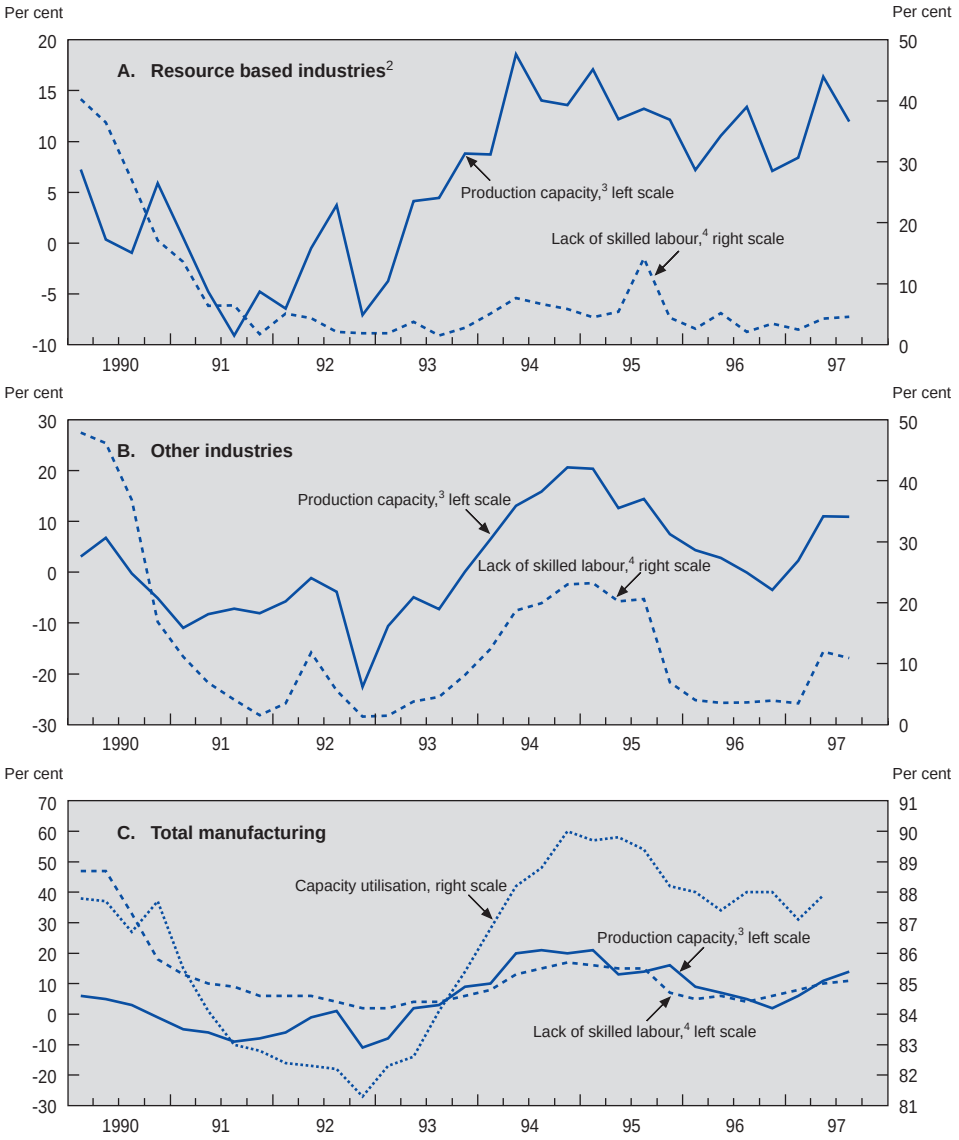
B. Swedish export market share in selected countries

	Per cent					
	Weight		1994	1995	1996	1997
	1993	1996				I
Germany	14.40	11.88	2.16	2.20	2.16	2.10
United Kingdom	10.25	9.69	2.72	2.90	2.82	2.52
Norway	8.15	8.57	18.28	18.59	21.12	19.07
United States	8.43	8.41	0.74	0.85	0.89	0.83
Denmark	6.64	6.39	11.76	12.02	11.89	11.28
Netherlands	5.08	5.63	2.64	3.11	3.51	. .
Finland	4.58	5.25	12.61	13.46	14.27	14.11
France	5.41	4.73	1.34	1.47	1.41	1.46
BLEU	4.16	4.49	2.48	2.19	2.47	2.39
NICs	1.45	1.67	0.18	0.19	0.21	. .
Other non-OECD	12.21	13.33	1.29	1.35	1.46	. .

Source: SCB; OECD.

Import penetration continued to increase from an average of 32 per cent of real GDP in 1996 to 34 per cent in 1997. The composition of demand explains roughly one percentage point of this increase, leaving import penetration about 1 percentage point higher than would be implied by conjunctural developments and unchanged import-sensitivity in the different demand components. As capacity constraints have increased since the beginning of 1997 (Figure 6), this development may partly be explained by domestic producers having short-term

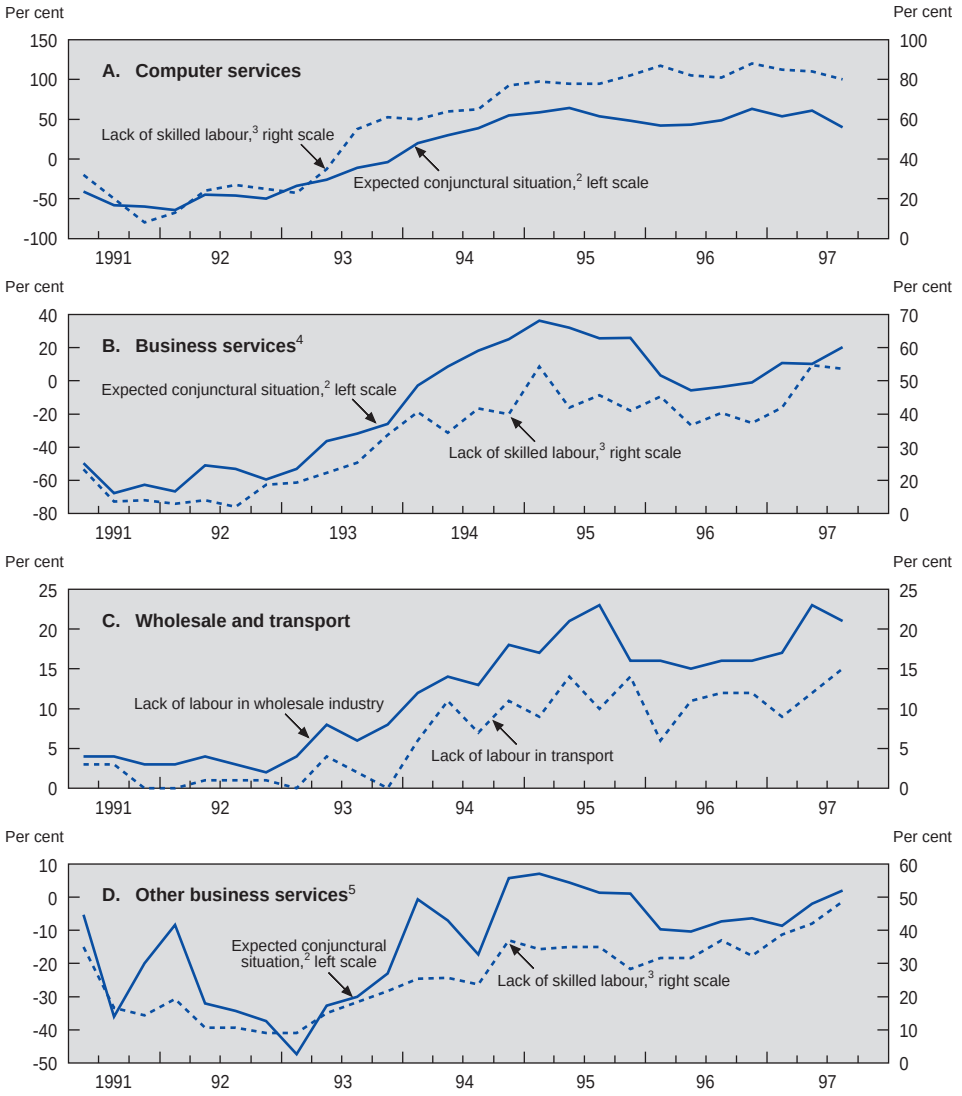
Figure 6. **RESOURCE USE IN THE MANUFACTURING SECTOR**[1]



1. Business survey data.
2. Wood, paper and pulp, minerals, iron and steel.
3. Balance between firms reporting increases (+) or decreases (-) in production capacity.
4. The share of firms reporting a shortage of skilled labour.

Source: National Institute of Economic Research (NIER); SCB; OECD.

Figure 7. **RESOURCE USE IN THE SERVICE SECTOR**[1]



1. Business survey data.
2. Balance between firms' optimistic (+) and pessimistic (-) answers regarding the conjunctural situation in the forthcoming semester.
3. The share of firms reporting a shortage of skilled labour.
4. Marketing, organisation and advertising services.
5. Consulting, accounting and legal services.

Source: NIER.

difficulties in meeting demand, and partly by the increase in demand for durable goods with a high import content, such as cars. With resources being far from fully utilised and ongoing investments expanding capacity, the manufacturing sector should on the whole be able to meet domestic and external demand increases even in the fairly short term. On the other hand, bottlenecks are emerging in some specific service industries with a demand for highly specialised personnel (Figure 7).

The favourable competitive position of Swedish exporters and the relatively weak domestic growth have led to exports growing more strongly than imports, with respect to both volumes and prices (Table 3). Both the trade and current account balances have improved (Table 4) to, respectively, 7 and almost 3 per cent of GDP in 1997. The improvement in the trade balance has been partly

Table 3. **Commodity composition of goods exports and imports**

Per cent growth

	Share in 1994	Volume			Prices[1]		
		1995	1996	1997[2]	1995	1996	1997[2]
A. Exports							
Total		12.3	6.4	10.5	7.9	–6.3	0.2
of which:							
Industry excluding electricity	99.4	12.2	6.1	10.4	7.9	–6.5	0.1
Intermediate goods	44.2	1.5	2.8	7.7	17.9	–8.9	0.3
Wood industry	14.2	–5.7	4.3	4.8	28.5	–16.4	0.7
Metal industry	7.9	1.9	1.6	10.0	16.0	–10.0	0.6
Other industry[3]	22.1	6.0	2.3	8.5	11.8	–3.8	–0.1
Investment goods	44.9	23.9	8.5	13.3	0.6	–4.9	0.1
Consumption goods	10.3	7.4	7.6	6.5	3.9	–2.5	0.5
B. Imports							
Total		10.5	2.6	9.5	4.9	–4.9	0.7
of which:							
Industry excluding electricity	96.8	10.8	2.3	9.7	4.7	–5.2	0.4
Intermediate goods	40.1	4.4	–1.2	3.9	11.0	–5.2	2.8
Wood industry	1.1	37.0	–17.6	14.6	18.2	–10.7	–0.7
Metal industry	5.4	3.9	–1.4	10.8	18.4	–14.5	1.5
Other industry[3]	29.0	5.2	–3.1	2.0	10.9	–5.0	3.1
Investment goods	38.5	25.3	4.9	13.9	–0.8	–6.1	–1.1
Consumption goods	18.2	–6.0	3.3	11.1	5.8	–2.6	0.3

1. Paasche price index.
2. Estimates.
3. Includes petroleum.

Source: Konjunkturinstitutet (National Institute of Economic Research – NIER).

Table 4. **Balance of payments**

SKr billion

	1994	1995	1996	1997[1]
Trade balance	67.1	108.5	118.6	100.4
Net services	–1.0	–3.6	–5.1	–5.6
Net investment income[2]	–45.8	–46.3	–50.3	–37.5
Unrequited transfers	–14.3	–21.0	–17.4	–12.7
Current account	5.9	37.6	45.8	44.6
Direct investment	–2.7	23.1	–3.1	5.7
Trade in shares	33.3	–50.4	–23.2	–78.3
Loans/bonds in foreign currency	–21.2	5.0	–20.0	–40.5
Loans/bonds in SKr	1.2	39.7	–7.3	–9.4
Changes in official reserves	–18.1	9.9	41.9	30.8
Others[3]	1.5	–64.7	–34.1	47.0
Financial account	–5.9	–37.6	–45.8	–44.6

1. First three quarters.
2. Including wages and salaries.
3. Including financial derivatives, other capital transations and net errors and omissions.

Source: Sveriges Riksbank.

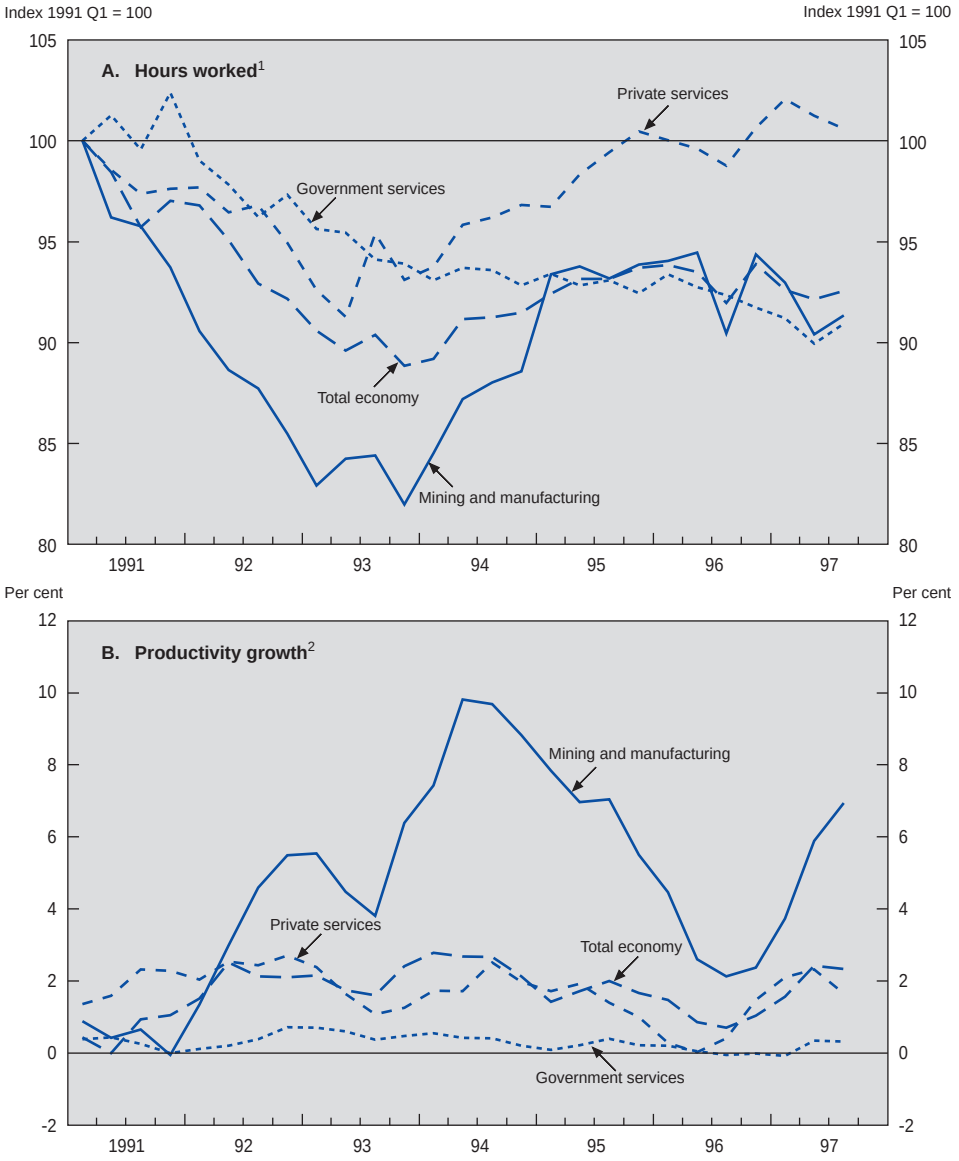
offset by higher interest payments on the external debt – which stood at 42½ per cent of GDP by end-1996 – and a further deterioration in the services balance, while transfer payments accompanying the EU membership still benefit from an introductory rebate. The capital account continues to reflect substantial portfolio changes, with large capital inflows from net foreign direct investment more than offset by large capital outflows in connection with Swedish purchases of foreign shares – mainly as the result of Swedish insurance companies being allowed to invest in foreign assets – and foreign investors selling Swedish shares. In addition, Swedish acquisitions of bonds denominated in foreign currencies have been much larger than foreign purchases of krona-denominated bonds.

The labour market and inflation

Continuing labour shake-out

Labour demand grew ¼ per cent in 1996 in terms of hours worked, but has fallen by about 1 per cent in 1997 (Figure 8, panel A), reflecting a decline in most

Figure 8. **HOURS WORKED AND PRODUCTIVITY GROWTH**



1. Seasonally adjusted.
2. Seasonally adjusted. 4 quarters moving average of productivity growth at annual rates.

Source: SCB; OECD.

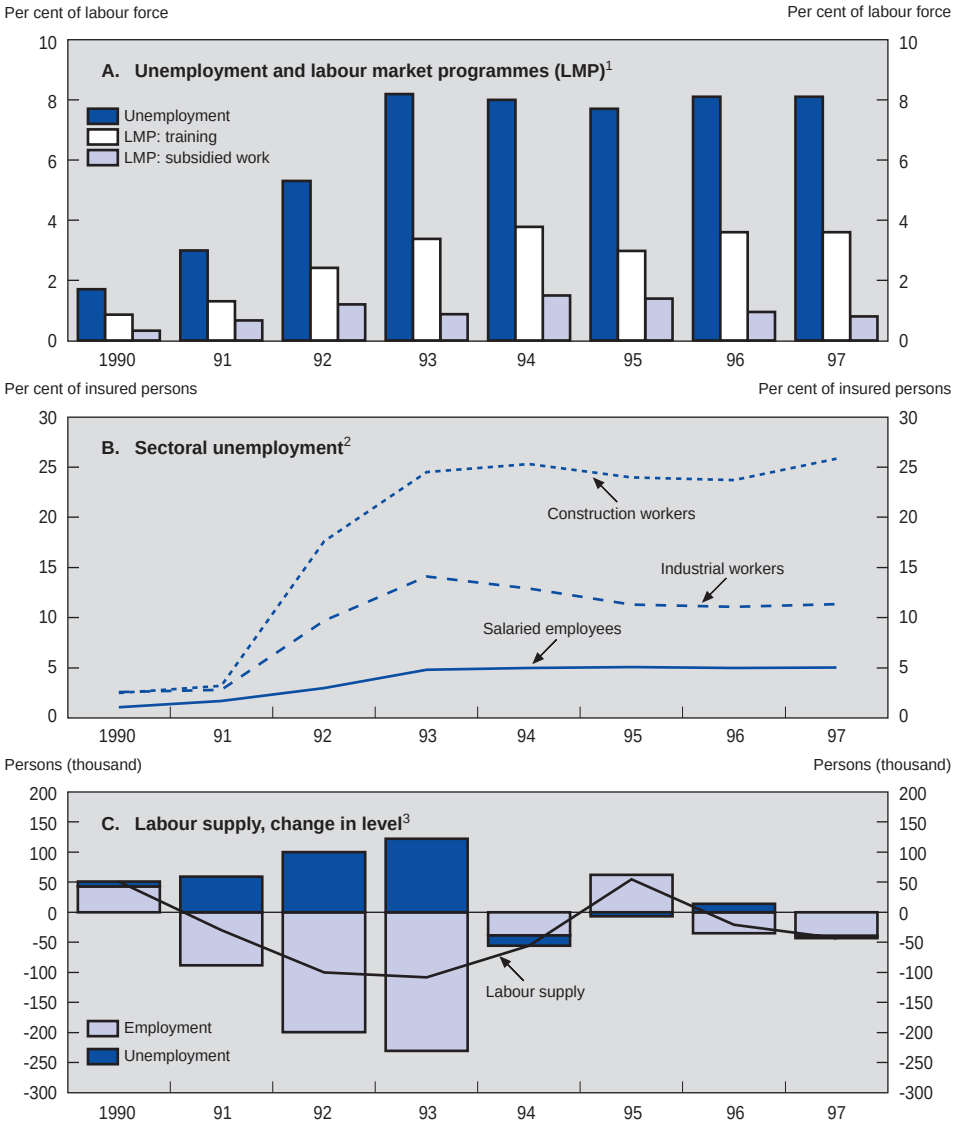
sectors: employment increased only in some utility and services sectors. The decline in labour inputs reflects the acceleration of productivity growth to 2¾ per cent in 1997 following a cyclical slowdown in 1996 (Figure 8, panel B). Average hours worked per employee grew through the early 1990s, but stagnated in 1996 and 1997, a continued fall in sickness leave being offset by a decline in overtime work.

Overall employment fell by 50 000 persons in 1996 and by 40 000 in 1997. However, the decline slowed somewhat during 1997, with employment in the fourth quarter only 20 000 lower than the year before, indicating that a cyclical trough had been reached. Unemployment was on an increasing trend throughout 1996 and the first half of 1997, but the rise was attenuated by a reduction of the labour force, explained by, among other things, an increase in the student population as the government continued to extend its education effort. The latter effect was particularly pronounced in late 1997, leaving unemployment at the end of the year lower than at end-1996, while the average for the year, at 8.0 per cent, was practically the same as in 1996 (Figure 9, panels A and B).

Faster wage growth and low inflation have led to considerable real wage increases

About 25 per cent of the work force were covered by new wage settlements in 1997 and for these the results were one-year agreements based on average wage increases of just below 3 per cent – indicating that lower inflation and inflation expectations are feeding into negotiated wages. For the rest, wage rates for the three-year period from early-1995 to early-1998 were set in the 1995 wage negotiations. The finalisation of these agreements was spread out through the year, so that recorded wage growth was artificially low in 1995 with a corresponding increase in early 1996. The wage agreements front-loaded wage growth for the period up to 1997. Consequently, there was a halving of annual wage growth to 4 per cent in the first semester of 1997 (Table 5), with particularly low wage drift. Wage increases have been roughly similar among different groups, reflecting the strong wage-wage links in Sweden. While the manufacturing sector has managed to balance real wage growth and productivity growth, the growth in real product wages has been well above the 2 per cent productivity growth in the private services sector. With producer prices falling for most of the period since the beginning of 1996, the private sector's profit share has fallen

Figure 9. **UNEMPLOYMENT, LABOUR MARKET PROGRAMMES AND LABOUR SUPPLY**



1. NIER estimates for 1997.
2. Insured unemployed persons as percentage of all insured persons in the relevant UI-fund. First semester data for 1997.
3. OECD estimates for 1997.

Source: NIER; SCB; OECD.

Table 5. **Wages and salaries in the private sector**[1]

Percentage change

	1995		1996		1997
	I	II	I	II	I
Hourly earnings					
Production sector[2]	4.5	6.2	7.8	5.5	4.0
of which:					
Metal industry	5.9	6.8	7.0	5.1	4.5
Construction	3.3	5.7	4.5	1.9	3.2
Trade and services	4.1	5.4	6.0	4.2	4.3
Transport	4.6	5.5	4.3	5.8	7.6
Financial sector	0.6	1.6	8.7	9.5	4.7
Total	4.1	5.8	6.6	4.8	4.1
Monthly salaries[3]					
Production sector[2]	3.8	4.4	8.7	8.3	3.8
of which:					
Metal industry	2.7	3.4	8.4	8.2	4.1
Construction	3.0	3.3	5.5	6.1	1.7
Trade and services	2.7	3.1	5.6	5.1	2.8
Transport	3.2	4.2	6.7	6.4	5.2
Financial sector	2.6	3.6	6.6	6.4	2.0
Total	2.7	3.7	7.2	7.0	3.4
Memorandum item:					
Consumer price index	3.1	2.7	1.4	0.2	0.3

1. Year-on-year changes, excluding retroactive supplements.
2. Mining, quarrying, manufacturing and utilities.
3. Including commissions and fees.

Source: SCB.

from almost 30 per cent in 1995 to an estimated 25 per cent. This is still relatively high in a historical context, but in an environment of weaker economic activity and higher unemployment, the shift from profits to labour income has served to prolong the period of labour shedding and prevented an easing of the imbalances in the labour market.

After peaking at 3.3 per cent in mid-1995, headline consumer price inflation decelerated rapidly through the rest of that year and 1996. Inflation moderation reflected both the appreciation of the krona and the fall in long-term interest rates as well as the slower growth in the economy. Slower CPI growth allowed the central bank to continue to reduce short-term interest rates, which also fed into lower recorded price growth, and in November 1996 the consumer price index

was actually below its level one year earlier. Consumer price inflation has since been steadily increasing as the above effects have worn off, reaching 1.9 per cent in December 1997. The renewed rise has been heavily influenced by tax increases that form part of the fiscal consolidation programme. Consumer prices net of indirect taxes and subsidies have been falling by an average of 1¼ per cent from 1996 to 1997. In 1998 there will be additional indirect tax increases and a further reduction of housing subsidies, although the effect on inflation should be only half of what was recorded in 1997.

The short-term outlook

Economic policy assumptions and the external environment

The current fiscal and monetary policy stance, combined with a benign external environment, provides a favourable setting for a continuation of the recovery (Table 6). The front loading of the fiscal consolidation programme implies that only a minor additional tightening remains to be implemented, and the government has committed additional resources to municipalities and to the education system as part of the pledge to lower open unemployment to 4 per cent by year 2000. Altogether, this implies that the fiscal policy stance will be broadly neutral with respect to its short-term demand impact over the projection period. Over the past eighteen months, low inflation and fiscal restraint have allowed short-term interest rates to come down by 1½ percentage points, in the process reducing the differential with Germany from 2½ to less than 1 percentage point. Long-term interest rates have declined by more than 2 percentage points over the same period, leaving current rates at 6 per cent, about ¾ percentage point above similar German rates. The OECD is projecting a further lowering of both short and long-term differentials with the European currency area, but this has to be set against the likelihood that European interest rates will be on an upward trajectory. The effective exchange rate is currently at the same level as at the beginning of 1996 and the projections are based on the technical assumption that there will no change over the projection period. The export sector will benefit from import growth among Sweden's main trading partners of 7 to 8 per cent per annum, although market shares may fall somewhat as a result of the recent appreciation of the effective krona exchange rate and pressure on unit labour costs from wage growth in excess of productivity growth.

Table 6. **Economic policy assumptions and the external environment**

	1996	1997	1998	1999
Fiscal policy				
General government financial balance (per cent of GDP)[1]	–3.6	–1.5	0.2	0.1
Change in general government financial balance (per cent of GDP)	3.4	2.1	1.7	–0.1
of which:				
Cyclically adjusted (per cent of GDP)	4.0	2.4	1.2	–0.2
Interest and exchange rates				
Nominal effective exchange rate (1991 = 100)	90.0	87.1	88.3	88.5
Kronor per DM	4.5	4.4	4.3	4.3
Three-month German rates	3.3	3.3	4.1	4.6
Three-month domestic rates	5.8	4.1	4.7	4.9
Ten-year domestic bond yields	8.0	6.7	6.5	6.8
Export-market growth for total goods (per cent)				
Total OECD	6.7	9.3	8.4	7.5
Germany	6.3	8.2	8.0	7.3
United Kingdom	5.9	9.1	8.5	7.6
Norway	5.1	6.9	7.2	6.2
United States	7.2	10.6	8.3	8.4
Finland	5.9	8.0	7.8	7.1
Unit labour cost growth in competitor countries (per cent)[2]				
Germany	–0.4	–1.7	–0.5	0.5
United Kingdom	2.3	2.9	3.2	2.8
Norway	3.6	3.5	2.8	3.1
United States	2.0	2.3	2.5	3.1
Finland	0.1	–0.8	1.1	2.2

1. Based on the principles applied by the European Union (''Maastricht'' definition).
2. Business sector unit labour costs.
Source. OECD.

Short-term outlook and risks

Economic growth will increasingly rely on domestic demand (Table 7). Private consumption growth should benefit from the replacement cycle in durable goods, and the fact that household disposable income will tend to increase in line with employment and real wage growth as the fiscal consolidation process is completed. While construction activity will be maintained by a prolongation of a tax relief for housing improvements undertaken, the recovery in house prices may still be too modest to stimulate housing investment. However, the growth in business sector investment should be stimulated by increasing capacity utilisation

Table 7. **Short-term outlook**

Percentage changes (1991 prices)

	1996	1997	1998	1999
A. Demand and output				
Private consumption	1.3	2.3	2.5	2.0
Government consumption	–0.2	–2.3	0.0	0.5
Gross fixed investment	3.7	0.5	6.0	6.4
Public sector	–8.0	–3.5	3.2	2.7
Private sector residential	13.1	–20.0	2.5	7.0
Private sector non-residential	4.7	5.0	7.0	7.0
Final domestic demand	1.3	0.7	2.4	2.4
Stockbuilding[1]	–1.1	0.2	0.0	0.0
Total domestic demand	0.1	0.9	2.4	2.4
Exports of goods and services	6.1	10.0	7.3	6.0
Imports of goods and services	3.7	9.5	8.0	7.0
Foreign balance[1]	1.2	1.0	0.4	0.2
GDP at constant prices	1.3	1.8	2.6	2.4
GDP at current prices	2.3	3.6	5.1	4.8
B. Memorandum items				
Private consumption deflator	1.2	2.1	2.2	2.2
GDP price deflator	1.0	1.8	2.4	2.4
Total employment	–0.9	–1.0	0.6	0.7
Unemployment rate	8.1	8.1	7.3	6.9
Current account, per cent of GDP	2.3	2.8	3.4	3.7
Industrial production	2.3	7.0	5.0	4.0
Household saving ratio	4.7	2.6	2.4	3.0

1. Changes expressed a a percentage of GDP in the previous period.

Source: OECD.

in the export-orientated sector and a pick-up in demand for domestically-orientated products. Low interest rates and continued substitution out of labour in the production process will also make for buoyant capital spending. Overall, domestic demand growth should reach 2½ per cent in both 1998 and 1999, with GDP expanding at a similar pace. Business sector employment should then grow by about 1 per cent each year, while employment in the general government sector will only stabilise in 1998. Employment growth could reach ½ per cent in 1998 and ¾ per cent the following year. Reflecting the extensive education programme, the labour force will be almost flat throughout the projection period and this should allow the unemployment rate to come down to 7 per cent by 1999. Participation in labour market programmes is still expected to involve 4¼ per cent of the work force in 1999, leaving the total of unemployed and ALMP participants above 11 per cent of the work force.

With import prices again increasing and interest rates starting to move upwards, the period of very low recorded price increases is over, but price inflation should remain close to the price objective of 2 per cent and safely inside the tolerance rage of 1 to 3 per cent. There is, however, the prospect that the wage agreements which take place in early 1998 will be dominated by claims for restoring wage relativities, leading to wage growth above productivity increases in the service sectors. If associated with increasing wage drift in areas of labour shortage, these wage-wage links would result in a lower employment growth than that projected. The upside risks are mostly related to higher-than-projected demand. Supply factors on the export side may allow exporters to benefit more than projected from the growth among Sweden's main trading partners. Likewise, private consumption may be higher than projected if the replacement cycle for durable goods proceeds faster than expected or if additional asset price increases induce a further lowering of the household savings rate.

II. Macroeconomic policies

The recent decision not to participate in the Economic and Monetary Union of the EU at its inception means that Swedish monetary policy will continue to be anchored by the objective of price stability. This framework, which in practice sets a target for price inflation of 2 per cent with a fluctuation margin of 1 per cent on either side, has gained increased credibility over the past two years as inflation expectations in both labour and financial markets have been contained within the tolerance range of 1 to 3 per cent. Confidence in the ability of the central bank to achieve these targets should be further enhanced by a comprehensive revision of the Central Bank Act which explicitly sets out price stability as the Riksbank's overriding objective and introduces several provisions to underpin its independence. At the same time, the Maastricht Treaty and the recent Stability Pact of the EU commit Sweden to continued stringency in fiscal policy, even without EMU participation. With public finances scheduled to move into surplus in 1998, Sweden is currently well ahead of most other EU countries in fiscal consolidation and the recently-adopted objective of 2 per cent surplus in public finances over the economic cycle bears witness to continued high ambitions in this field. The present chapter discusses in more detail the progress made in improving the credibility of monetary and fiscal policies, while assessing the requirements for ensuring that progress on both these fronts is maintained.

Monetary policy

Strategy and implementation 1996-97

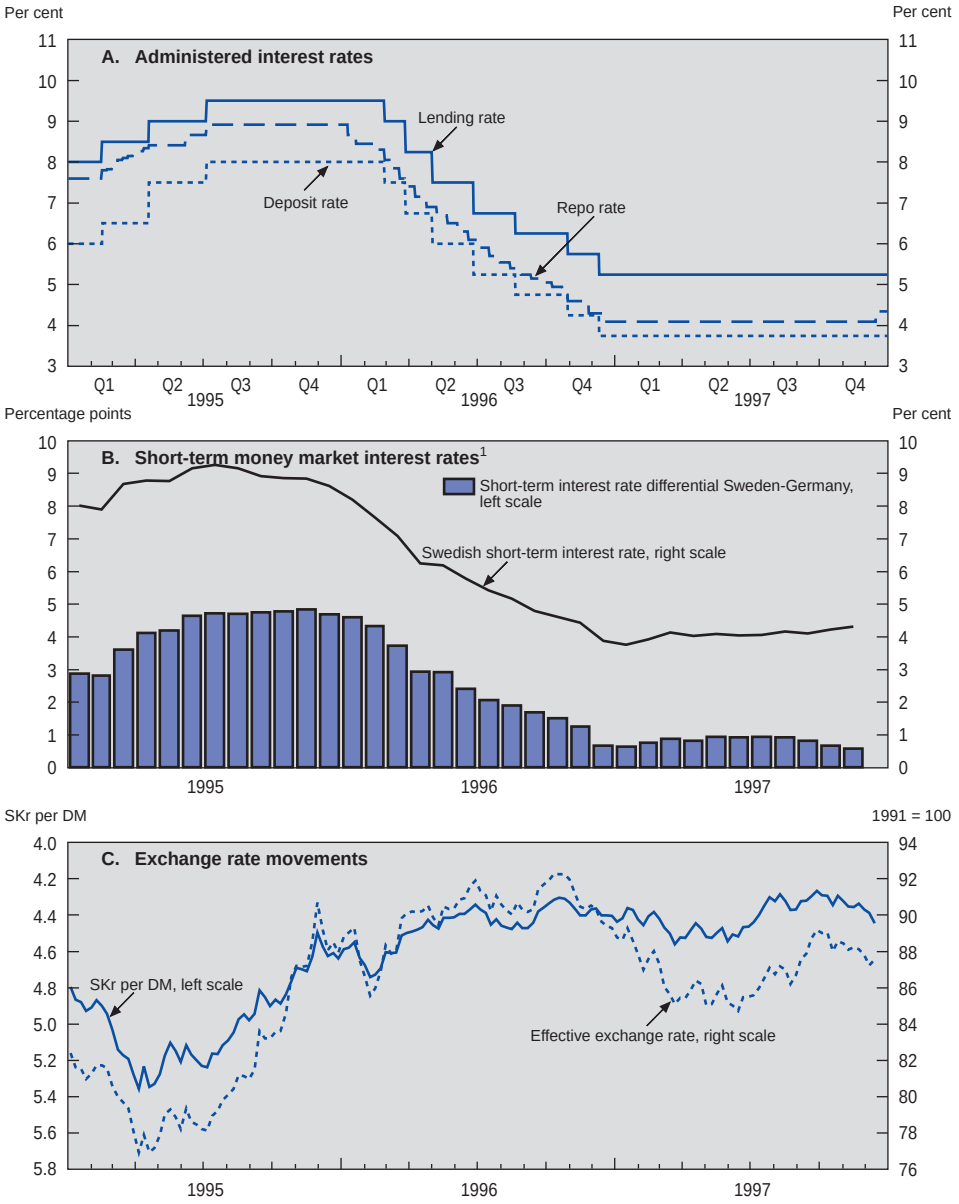
In general, on account of the time lag before monetary policy measures affect inflation, the central bank pursues its inflation objective based on assessment of trends over the coming two years. The assessment is made in the light both of monetary and real economic factors, including GDP growth and any

associated capacity shortages. The latter entail estimates of potential output and the ''output gap'', which are difficult to pin down empirically. The former include the information given by movements in exchange-rate and bond markets, but also, more concretely, data on inflation expectations.[3] These are potentially important for wage settlements, which are the main component of domestic inflation. Indeed, an important cause of the unduly high settlements of 1995, discussed in the previous *OECD Economic Survey of Sweden*, was possibly the fact that the future inflation expected by labour market participants at that time was considerably higher than the level targeted by the Riksbank, the central bank.

Against this background, and the central bank's own projections at the time of price increases remaining in the upper end of the tolerance range, the key signal rate of the central bank, the repo, was kept unchanged at 8.91 per cent from mid-1995 to early January 1996 (Figure 10). As financial-market confidence increased, leading to an effective appreciation of the Swedish krona, and as economic growth slowed, inflation prospects improved considerably and the central bank was able to embark upon a series of gradual reductions in the repo rate which lasted through 1996. The repo rate fell to 4.10 per cent by mid-December 1996, while the corridor for short-term market rates, defined by the deposit and lending rates of the central bank settled at between 3.75 and 5.25 per cent. In the following twelve months, no adjustment to interest rates occurred. The krona underwent a general depreciation up to May 1997, and with economic prospects implying above-trend economic growth over the central bank's policy horizon, further interest rate cuts were put on hold. Having appreciated up to October, the krona again depreciated, while the economic cycle appeared to be strengthening more than anticipated with a projected closing of the output gap during 1999. Consequently, the Central Bank increased the repo rate by ¼ percentage point in mid-December 1997 to ensure a configuration of interest rates consistent with its inflation objectives.

The reduction in headline inflation during 1996 and into 1997 owed much to lower interest rates and the preceding appreciation of the krona. Measures of underlying inflation which exclude the effects of these two factors through housing costs and import prices show a relatively stable pattern of domestic inflation in 1996, with a tendency to rise in 1997. This tendency towards acceleration, however, has come largely from administrative prices and indirect taxes. Most importantly, since late 1995 survey-based inflation expectations among the

Figure 10. **SHORT-TERM INTEREST RATES AND THE EXCHANGE RATE**



1. For Sweden: 3-month Treasury discount notes; for Germany: 3-month Fibor.
Source: Sveriges Riksbank ; OECD, *Main Economic Indicators.*

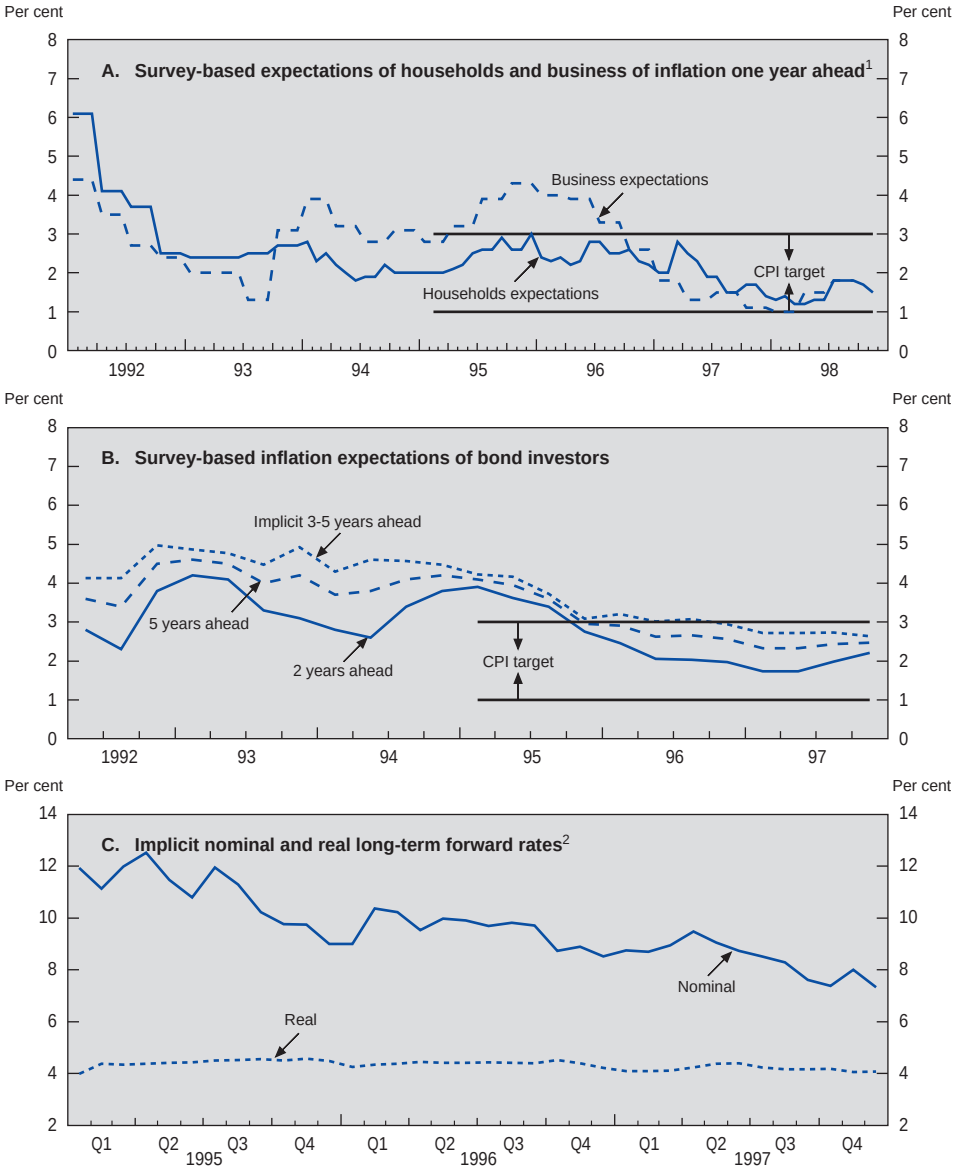
labour market organisations, business and households have been contained within the tolerance range set by the central bank (Figure 11, panel A). Inflation expectations of bond investors have been similarly contained, while implied forward interest rates derived from bond-market pricing have continued to fall at all maturities (panel C). Insofar as the pricing of indexed bonds has implied a fairly constant real interest rate, slightly above 4 per cent for a maturity of ten years, the fall in nominal yields can be attributed to the inflation component of nominal bond yields.

The decline in Swedish bond yields has continued – with some aberrations – through 1997, reaching 6 per cent at the end of the year (Figure 12). In the process, the differential against German long-term rates has fallen from about 1½ percentage points in September 1996 to about ¾ percentage point in December 1997. This decline cannot match that of Italy and Spain, which benefit from the expectation of participation in the European Monetary and Economic Union, and where long-term rates are now below Sweden; however, the differentials *vis-à-vis* the rates of the other two main EU countries who intend to stay outside the EMU at the outset, Denmark and United Kingdom, have fallen strongly. In this respect, the strong improvement of public finances in a low-inflationary environment has served to underpin financial-market confidence.

A new accord on central bank independence

While the established credibility of the low-inflation framework owes much to fiscal policies and structural changes, such as internationalisation and deregulation which have enhanced competition, the effectiveness of inflation targeting has been underpinned by the fact that the Riksbank has been able to draw upon its independence in relation to the political authorities. This independence was bolstered in May 1997 by an agreement between the governing Social Democratic Party and the four centre-right opposition parties in Parliament, followed up by a government bill to Parliament in November containing the necessary legislative measures.[4] Reflecting the requirements of the Maastricht Treaty, the legislative proposal embodies *i)* the over-riding commitment to price stability to be explicitly enshrined in the Central Bank Act; *ii)* an amendment to the constitution that prohibits any government authority from instructing the central bank with respect to its execution of monetary policy (the present requirement to consult with government is modified to one of informing it ahead of important

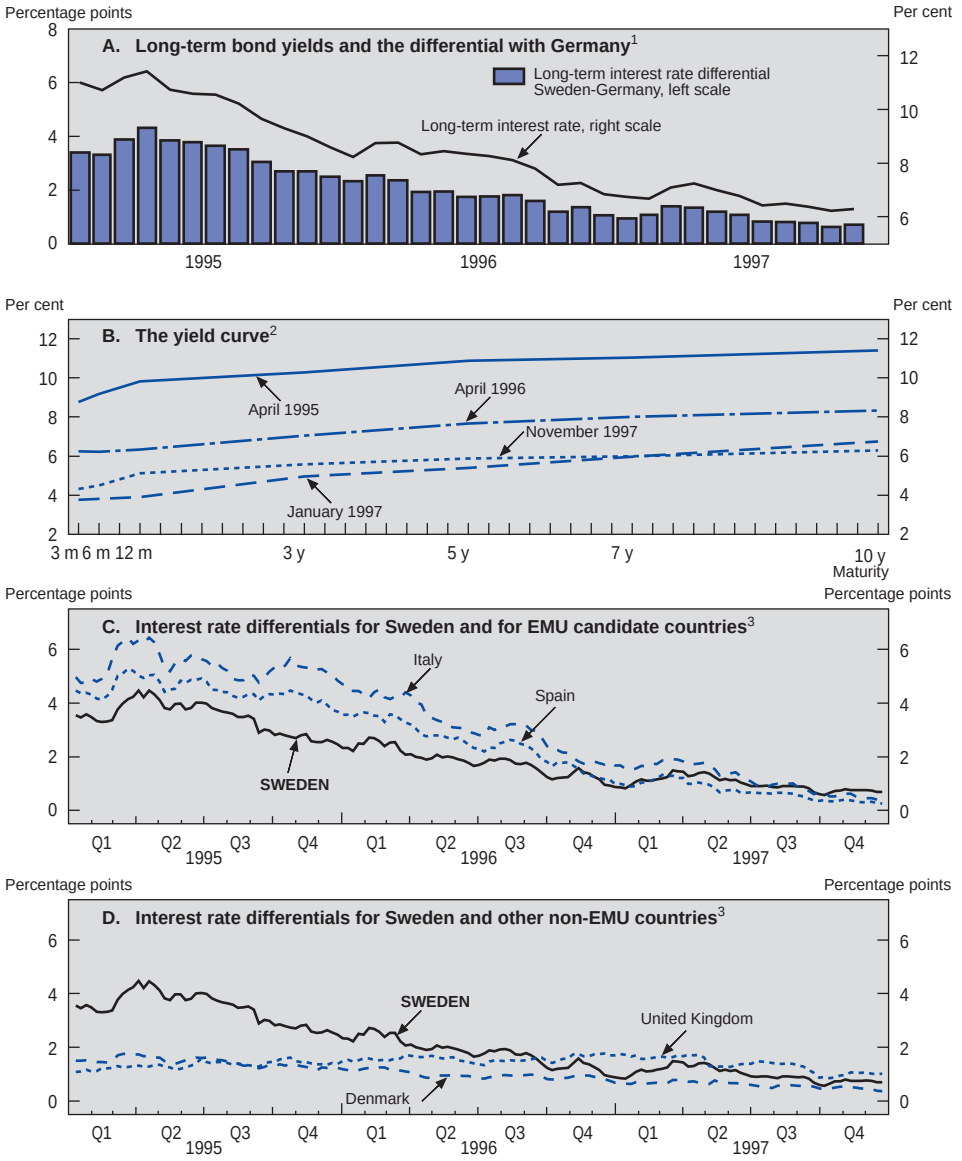
Figure 11. **INFLATION EXPECTATIONS**



1. Graphs have been shifted 12 months forward to coincide with the point in time to which expectations refer.
2. 10 years. Real rates are calculated from indexed bonds.

Source: Sveriges Riksbank.

Figure 12. **BOND YIELDS**



1. For Sweden: 10-year Central government bonds; for Germany: Benchmark government bonds (10 years).
2. Effective yields on 3-6-12 months Treasury bills and 3-5-7-10 years government bonds.
3. *Vis-à-vis* Germany. 10-year government bonds.

Source: Sveriges Riksbank; OECD, *Main Economic Indicators.*

decisions in monetary and exchange rate policy); and *iii)* enhancement of the independence of the central bank governor so that he/she can only be removed in cases of serious breach of duties or incapacity to execute these. The structure of governing bodies in the central bank will also be changed, with the present board, appointed directly by and often among members of parliament, retaining primarily supervisory functions and having no influence on monetary policy. The latter functions will be the responsibility of a new Executive Board whose appointees have to be full-time employees of the central bank and can only be relieved from office under the rules applying for the governor.

The Swedish central bank has been unusual among central banks in also having responsibility for exchange rate policy. In line with the division of responsibilities elsewhere in the European Union, the choice of exchange rate regime is formally to be a matter for Government decision, while all other elements of exchange rate policies – such as managing a floating currency or the choice of a central rate and the width of fluctuation bands in a fixed-exchange rate regime – remain the responsibility of the central bank. The Government announced in early June 1997 that Sweden will not participate in the European Economic and Monetary Union (EMU) at its inception on 1 January 1999, but would consider joining at a later stage, subject to public approval either through a referendum or a general election. Against this background, inflation targeting with a floating exchange rate can be expected to remain the framework of monetary policies, with a participation in ERM 2 being only a prelude to full participation in the EMU.[5]

Policy considerations in the current conjuncture

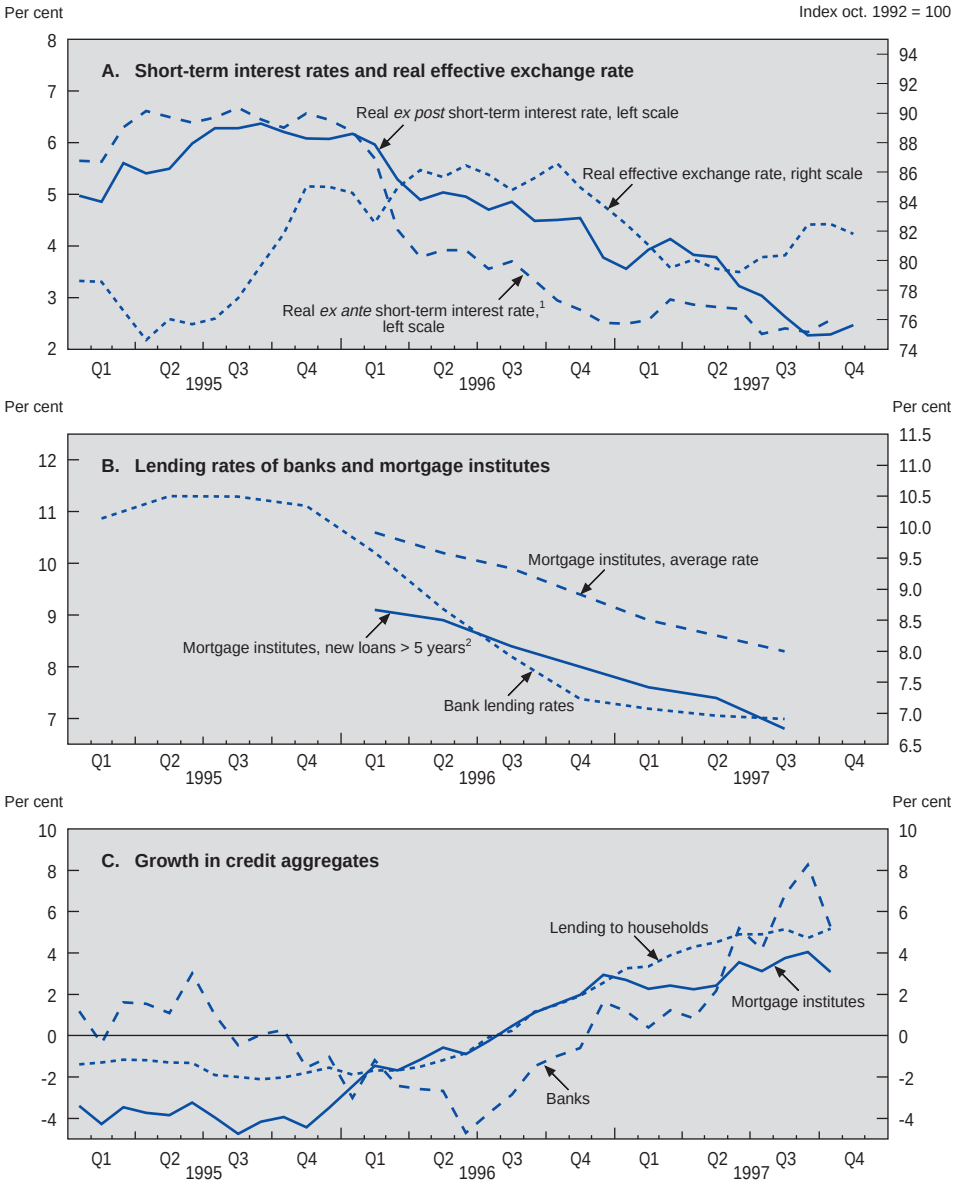
A major test of inflation targeting will be the outcome of the 1998 wage bargaining round, already under way, which will take place in a situation where inflation expectations have stopped falling and the output gap is narrowing because of increasing demand for labour and of policy measures which are acting to restrict labour supply (see chapter III). At the same time, the establishment of the EMU might affect the configuration of exchange rates and interest rates internationally, with repercussions for monetary conditions in Sweden. Monetary policy thus faces several challenges in reinforcing the credibility built up thus far and ensuring the compatibility of continued growth with stable inflation expectations.

Maintaining an appropriate monetary stance

Monetary conditions have eased substantially since early 1996. The real effective exchange rate has depreciated by 5 per cent, with most of the fall taking place over the past year and the real short-term interest rate has come down by close to 4 percentage points, with most of the fall in the *ex-ante* rate taking place in 1996, and in the *ex-post* rate in 1997 (Figure 13). Lending at fixed interest rates – an important source of finance in the Swedish economy – has become far more favourable with the strong fall in long-term interest rates. Stimulated by a fall in the lending rates of mortgage institutes by 2 percentage points and of banks of 3 percentage points since early 1996, credit aggregates are increasing by between 3 and 5 per cent, the most expansionary part being credit to the household sector. With domestic demand now picking up, the most liquid component of the money supply, M0, is growing at an annual rate of 3 per cent, while the growth in the broader aggregate M3 has slowed from 11½ per cent in 1996 to 1 per cent, reflecting lower overall financial savings in the private sector and higher investment in financial assets outside this aggregate.

On the most recent assessment of the central bank, the current monetary policy stance may be seen on the whole as well adapted to the achievement of its inflation goals. Moreover, if assessed against an admittedly simplistic policy rule (the so-called Taylor rule) which gives some weight to output stabilisation,[6] Swedish short-term rates are now fairly close to those warranted by conditions in the real economy (Figure 14). However, there is emerging evidence of high capacity utilisation and lack of skilled labour in some private services sectors (see Chapter I), and the formulation and direction of labour market programmes have not been able to anticipate these trends and fend off possible bottlenecks. Using a number of different approaches and methodologies, the central bank has estimated the output gap in early 1997 to be between ½ and 4 per cent. The above-mentioned trends may indicate that the gap is on the lower side of the estimates available, in which case the 3 per cent annual growth estimate of the central bank for 1998-99 would mean that the gap may close earlier than projected. Moreover, persistently high unemployment over several years may have raised the NAWRU to a point where there is only a small margin of spare capacity in the economy, requiring a tighter monetary policy even in the case of somewhat slower growth. With the output gap closing, the question arises as to how to set monetary policy on a stricter course.

Figure 13. **INDICATORS OF MONETARY CONDITIONS**

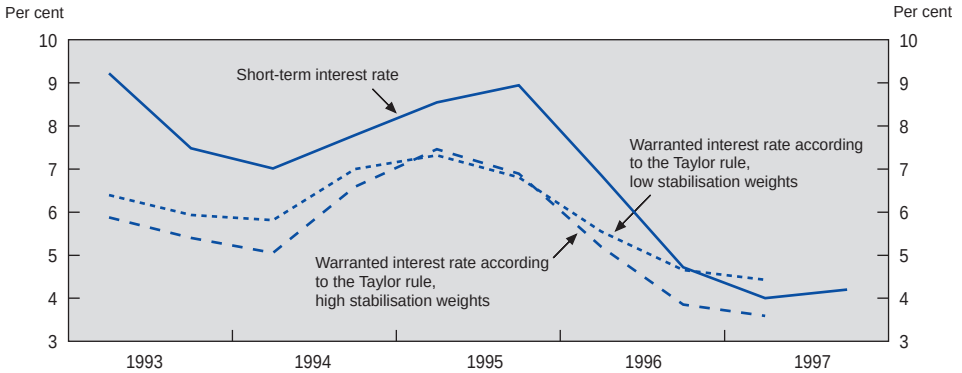


1. The rate for three-month treasury bills adjusted for the CPI change expected by households in the coming year.
2. More than 5 years between interest rate adjustments.

Source: Sveriges Riksbank ; OECD.

Figure 14. **THE TAYLOR RULE**[1]

Per cent



1. The rule indicates the warranted short-term interest rates on the basis of *i)* a real short-term rate consistent with neutral monetary conditions, set to 4 per cent for Sweden; *ii)* expected price inflation; *iii)* the deviation between current inflation and the inflation target, the latter 2 per cent for Sweden; *iv)* the deviation between actual and potential GDP. Calculations are carried out with two different sets of stabilisation weights.

Source: OECD.

An adjustment towards tighter monetary conditions in Sweden will take place in the context of the pick-up in activity in Europe and the extensive integration of financial markets with the advent of the Economic and Monetary Union from 1 January 1999. According to the latest OECD projections, short-term rates in Europe may be expected to increase further through 1998 as the Bundesbank increases rates ahead of the introduction of the Euro. Continued improvement in Swedish public finances may allow short and long-term interest rate differentials towards European currencies to continue to fall, but some upward adjustment of Swedish short-term rates may still be necessary to avoid an inflationary effect through a depreciating krona towards the Euro area (the eleven most probable participating countries account for more than 50 per cent of Swedish exports and imports). In such a scenario, the adjustment required for internal reasons might well be effected while adapting to an external shock.

... and meeting the challenges of EMU non-participation

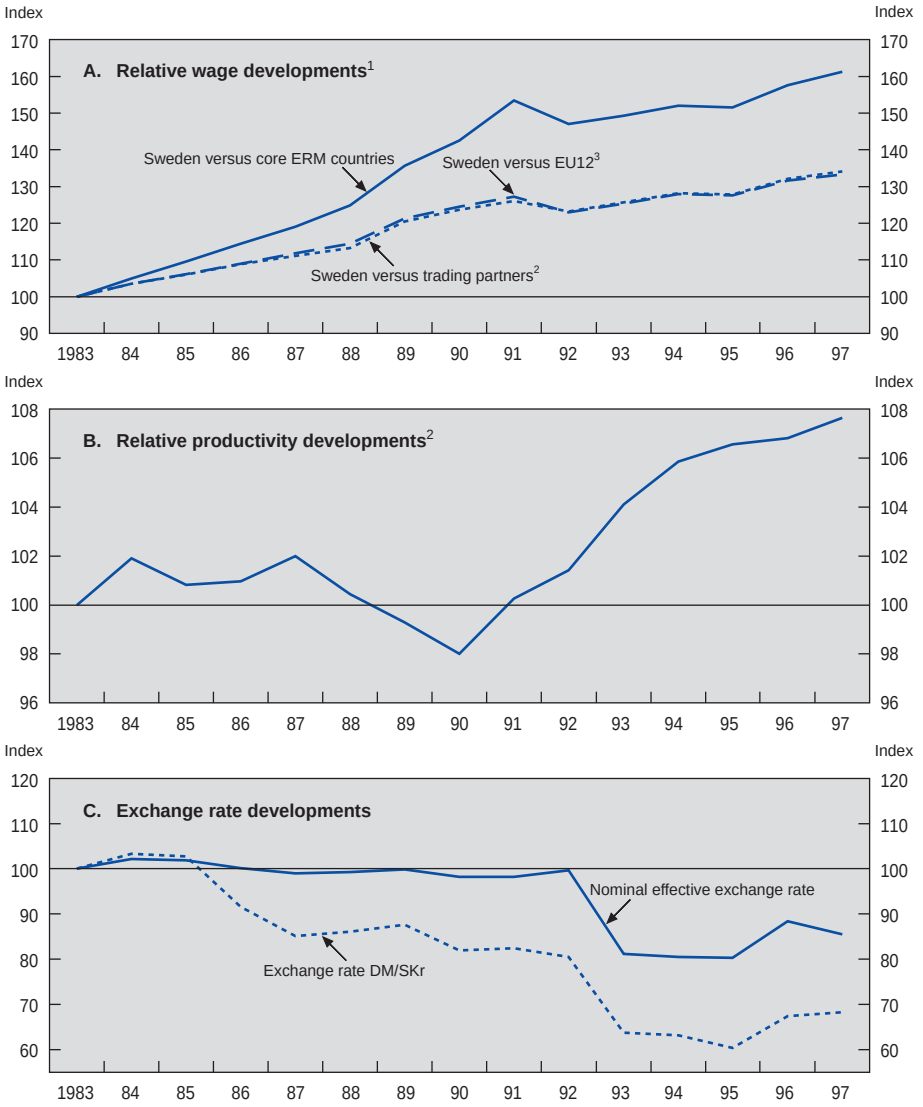
As noted in the previous *Economic Survey of Sweden*, the overall improvement in economic performance over recent years, with respect to both budget

deficits and price inflation, would have allowed Sweden to meet the respective criteria for entry into the EMU on a par with other EU countries. Financial-market structures would also make for an EMU-compatible transmission mechanism of monetary policy, insofar as lending to households and enterprises is conducted at fixed interest rates as in the majority of continental European countries. Moreover, while in the past the economic cycle may have been quite divergent as between Sweden and continental Europe, the main reason for this is to be found in the formulation of economic policies which has triggered out-of-phase cycles in the Swedish economy, whereas the cyclical divergencies stemming from differences in industrial structure have not been of particular consequence since the 1970s. The Government-appointed commission of independent experts evaluating the effects of EMU on the Swedish economy[7] nevertheless pointed to the ratcheting up of unemployment in Sweden over the 1990s and the underlying structures that make wages rather insensitive to imbalances between supply of and demand for labour. In the assessment of the OECD, there is in the absence of a domestic monetary policy lever a risk that asymmetric shocks to economic activity could entail a further upward spiralling of unemployment while labour-market structures remain as they are.

Overall, however, the room for monetary manoeuvre to deal with such shocks remains limited even if Sweden is not to join the EMU. As recognised in the recent Government Bill outlining the effects of the EMU, the decision to stay outside, but to keep options open to join at a later stage, makes the requirements of Swedish macroeconomic policy, both of fiscal and monetary stance, all the more stringent. In terms of monetary policy, the requirements of an irrevocably fixed exchange rate and of inflation targeting are the same over the longer term if the credibility of the inflation target is to be maintained. In this respect, the domestic policy environment has shifted perceptibly from the 1980s to the 1990s. While the former period was characterised by wage growth clearly above both trading partners and core ERM countries without any offsetting effect on the productivity side, in the more recent period high relative wage growth has been offset by relative high productivity growth (Figure 15). Indeed, with the increased credibility of the inflation targeting regime, this underlying trend has also showed up in larger stability of the nominal exchange rate. The challenge to be dealt with over the period up to a possible EMU participation in some years' time is to make such stability consistent with higher employment growth and

Figure 15. **COMPONENTS OF RELATIVE UNIT LABOUR COST**

Index 1983 = 100



1. In the business sector. In national currencies using OECD competitiveness weights. From 1991 overall Germany.
2. Versus OECD trading partners excluding Czech Republic, Hungary, Iceland, Ireland, Mexico and Turkey.
3. Excluding Ireland.

Source: OECD.

Table 8. **Budget plans and outcomes**[1]

General government, per cent of GDP

	1994		1995		1996		1997		1998
	Budget	Outcome	Budget	Outcome	Budget	Outcome	Budget	Estimated outcome	Budget
Net lending	–11.1	–10.3	–10.2	–7.9	–4.6	–2.5	–2.6	–1.9	0.6
Central government	–13.1	–11.7	–10.9	–8.8	–5.8	–3.2	–2.8	–2.3	–0.5
Social security	1.6	1.6	1.0	1.2	1.2	0.9	0.9	0.7	1.3
Municipalities	0.4	–0.2	–0.3	–0.2	0.1	–0.3	–0.7	–0.3	–0.3
Memorandum item:									
Net lending (SKr billion)	–165	–158	–161	–129	–79	–42	–47	–33	10
Revenues and expenditures									
Direct taxes	21.7	22.2	22.7	23.6	23.4	25.0	24.1	25.8	25.3
Indirect taxes	13.7	14.4	13.2	12.9	13.9	14.5	14.0	13.6	13.6
Social security contributions	13.7	13.3	14.4	13.6	14.6	14.7	14.5	14.3	14.2
Transfers	33.5	33.0	32.6	31.9	29.7	30.6	29.4	29.7	28.4
Consumption and investment	29.9	30.2	30.2	28.8	28.8	28.3	28.3	27.9	26.6
Other items, net	3.2	3.1	2.3	2.8	2.0	2.1	2.5	1.8	2.3
Gross government debt[2]	..	79.3	87.9	78.2	79.1	77.8	78.0	77.1	73.9

1. For 1994 to 1996, budget plans are based on assessments on a national accounts basis in November/December ahead of the calendar year, for 1997 and 1998 according to the Finance Bills of September.
2. Maastricht definition.

Source: Ministry of Finance, *The Swedish Economy* (various issues).

falling unemployment, the key to which lies in wage formation outcomes which prevent labour-shedding and reflect underlying productivity developments and employment prospects (see chapter III for a more detailed exposition of structural reform in the labour market).

Fiscal policy

Fiscal retrenchment has played a fundamental role in rebalancing the economy and establishing the conditions for renewed growth. The fiscal consolidation programme being implemented over the years 1995-98 will amount to 8 per cent of GDP for the central government budget. Supported by a restructuring of municipal finances, the general government budget is scheduled to move into surplus in 1998 from a 10 per cent deficit in 1994. The Swedish parliament has adopted a longer-run goal of achieving a budget surplus of 2 per cent of GDP over the cycle, to be gradually approached via surpluses of 0.5 per cent in 1999, 1.5 per cent in 2000 and 2.0 per cent in 2001. The objectives thus go well beyond the commitment to the underlying principles of price stability and sound public finances set out in the Maastricht Treaty and the Stability Pact. The objectives should, however, be seen against the background of the very high sensitivity of the budget to the economic cycle and the concomitant exposure to adverse financial-market sentiment in the event of their not being reached. Moreover, with a high level of taxation following from high public consumption and generous transfer schemes to the working-age population, substantial challenges remain in the formulation and implementation of fiscal policy.

Eliminating the public sector budget deficit

Reasserting control over public finances has been the main priority of fiscal policy since budgets plunged into severe deficit following the recession in the early 1990s (Table 8). The basic strategy for re-establishing balance was adopted in 1994 and 1995, with discretionary measures embodying both tax increases and cuts in central government transfers covering the years 1995 to 1998, intended to result in an overall budget deficit improvement of 7 per cent of GDP. The Swedish convergence programme presented to the European Union in June 1995 committed the Swedish government to eliminating the budget deficit by 1998 and to reducing the gross debt of the consolidated general government as a ratio of

GDP from 1996 on. At the same time, a follow-up process was instigated, based on biannual reviews of the progress made towards achieving these objectives. The measures undertaken, being fully detailed and set in a medium-term perspective, contributed to the easing of financial-market pressures as noted above, with long-term interest rates falling and the exchange rate appreciating strongly through late 1995 and early 1996.

With growth prospects weakening in early 1996, the spring 1996 review triggered an additional fiscal package to ensure that the overall objectives would be complied with. The fiscal consolidation programme was thus augmented, the permanent savings component reaching 8 per cent of GDP and additional temporary measures of SKr 2 and 4 billion respectively coming into effect for 1997 and 1998. The programme was heavily front-loaded, with 70 per cent of the budget impact to be effected by 1996. While tax increases were implemented more rapidly than expenditure cuts, roughly the same reliance was placed on each in the programme as a whole.

By late 1996 it became evident that the pace of budget consolidation was faster than set out in the convergence programme. Central government revenues from taxation of real estate and corporate taxes – the latter the most volatile of revenue components – increased more than projected, reflecting a new tax assessment of houses and the exhaustion of the possibilities for enterprises to carry forward losses. Following a decline in the birth rate from its 1992 peak, grants for parental leave declined, while sickness benefits fell due to a falling take-up rate. In total, such effects on the expenditure side were more than sufficient to offset higher unemployment benefits. Costs of debt-service also declined more than foreseen in the convergence programme, which incorporated a decline in long-term (five-year) interest rates of only about 1 percentage point from mid-1995 to mid-1997 whereas the actual fall amounted to about 4 percentage points. As a secondary effect, lower long-term interest rates also reduced the outlays necessary for housing subsidies while the fall in short-term rates fed into the Consumer Price Index and lowered price-indexed transfers and tax brackets (Table 9).

The improvement in public finances envisaged in the convergence programme also included lower expenditures in the municipal sector, the overall claims on resources being projected to fall by 3 per cent of GDP over the 1994-98 period. According to the 1996 Spring Bill, this was meant to be

Table 9. **Contributions to budget consolidation**

Per cent of GDP

A. Cumulative change 1994-1998 as measured in the Convergence Programme and in the 1998 Finance Bill	Convergence Programme	Finance Bill 1998
Revenue	1.9	2.2
Taxes	1.8	3.4
Capital income	–0.1	–1.2
Expenditure	–7.6	–8.1
Household transfers	–4.8	–3.5
Interest payments	–1.1	–0.7
Consumption and investment	–3.1	–2.8
Net lending	9.5	10.3
B. Contributions to the elimination of the public sector deficit 1994-98		
Increased revenue		2.3
Consolidation programme		3.8
Others		–1.5
Lower expenditure		8.0
Consolidation programme		4.2
Local government expenditure		1.7
Interest payments		0.7
Others		1.4
Net lending		10.3
of which:		
Consolidation programme		8.0
Employment creation 1997-98		–1.5
Others		3.8

Source: Ministry of Finance.

accomplished by a combination of nominally unchanged block transfers to municipalities, supported by agreements – later converted into legal requirements and the formulation of the block transfer scheme – which committed both counties and municipalities to balance their books by 2000 and to maintain tax rates unchanged in 1997 and 1998. While the envisaged down-scaling of municipal activities has been partially reversed (see below), it is still likely to contribute to the overall budget improvement by 1¾ per cent of GDP by 1998.

The borrowing requirement as defined by the financing operations of the Government Debt Office has fallen even more strongly than the net lending of the central government. This reflects a number of one-off payments and purely financial transactions (Table 10). The main items have been privatisation

Table 10. **The central government borrowing requirement**

SKr billion

	1995	1996	1997	1998
Net lending	–146	–54	–40	–9
Borrowing requirement	139	21	0-10	–10-0
Extraordinary reinforcements, one-off factors, etc.				
Government Debt Office	..	65	45	25
of which:				
Financial transactions		42	12	6
Privatisations, extraordinary dividends		5	31	15
Tax collection periods, etc.		16		3
National Audit Office	28	90	50	18
of which:				
Financial transactions	7	33	9	–2
Privatisation, extraordinary dividends	9	11	21	5
EU effects	–4	4	1	0
Change in tax collection periods, etc.	0	17	0	0
Reorganisation of housing finance	7	14	6	0

Source: Ministry of Finance; National Audit Office; Government Debt Office.

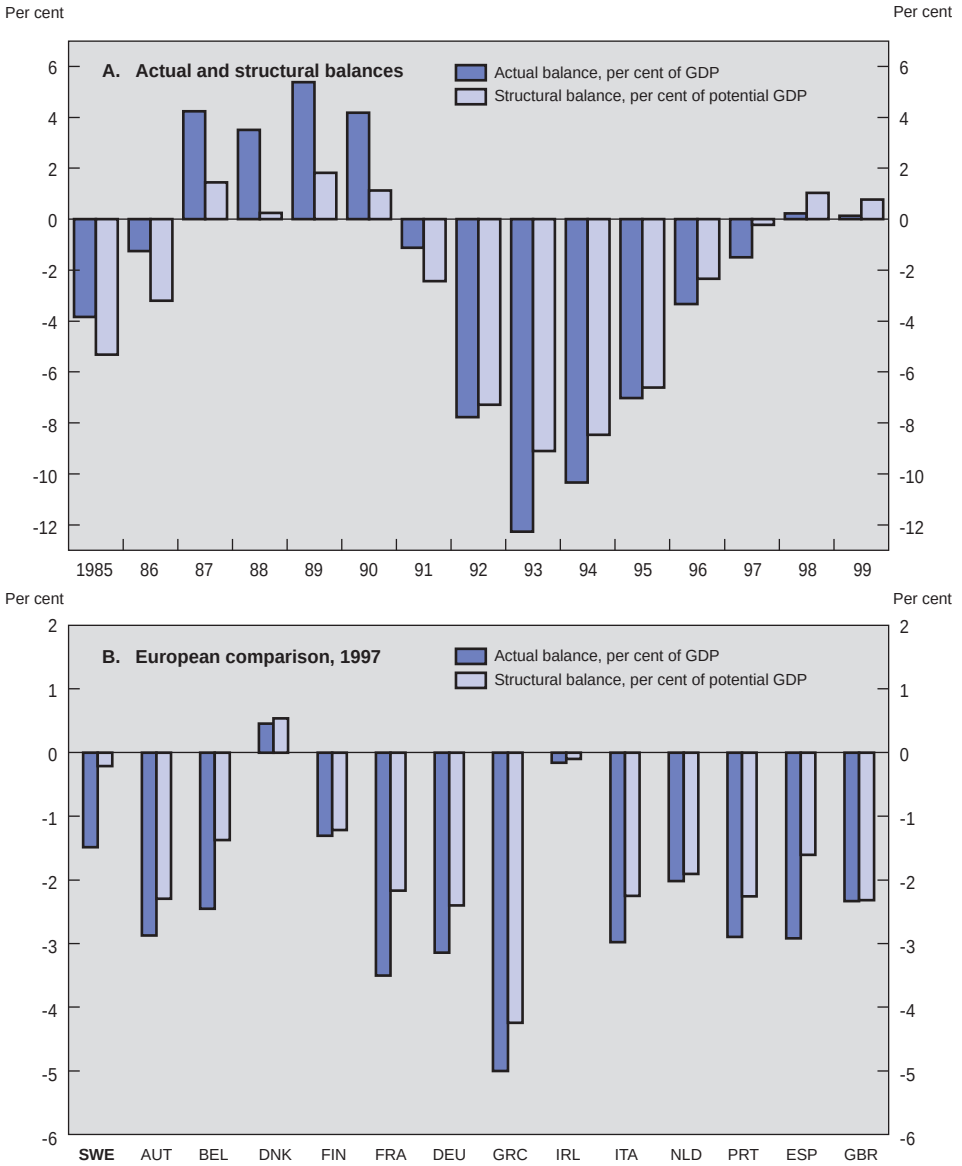
proceeds, and extraordinary dividends from government-owned enterprises, *inter alia* in the real estate sector.[8] In total these two components amounted to more than SKr 30 billion in the 1995-97 period, with another SKr 20 billion of privatisations still scheduled. Reflecting these effects and those following from the appreciation of the Swedish krona, gross government debt peaked in 1994 and has since been on a downward path. Net government debt, also affected by the appreciation effects but otherwise reflecting net government lending, peaked only in 1997. With the budget moving into surplus in 1998, debt is now scheduled to fall in absolute terms, implying a substantial decline relative to GDP (Figures 16 and 17).

Will progress be maintained?

As budget balances have improved more rapidly than envisaged, there have been pressures to use the added room for manoeuvre for employment creation. In the 1996 spring Budget Bill and the ensuing Employment Bill a number of measures were introduced to take effect from 1997, to be financed in the short term out of higher energy taxes and a transfer of funds from Active Labour Market Policy Programmes. These efforts, subsumed into the so-called five-point

Figure 16. **GENERAL GOVERNMENT BUDGET BALANCES**[1]

Per cent of GDP

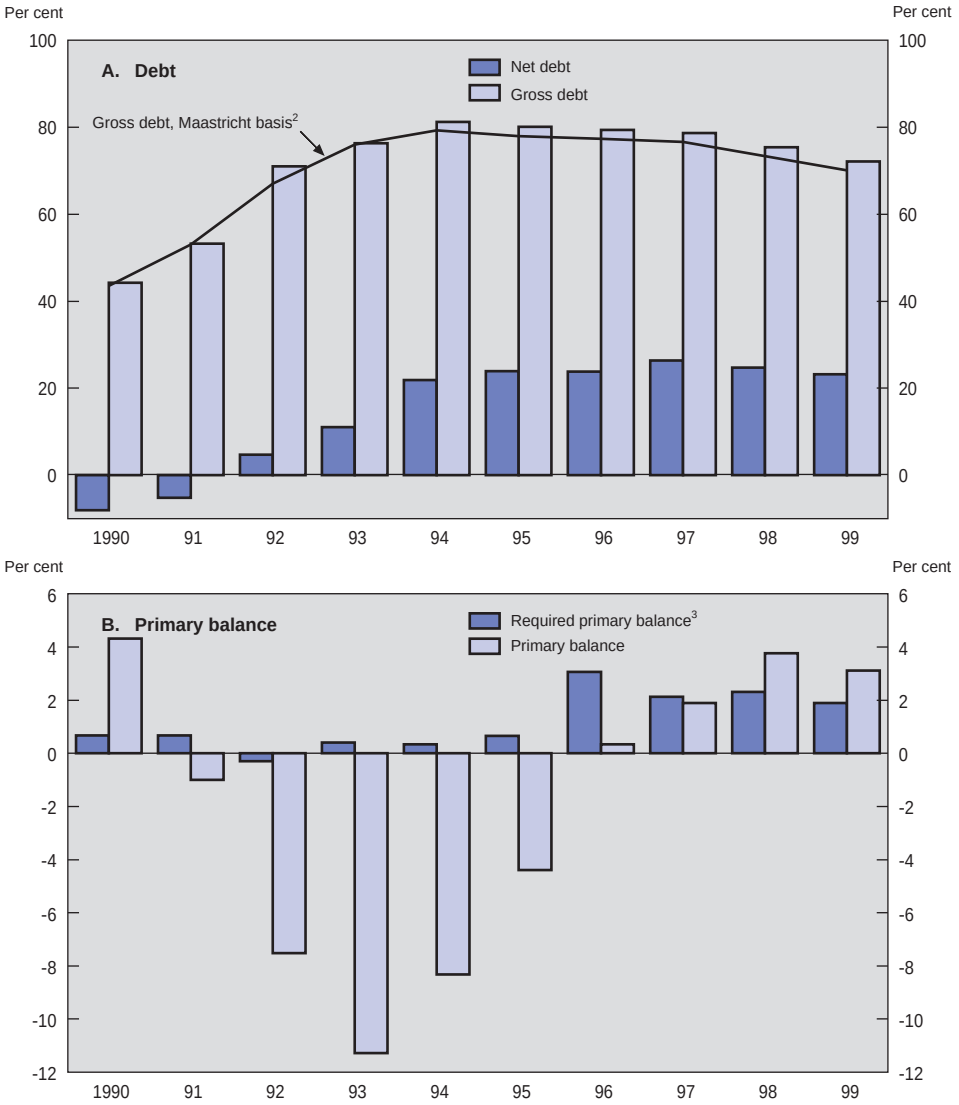


1. OECD estimates for 1997-99.

Source: OECD.

Figure 17. **GENERAL GOVERNMENT DEBT AND THE PRIMARY BALANCE**[1]

Per cent of GDP



1. OECD estimates for 1997-99.
2. Data are provided by the Commission Services of the European Union for 1990 to 1996.
3. Primary surplus needed to stabilise the net debt ratio. Revaluation effects on the debt are excluded.

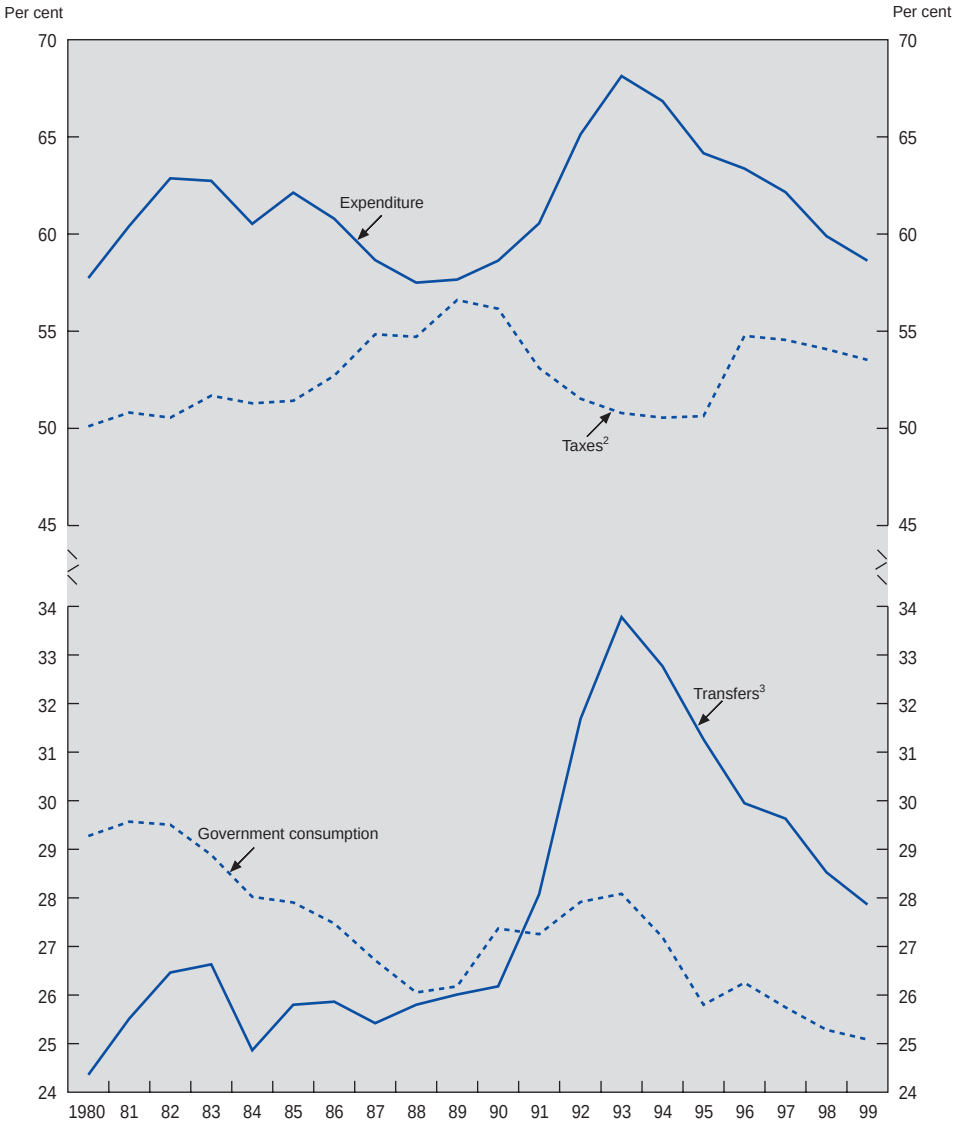
Source: OECD.

programme in the Spring 1997 Budget Bill, encompass *i)* higher block grants to municipalities to increase employment in health and elderly care and schools; *ii)* an expansion of adult and tertiary education; *iii)* an investment programme for improvement of the environment; *iv)* selected tax reliefs for small and medium-sized enterprises and energy-intensive enterprises; and *v)* an array of labour-market programmes for older workers and more resources for labour market training. The scope of the programme, basically affecting the expenditure side of the budget, has been extended on two occasions, most recently in the 1998 Budget Bill, when further resources were added for the years 1998 to 2000. For these years only a small part of the expenditure increase is financed from savings elsewhere in the budget; rather, this part of the programme is financed out of the savings arising from the fiscal consolidation programme and the strongly improved growth prospects embodied in official projections. Indeed, the employment-creation measures in the public sector have been allocated resources amounting to about 1 per cent of GDP out of the margin created by the potential overshoot of the 1998 budget target. The scope for transferring resources back to households in 1999, envisaged as tax cuts in the spring 1997 Budget Bill has, in the process, been reduced from SKr 15 billion to SKr 2.6 billion.

The objective of taking the budget into a surplus of 2 per cent in 2001 has been retained, but this is now contingent upon a growth trajectory which by 2000 raises the level of GDP by ¾ percentage point above that underlying the Parliament's decision in spring. In this respect, recent fiscal policy decisions imply a *de facto* lowering of ambitions. Given the strong sensitivity of the budget balance to GDP (each 1 per cent increase in economic activity gives rise to a cut in the deficit of ¾ per cent), an additional cyclical improvement of ½ per cent of GDP is incorporated in the officially-projected budget balance by 2000. An unchanged level of ambition with respect to discretionary expenditure restraint would imply that the budget surplus would reach 2 per cent one year ahead of schedule, leaving a larger buffer for public finances if and when the cycle turns down. In view of the high cyclicality of the Swedish economy and the large automatic stabilisers operating on the budget (Figure 18), achieving a budgetary surplus of 2 per cent over the cycle would require a significantly higher surplus at the cyclical peak, so as to obviate the need for pro-cyclical action to correct the situation if and when the economy turns down. This requirement poses a particular challenge in terms of the need for continuous fiscal discipline, to ensure that

Figure 18. **GENERAL GOVERNMENT EXPENDITURE AND TAX RATIOS**[1]

Per cent of GDP



1. OECD estimates for 1997 onwards.
2. Direct taxes, indirect taxes and social security contributions.
3. Total transfers paid and subsidies.

Source: OECD.

high surplus in periods of buoyant activity are not used to increase public spending. To this end, the Swedish medium-term fiscal strategy is to allow the deficit for individual years to vary with the cycle, subject to the overall objective of a 2 per cent surplus over the cycle as a whole, thereby helping to avoid the danger of pro-cyclical budgeting. However, the framework remains sensitive to the underlying assessment of the potential growth rate of the economy and the evolution of output gaps. If by 2001 the economy may be close to the peak of the cycle as is implied in OECD projections, further fiscal measures may be needed to achieve the overall surplus objective.

Progress in expenditure control

Improving the budget process

Over recent years Sweden has adopted a number of changes in its budget process, designed to improve spending control. Described more fully in the previous *OECD Economic Survey of Sweden*, reforms began with an extension of the parliamentary period from three to four years, and a strengthening of the Select Finance Committee in parliament at the expense of other committees. A two-tier budget process was introduced, with three-year ceilings for 27 major expenditure areas to be adopted in the spring budget and consequent detailed appropriation proposals for the coming budget year to be voted in the autumn.[9] Whereas the Swedish budget management process had previously been relatively weak and fragmented, the reforms equipped Sweden with what, if fully utilised, could prove to be a very efficient models of budget control.

The process has been in effect for close to two years, but it has not been tested in full, insofar as spending ministries and Parliament have not had to face decisions which required further cuts on a broad scale. The main focus of policy-making has been on the overall central government expenditure ceiling, rather than on the separate ceilings for the major expenditure categories of the budget. The total ceilings adopted in 1996 for the years 1998 and 1999 have been maintained also in 1997, even though planned block transfers to municipalities have been increased. In some cases, as with the recent employment-creation measures, a large reduction in unallocated reserves has been needed to preserve the overall ceiling (Table 11). Moreover, it would also appear that once the budget year is well under way, the ceiling loses some of its significance, as in 1997, when overruns of labour market expenditure ceilings were financed out of

Table 11. **Expenditure ceilings 1997-2000**

SKr billion

	1997		1998		1999		2000	
	First proposal	Estimated outcome	First proposal	Present	First proposal	Present	First proposal[1]	Present
Central government budget								
26 specified categories, total	570.5	584.7	565.1	578.7	570.3	590.4	577.2	590.5
of which:								
Old-age pensions	64.3	63.1	64.5	62.7	65.5	62.9	63.6	63.5
Family transfers	36.0	32.9	37.2	35.8	37.9	36.3	34.1	36.8
UI benefits	36.1	43.3	36.1	42.7	34.6	33.5	25.3	25.3
ALMPs, etc.	46.8	52.2	43.9	47.5	41.9	49.4	48.8	48.6
Education	25.6	26.0	26.4	27.1	27.6	28.1	29.4	29.4
Housing, etc.	34.8	30.9	27.7	22.8	23.7	20.4	17.4	17.6
Municipalities	79.8	87.7	79.9	93.0	80.0	100.3	93.8	101.8
Social security outside central government	132.6	129.5	136.7	132.7	142.2	137.9	143.2	144.4
Allocations transferred	4.5	..	0.0	5.0	0.0	3.0	3.0	3.0
Unallocated reserves	15.0	8.9	18.0	3.6	22.0	3.6	20.0	6.2
Total	722.6	723.1	719.9	720.0	734.5	735.0	743.4	744.0
Memorandum item:								
Outside ceilings: interest payments	109.6	95.3	106.3	109.1	102.5	98.2	89.4	93.4

1. In April 1997.

Source: Ministry of Finance, *The Fiscal Bill* (various issues).

lower interest payments, and did not set off a review of expenditures in the same area. In general, the three-year ceilings for each separate expenditure category have not been used actively to restrain cost overruns by requiring compensating expenditure cuts in the same area. No proposals for modifications of entitlement (transfer) programmes have been tabled when trends have not been consistent with the expenditure ceiling adopted. The principle by which overruns should be covered by drawing down other programmes in the same area, or by corresponding reductions in following years, has proved difficult to enforce. The preliminary and in some cases contingent character of expenditure ceilings beyond the budget year militates against using such ceilings in a mechanical way, but their more explicit use could improve long-term consistency in the budget process.

Containing tax expenditures

Tax expenditures are still significant in Sweden, amounting to SKr 199 billion in 1997 on an outlay equivalent basis and to SKr 136 billion or close to 8 per cent of GDP on a net budget basis.[10] The erosion of the tax bases are in some cases substantial, at about 45 per cent for excise taxes and duties and 20 per cent for the value added tax and even for income taxes almost 10 per cent. Bearing in mind that efficiency losses from taxes increase more than proportionately with the level of tax rates, further efficiency gains could be reaped in the Swedish case if tax bases were broadened and tax rates brought down, in the process paving the way for a less complex tax code. So far, the evidence is that tax expenditures are increasing; the estimated 1997 level is up SKr 20 billion from 1995. In a few cases this reflects initiatives to simplify the tax code and ease taxation of small and medium-sized enterprises, but such initiatives should on the whole have been possible to effect within the scope of unchanged nominal tax expenditures. A closer integration of tax expenditures within the expenditure ceiling system is thus called for.

Facing long-term challenges

As noted above, the fiscal consolidation over the 1994-98 period has had to rely as much on tax increases as expenditure restraint, and in the process the tax ratio has been brought close to its peak level of 1989-90, at about 55 per cent of GDP (Figure 18). Achieving effective control of public spending is all the more important not only to prevent taxes from rising but to lay a foundation for lower tax levels in the future. The high tax level – matched only by Denmark among

Table 12. **Public sector revenues and expenditures in selected OECD countries**[1]

Per cent of GDP at market prices

	Sweden	Denmark	Finland	France	Germany	Netherlands	Norway	United Kingdom	United States
Revenue									
Taxes on income, profits and capital gains	**21.1**	29.6	20.1	8.3	10.8	11.7	15.0	13.0	12.6
Personal[2]	**18.1**	26.1	17.3	6.6	9.4	7.6	10.7	9.6	10.0
Corporate[2]	**2.9**	2.3	2.8	1.7	1.4	4.1	4.3	3.3	2.6
Social security contributions	**15.3**	1.6	12.5	19.5	15.6	17.4	9.5	6.2	6.9
Taxes on property, etc.	**1.9**	1.6	1.1	2.3	1.1	1.9	1.2	3.7	3.1
Taxes on goods and services	**11.9**	16.3	14.5	12.5	10.6	12.4	15.7	12.2	4.9
Total taxes	**51.6**	49.4	48.3	45.6	38.2	43.5	41.4	35.1	27.6
Other revenue	**9.4**	6.2	5.8	4.6	7.1	3.8	9.1	2.3	3.4
Total revenue	**60.9**	55.6	54.1	50.3	45.3	47.3	50.5	37.4	31.0
Disbursements									
Final consumption expenditure	**26.2**	25.4	21.9	19.7	19.8	14.0	19.8	21.2	15.7
Interests and dividends	**7.2**	7.2	5.8	4.1	3.7	5.6	2.6	3.7	2.1
Subsidies	**4.8**	2.6	2.9	2.0	2.0	1.9	3.4	1.1	0.3
Other current transfers	**25.7**	22.3	25.4	25.7	21.1	26.6	17.2	14.7	13.8
Total current disbursements	**63.9**	57.5	55.9	51.5	46.6	48.1	43.0	40.6	32.0
Capital outlays	**0.3**	–0.8	1.5	3.3	2.3	1.5	1.6	2.4	1.0
Memorandum items:									
Gross social expenditures[3]	**31.8**	25.4	28.5	21.5	21.8	23.4	22.5	17.7	9.8
Gross to net social expenditure adjustment[4]	**–8.3**	–7.2	n.a.	n.a.	–3.7	–8.1	n.a.	–1.9	1.0

1. 1996, except United States 1995.
2. Excluding taxes not entirely allocable to either sector.
3. 1993 excluding health, comprising both transfers and services in kind.
4. 1993. Includes effects of differences in mandatory social programmes, tax treatment of benefits, indirect taxes and tax expenditures for social purposes.

Source: OECD, *National Accounts and Revenue Statistics*; OECD, SOCX database; Adema *et al.* (1996), *OECD Labour Market and Social Policy Occasional Papers*, No. 19.

OECD countries – reflects both a higher level of services provided by the public sector than in other countries and a higher level of transfers, even when the Swedish practice of providing transfers gross-of-taxes is taken into account (Table 12). While the ratio of expenditure to GDP has fallen strongly in recent years, with the adoption of the fiscal consolidation programme and the upturn of the economic cycle, resulting in a fall in transfers and subsidies of 5 per cent of GDP since their peak in 1993, it is still 3 per cent higher relative to GDP than in 1990. However, efforts to cut costs and improve efficiency in public services have resulted in declining government consumption since the early 1990s, and the decline in employment in local government – the main provider of these services – has been somewhat more pronounced as municipalities are starting to outsource some of their activities.

Over the longer run a distinct trend appears, with government consumption falling relative to GDP (partly owing to demographic trends) and transfers increasing, the latter partly due to the higher unemployment level. This should be seen as being in accordance with deep-seated priorities in the Swedish public, as the safety net of the welfare state has been extended to comprise ever larger sections of the working-age population. Nevertheless, the 1997 *OECD Economic Survey of Sweden* pointed to the need to reassess the emphasis of social aims over incentives in the light of the low redistributive effect of the Swedish tax and transfer system. About two-thirds of transfers reflect redistribution over the life-cycle of the individual, whereas one-third reflects inter-personal redistribution. Even if the old-age pension is excluded, more than 50 per cent of unemployment, sickness and early retirement benefits pertain to redistribution over the life-cycle.[11] This reflects the universal nature of the Swedish social insurance system administered by the government. However, the relation between contributions paid and benefits received is not evident to individuals insofar as benefits are capped while the main part of contributions (payroll taxes) is not. Insofar as also moral hazard effects may be strong, *inter alia* in unemployment and sickness insurance, the system would seem to imply a poor trade-off of equity against incentives. Savings for foreseeable contingencies might to a far larger extent than presently be left to individuals themselves or to labour market organisations and thereby establish a clearer link between premiums (contributions) and future payments than exists under the present tax-financed PAYG arrangement, while also increasing the robustness of government finance and the social security system.

The need for a reappraisal is all the more urgent in that long-term demographic developments indicate an increase in the dependency ratio (those above 65 years of age relative to those 16 to 64 years) from 28 per cent in the mid-1990s to 35 per cent in 2030 – a demographic shift of the same magnitude as over the 1950-95 period. A reversal of both the overall benefit-dependency and the declining effective retirement age will be necessary in order to meet these developments.[12] The current commitments of the Swedish government in the area of old-age pensions are surpassed by few other countries and Sweden stands out as already having a relatively high share of retired persons, with a sharper increase in that share in the coming decades than others. While the supplementary earnings-related pension scheme is funded, it has not yet reached maturity in terms of commitments but is nevertheless in a position where the returns of the funds accumulated are used in part for financing current pension payments. The problem should thus be seen as already rather acute.

A blueprint for pension reform was agreed upon by the centre-right parties and the Social Democrats in 1994, with the detailed proposals for reform settled only in early 1998. The emphasis in the reform is on maintaining the generosity of the present system given an underlying growth in real incomes per capita in the area of 2 per cent. The main innovations to emerge were therefore mechanisms that made the pension system more robust *vis-à-vis* adverse macroeconomic and demographic developments, with benefits to be capped in such cases, but likewise to be adjusted upwards if underlying trends turn out to be more favourable. While the system is to be fully funded, with revenues now going to the central government being transferred to the social insurance system, it will still contain important non-actuarial components, such as a universal flat-rate (guaranteed) pension, and pension entitlements will be earned also when caring for one's own children, during military service and education. Income-related pensions will increase in importance over time, *inter alia* because the guaranteed pension is to be kept unchanged in real terms and is not to benefit from higher overall productivity in the economy. But disincentives to private savings arising from the less than complete link between contributions and benefits noted above will remain and the partial individualisation of contributions (14 per cent are to be handled by competing funds) can only imply a partial offsetting effect. In view of the continued extensive role played by the public sector in the provision of pension insurance, and the negative implications for the Government's true debt position and the future burden imposed by high transfer payments to the working-age population, a higher level of ambition in fiscal policy would appear warranted.

III. Implementing structural reform: a review of progress

Introduction

Within the framework set out by the *OECD Jobs Study*, the 1997 *OECD Economic Survey of Sweden* provided a set of detailed policy recommendations[13] for improving labour market flexibility and reducing structural unemployment. Reflecting the relatively recent shift from low to high unemployment, the Swedish labour market has yet to display some of the features of segmentation common to other OECD countries, particularly insofar as long-term unemployment is relatively low and the exit from unemployment has not varied significantly between highly-qualified workers and those at low levels of educational attainment. On the other hand, entry into ordinary jobs is difficult for a large segment of youth cohorts, prompting a wide array of publicly-defined avenues into the labour market and a large influx into the education system. Relatively inflexible real wages and compressed wage differentials prevent wage growth and relative wages from adapting rapidly to imbalances between labour demand and supply and deter individuals from investing in human capital. The highly regulated Swedish labour market would also appear to impair the expansion of small and medium-sized enterprises, which is important for raising the employment-generating capacity of the economy.

Recognising the need to move on a broad front to prevent high unemployment from becoming deeply entrenched in the Swedish economy, the OECD proposals called for a far-reaching reappraisal of existing structures and policies in the labour market proper, product markets and with regard to the entrepreneurial climate:

– *Increased wage and labour cost flexibility:* lower taxes on labour would reduce labour costs and stimulate labour demand, particularly for low-skilled workers, while lower replacement rates in social insurance

schemes would cut reservation wages and encourage job search. Wage bargaining should allow for increased scope for local adaptation of wages, while wages in general would be more responsive to labour market developments if the social partners were to carry a larger share of the financing of Unemployment Insurance benefits and Active Labour Market Policy programmes (ALMPs).

- *Reform of unemployment benefit and related benefit systems:* a lowering of replacement rates should be accompanied by stricter rules for requalifying for benefits, discontinuing the practice of using ALMP participation to restore benefits which are about to expire.
- *Increased working time flexibility and reform of employment security provisions:* employment protection legislation provisions affecting hiring costs should be reviewed, while the scope to vary the terms and conditions of employment and working time at enterprise level should be increased. Reducing the scope for secondary action now allowed in the industrial relations legislation would also increase the capacity of small and medium-sized enterprises to adjust.
- *Enhanced effectiveness of labour market policies:* efforts to re-introduce job seekers into ordinary jobs should focus on the hard-to-place.
- *Improved labour force skills and competencies:* a strengthening of the vocational content and labour-market relevance of both secondary and tertiary education would make for a more efficient school-to-work transition.
- *Enhanced product market competition:* municipally-owned enterprises should be made to compete on equal conditions with private enterprises to make the markets for services fully contestable; deregulation of the sheltered sectors should continue, and the privatisation of public utilities should be completed, with natural monopolies being separated from competitive activities.
- *Nurturing an entrepreneurial climate:* small and medium-sized enterprises would benefit from a simplification of the tax code and a rationalisation of the wide array of government support schemes and provisions regulating their activities.

The recommendations in the *1997 Survey* were made against the background of substantial modifications to transfer schemes and tax schedules arising out of the 1994-98 fiscal consolidation programme and of the programme set out

in the 1996 Employment Bill to halve open unemployment to 4 per cent by 2000. The employment programme places heavy reliance on efforts in the spheres of adult and tertiary education to upgrade the qualifications of the workforce, on re-focusing resources going into ALMPs and on measures to strengthen small and medium-sized enterprises. The main emphasis with respect to the labour market itself is on securing aggregate wage growth in line with Sweden's main competitors. Reflecting a strong attachment to income-distribution objectives even in the short run, measures which would contribute to increased wage differentials in order to promote job creation have been ruled out.

Against this background, the education effort takes on a crucial importance for the success of the government's present employment strategy, and emerging issues and challenges now facing the Swedish education system are treated in detail in Chapter IV below. By necessity, the effort to promote human capital formation will influence employment outcomes only in the medium term, while the labour supply and therefore open unemployment will fall in the short run as more people move into education. The capacity of the labour- and product markets to adapt to new opportunities for employment and to transfer resources from declining to expanding sectors remains central, however, to employment creation (and to the reintegration of those leaving the education system over the years to come), so that the remainder of this chapter is devoted to an overall stock-taking of the *Jobs Strategy*. For the 1998 OECD Ministerial Council Meeting, issues related to entrepreneurship have been singled out for in-depth examination, with Sweden as one of five countries selected for case studies, and this assessment is set out separately in chapter V below.

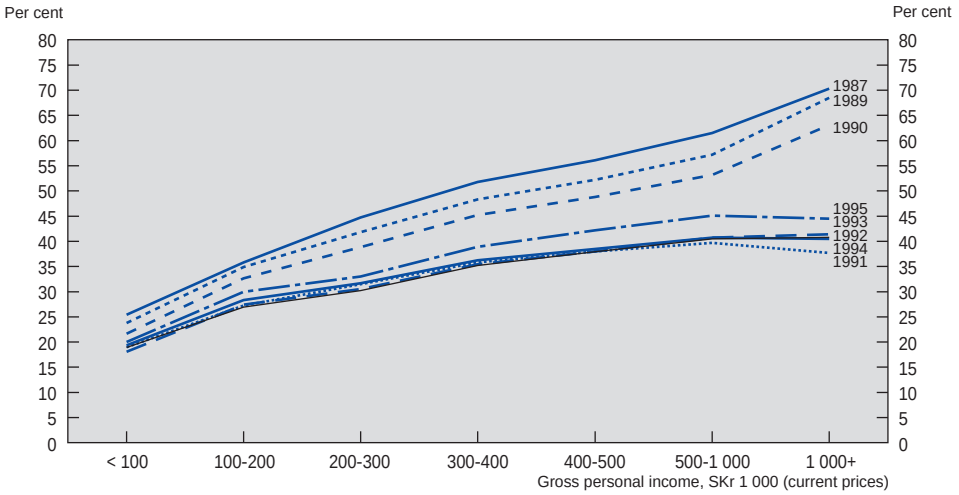
Developments in structural reform

Increased wage and labour cost flexibility

Reflecting the positive effects on *lifetime* earnings of securing a job in the ordinary labour market at an early stage and of on-the-job experience, the above recommendations place a strong emphasis on improving work incentives. In Sweden, work incentives improved significantly in the early 1990s, with the implementation of the 1990/91 tax reform (Figure 19) and the gradual reduction of replacement rates in social insurance from 90 to 75 per cent. But reflecting the emphasis derived from the underlying principles of the consolidation process,

Figure 19. **TAXATION AT DIFFERENT INCOME LEVELS**[1]

Per cent



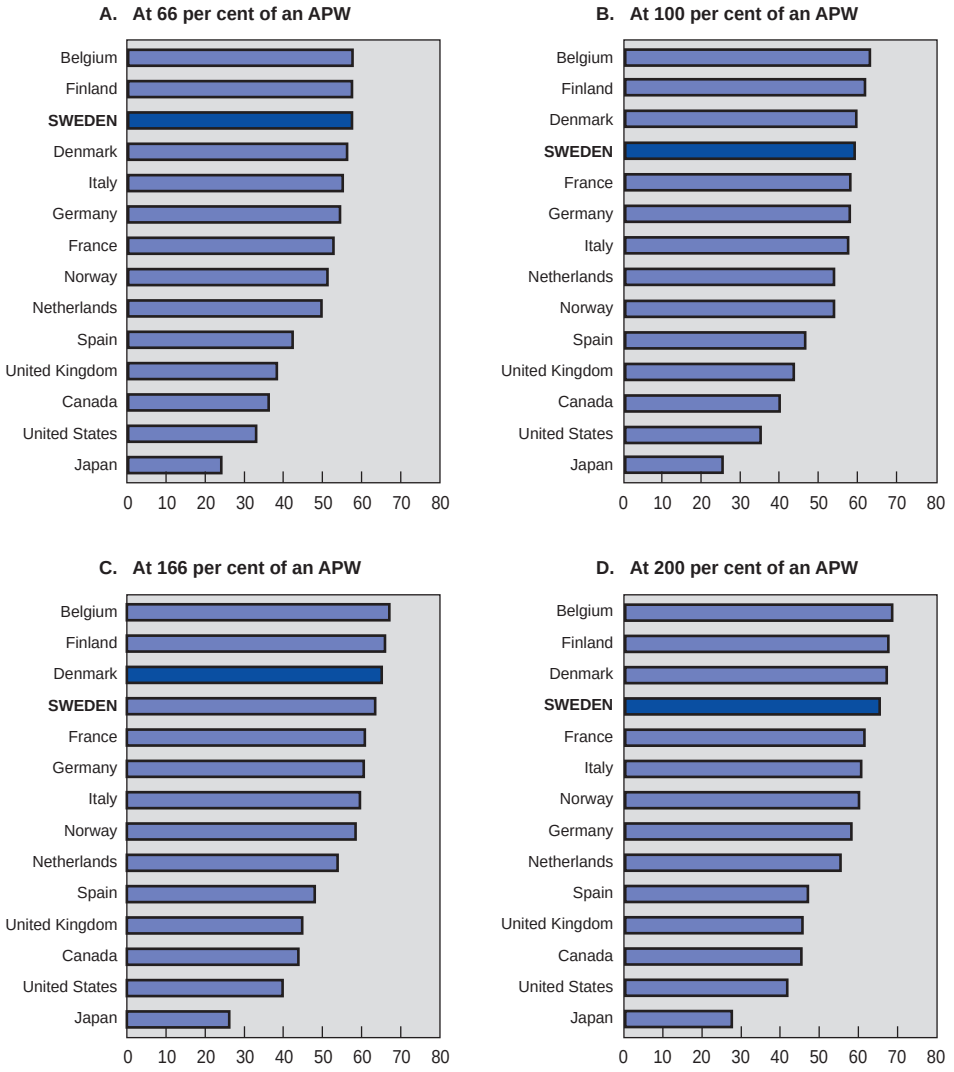
1. Taxes paid as per cent of taxable income.
Source: SCB.

that expenditure cuts and tax increases be balanced to ensure that sacrifices are shared equally by the population, this trend was reversed with the introduction in 1993 of an employees' social security contribution, rising to 7 per cent by 1998, and the 5 per cent surcharge on higher incomes in effect from 1995 to 1998. Moreover, the Employment Bill proposed a partial reversal of the fall in replacement rates, to a level of 80 per cent. Incentives to effective job seeking were nevertheless to be maintained by the introduction of upper limits to the duration of unemployment benefits, by stronger sanctions to ensure job availability, and by a concentration of the resources of Employment Offices on the long-term unemployed. Employment protection legislation was also to be modified to facilitate enterprises' hiring decisions. However, hiring and firing costs and high reservation wages are set to remain severe impediments to employment creation in Sweden.

The high overall tax level implies higher tax rates for low-income workers in Sweden than in most other countries (Figure 20). Forty per cent of full-time employed now face marginal rates above 50 per cent. While the higher marginal

Figure 20. **EFFECTIVE TAX RATES AT DIFFERENT INCOME LEVELS**[1]

Per cent



1. In 1995. Employees' social security contributions and personal income taxes as a percentage of gross labour costs (incl. employers' social security contributions) and taking into account price level effects of consumption taxes. Income levels are given relative to the earnings of a single Average Production Worker (APW).

Source: OECD, *The Tax/Benefit Position of Employees*.

tax rate was to be removed after 1998, recent signals indicate that it will remain, with a more limited coverage. No plans for a major decrease in overall taxation has been presented; however, the tax wedge in labour income is being reduced as a part of the pension reform insofar as benefits are being aligned more closely with contributions paid. It should be noted that for a given tax take, modifications of the income tax schedule so as to shift the tax burden from low to middle and high incomes cannot be expected to improve labour market outcomes. Fifty per cent of income tax revenues come from taxable incomes below the wage income of the average production worker, so that lower marginal tax rates depend critically on a lowering of the overall tax level. The interaction between tax reform and labour market outcomes is discussed further in Box 1.

The 1997 *Survey* argued that the most efficient way of eliminating differences in unemployment risk would be to allow wages to reflect differences in productivity, enterprise finances and local employment conditions. This would require a further decentralisation of wage-setting in Sweden. Currently, the three-year horizon of most major wage agreements, stretching from 1995 to 1998, mean that no immediate change is to be expected on this front, and it is evident that there are strongly conflicting views between the organisations of employers and employees as to future developments in this area. The majority of unions wish to maintain bargaining at industrial level. The practical focus has therefore been on developing mechanisms that ensure a lowering of aggregate nominal wage growth, with the manufacturing sector taking the initiative by prolonging the period of renegotiation and by giving jointly-agreed mediators powers to curtail industrial action and to ensure progress in the negotiations. Initiatives from public authorities to broaden or institutionalise such arrangements await *inter alia* a final expert report to be delivered by November 1998 on how to strengthen the role of the public authorities in this field and other ways to improve the wage bargaining system.[14]

Pressures to modify wage structures in response to the major job losses which have occurred in the 1990s have, in general, been impeded by recent social-security initiatives. The increase in the unemployment-insurance replacement rate from 75 to 80 per cent entered into force in September 1997, with other social insurance schemes to follow from 1 January 1998, and will feed into higher individual reservation wages.[15] A higher contribution from UI funds to cover some of the cost, combined with an ability to vary members' fees,

Box 1. **Tax structure and labour-market outcomes**

The tax reform of 1990/91 simplified the Swedish tax structure, resulting in a flat tax schedule for labour income for the large majority of wage earners. However, income taxes had to be raised to finance the substantial deficit in public finances that emerged in the first half of the 1990s. In the process, the original income tax schedule of a municipal tax averaging 31 per cent and an additional central government tax of 20 per cent on high incomes, has been transformed into an average municipal tax close to 32 per cent, an additional central government tax of 25 per cent (in effect up to 1998) and an employees' social security contribution of about 6 per cent (7 per cent from 1998) which is deductible from the base for the other taxes. At the same time, the real value of tax brackets has also been reduced, as indexation clauses were partially abandoned as part of the overall effort to improve public finances. As a consequence, a tax rate of 36 per cent becomes effective on annual wage incomes above 5 per cent of the wages of the Average Production Worker. When the employers' social security contribution of 33 per cent is included, the total tax relative to wage costs amounts to about 60 per cent for even the lowest paid. Indeed, a high level of transfers and free public services is necessary to secure a decent standard of living for low-paid workers: the net wage income (after tax but before transfers) at 50 per cent of the APW wage is about 85 per cent of the standard social assistance rate. As noted in the *1997 Economic Survey of Sweden*, the wage, tax and benefit structures lead to a potentially significant benefit-dependence becoming embedded in the economy.

To improve these structures it has been proposed that *i)* the cost of labour should be reduced by an across-the-board reduction in employers' social security contributions; *ii)* the tax schedule be modified by increasing the basic income deduction, which would primarily lower the effective tax on lower incomes but leave marginal rates at higher income levels unchanged. The first proposal has been linked to an offsetting reduction of government support to business so that it would be neutral from the outset. However, in the long run, the overall tax on labour is on the whole independent of its composition between income taxes and employers' payroll taxes and is borne fully by wage earners. To avoid wage growth picking up to fill the ''gap'' created by the fall in payroll taxes, it is imperative that labour market flexibility improves in tandem with the lowering of taxes.

The rationale for proposing a targeted lowering of taxes on low-paid workers is to improve their competitiveness *vis-à-vis* other workers. But in the Swedish case, the floor set to wages by collective bargaining often prevents lower taxes from feeding into lower wage costs for employers. In this respect, a general reduction of the taxation of labour combined with increased scope for local determination of wage structures, would serve to increase the demand for labour in general and even out differences in unemployment risk. A targeted reduction of labour costs for certain groups of workers may only serve to compensate for wage-setting practices which do not internalise the costs of unemployment. Moreover, the economic signals as to choices of higher education or not may also be further blurred and in consequence differences in labour market outcomes perpetuated.

should work to clarify the trade-off between wage increases and consequent (un)employment facing trade unions. But the share of expenditures for unemployment insurance benefit covered by the social partners, now at 7 per cent of total expenses, is still very small. Incentives to modify structures generating unemployment have also been dampened by a programme initiated in 1996 to establish 40 000 public sector jobs for 55 to 64 year olds registered as unemployed for more than two years, and an offer available to 60 to 64 year-olds during the last six months of 1997 under which those unemployed for more than six months can retire with a benefit equivalent to the corresponding unemployment insurance.[16] During the first eight months of 1998, an offer will be available for workers aged 63 to 64 years who – subject to approval from their employer and their local union – will be allowed to retire with a benefit equal to 80 per cent of their previous wage if they are replaced by long-term (six months or more) unemployed persons of 20 to 35 years of age, the costs to be borne partly by public authorities (75 per cent) and partly by the employer (25 per cent). In a parallel case, a doubling of the period of sickness benefit paid by the employer from two to four weeks during 1997 might have stimulated enterprises to act to lower the incidence of sickness absence; in the end, this measure was repealed from 1 January 1998 after pressure from the social partners.

Reform of unemployment benefit and related benefit systems

The Employment Bill included a reform of the Unemployment Insurance System, due to enter into force from 1 January 1997, which tightened eligibility criteria and introduced an upper limit of three years on the duration of UI benefit. The intention was to offset the effect of the increase in the replacement rate. However, the act was repealed before it was to enter into force, and the revised proposal forwarded in March 1997 implied several modifications. While the increase in the replacement rate was maintained, the qualifying period was increased from five to six instead of nine months. Most importantly, the proposal for an upper limit of the duration was dropped. In addition, the underlying structure was changed, the UI benefit now consisting of an income-related part and an income-independent part and thereby integrating the separate cash benefit (KAS) for unemployed not yet UI-fund members. As maximum UI benefit was also raised after having been kept unchanged in nominal terms during the fiscal consolidation, UI benefits were increased by between 3 and 7 per cent for

different categories of job seekers. As noted above, these changes have been accompanied by an increase in the refund paid by UI funds to the central government, and UI funds have been allowed to set separate contribution rates. The combination of replacement rates well above its neighbouring European countries and the *de facto* unlimited duration of unemployment benefit – benefit eligibility is restored conditional upon ALMP participation – remains a major obstacle to the flexible functioning of labour markets.

The high reservation wage implied by social assistance, which is available to low-income households without social insurance, have traditionally been counterbalanced by strong means-testing, which requires recipients to sell their capital assets. Moreover, wide discretion has been allowed to municipalities to adapt rates to local circumstances, serving to limit its use and impact on the local labour market. In a recent amendment to the Social Services Act the discretionary powers of municipalities have been restricted, with a significant part of the benefit being based upon nationwide norms for expense items used in its calculation. Municipalities are thus allowed only to apply local cost norms to a limited number of items. It may thus be inferred that social assistance could now act to establish a floor for wages, at least in some parts of the country. On the other hand, the powers of social-security offices to impose requirements on assistance recipients have been strengthened somewhat, as those below 25 years of age may be required to undertake training or education to reduce their future benefit-dependence.

Increased working time flexibility and reform of employment security provisions

After the social partners failed to agree on reforms to the employment security provisions in mid-1996, the government tabled a set of proposals which were adopted by Parliament in late 1996, to enter into force during 1997. Of particular importance are: *i)* the length of notice periods is to be determined on the basis of tenure and not of age, implying that the costs of hiring older workers will fall relative to other groups; *ii)* enterprises' rehiring obligation *vis-à-vis* laid-off workers will expire after nine instead of twelve months; *iii)* twelve-month fixed-term contracts with no restrictions applied to the nature of the work carried out has been introduced, with all enterprises regardless of size being allowed to employ up to five persons on such contracts and new establishments being

allowed to extend them to eighteen months; *iv)* part-time workers are to be given priority for vacant full-time positions; and *v)* workers on replacement contracts for three years within a five-year period are to receive permanent contracts.

On balance, the above measures give enterprises somewhat more flexibility, but the last-in-first-out rule remains the basic principle of the law and the length of probationary periods is unchanged. A wide-ranging decentralisation of powers from the central organisations to the enterprise level proved impossible to carry out. For some provisions – such as fixed-term contracts and the scope of the last-in-first-out rule and the rehiring obligation – it is left to the labour market organisations themselves to decide at which level powers should be located. Due to resistance on the union side, there are so far no signs that this modification to the act has served to shift decision-making downwards to the enterprise level. Equally important, the provisions of the Working Time Act are fully applicable to small and medium-sized enterprises, while other legislation in the area of industrial relations has been left unchanged.[17] The legal restrictions on the hiring decisions of such enterprises thus remain more severe than in most other countries.[18]

Enhanced effectiveness of labour market policies

The number of participants in active labour market policies (ALMPs) has been scaled down somewhat to about 4½ per cent of the labour force from a peak of 5.3 per cent in 1994, and the Employment Bill aims to refocus the activities of the Public Employment Service (PES) towards job-matching in the labour market, with vacancies to be filled more rapidly and long-term joblessness to be reduced. Insofar as ALMPs have been used not only to maintain and improve the employability of those affected by high unemployment but also explicitly to reinstate benefit eligibility, the return on resources thus employed should thereby increase. Job-placement efforts are underpinned by individual action plans for job seekers, detailing the measures to be taken on the part of both the unemployed and the local Employment Office to promote reintegration in the labour market, giving ALMPs and other education and training efforts a clearer direction. The progress in improving the matching process and lowering long-term unemployment has been slower than called for by the objectives set out, and while long-term unemployed are getting higher priority on ALMPs, their share in ALMPs, at 34 per cent, is more or less in proportion to their share in overall registered

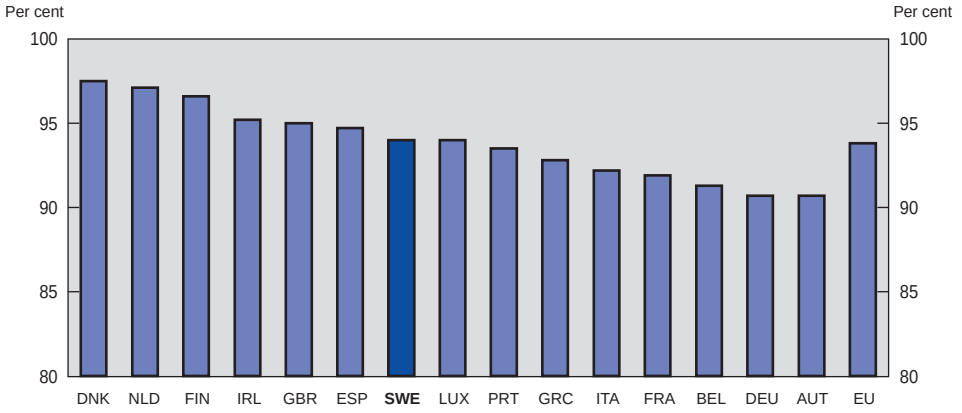
unemployment (27 per cent). Moreover, the recent decision to maintain the unemployment insurance benefit as a *de facto* open-ended income support could impede efforts to reintegrate the unemployed into gainful work as rapidly as possible. A more detailed analysis of labour market training, together with recommendations for the further development of this central ALMP programme, is contained in chapter IV.

Enhanced product market competition

Reflecting a long-standing commitment to free trade in goods and services, Swedish tariffs and non-tariff barriers to trade have been among the lowest in the OECD area and foreign direct investment outflows were early made subject to a benign regime. With the dismantling of all major support schemes to manufacturing in the mid 1980s, government aid to industry has been relatively low in a European context, and after its accession to the European Union from 1 January 1995 Sweden rapidly implemented all the major directives of the internal market programme and is now ahead of many other countries (Figure 21). Over the past few years, however, progress in completing product market reform has

Figure 21. **STATE OF IMPLEMENTATION OF THE EU DIRECTIVES ON THE INTERNAL MARKET**[1]

Per cent



1. Directives implemented, either by notification of measures or no measures necessary, relative to the relevant directives. As of 15 September 1997.

Source: Commission Services of the European Union.

appeared to be slow. Even if the regulatory framework for public utilities has been extensively liberalised in sectors such as air transport, postal and telecommunications services, the incumbent still exerts significant market power. EU membership implies *inter alia* that support to agriculture has again been stepped up after Sweden had set up one of the most market-oriented agricultural regimes of industrialised countries in the early 1990s. While support to ailing industries has been abandoned and the motivations for support are couched in more general terms, the scope for further reductions in government aid to non-agricultural businesses has not been utilised; in nominal terms, government support has thus increased by close to 50 per cent since its trough in 1990. Moreover, the business sector benefits from tax reliefs extended under a host of tax schemes, the tax expenditures thus incurred amounting to close to SKr 40 billion on a net basis in 1997, a 40 per cent increase since 1995 and more than five times the amount cited in a recent official report on direct support to business.[19]

Tax expenditures have thus taken on an added significance in the energy, transport and environmental field, as the internationally exposed manufacturing sector has been cushioned from higher overall taxes on energy by a differentiation of tax rates. This problem has been accentuated by the recent decision to start dismantling Sweden's nuclear power-generating plants from 1998 onwards, even if all of the twelve reactors have an economic and technical life-span stretching well into the next century. Decommissioning these plants and turning to other energy sources will entail costs to the Swedish economy as a whole, and sheltering energy-intensive industries from rises in energy prices will impede the required adaptation of the industrial structure towards lower energy-dependence and the public finances will suffer. Some offset could be found in the continued integration of the Nordic electricity market, which now entails a full integration of the Swedish and Norwegian wholesale spot markets for electricity, which also determines central references for prices in long-term contracts for delivery of electricity. Although highly efficient in a European context, a number of features of this market and the Swedish retail market still work to restrain competition. Through vertical integration, the major generators have acquired both transmission and distribution networks and households are still meeting significant obstacles in shopping around for electricity. The Swedish market would benefit from a full separation of transmission – the monopoly part – from the other activities, more transparent price-setting for the transmission grid and a reduction of the extensive vertical integration between generators and distributors.

Regulatory reform also remains important for the airline and telecommunications sectors. In the former case, the 1992 deregulation measures have still to produce lower prices and a transparent price structure, the position of the dominant supplier being underpinned by a shortage of peak slots (take-off and landing rights) at the central domestic hub (Stockholm airport). A differentiated pricing of slots would improve overall capacity use and be of particular importance for the opening up of the national market to international airlines (cabotage) from April 1997 to have effect. Following the 1993 deregulation, the telecommunications market was transformed from an unregulated monopoly to a state of regulated competition. The incumbent, the 100 per cent government-owned Telia, has nevertheless managed to hold on to a strong position in the face of several new entrants in the market, again reflecting its ownership of the fixed telephony network and strongholds in both networks for mobile telephony. Telia has therefore a central influence on market developments through the prices charged to competitors having to use its fixed network and through the possibility of cross-subsidising activities exposed to competition. A new telecommunications act which entered into force in mid-1997 seeks to address some of the market imperfections by obliging the network owner to allow other operators access at cost-based prices and by allowing government regulations to impose price caps to prevent it from exploiting its monopoly position. While the Swedish telecommunications market, particularly in mobile telephony, appears competitive in a European context, productivity is still low, and further impetus to competition and lower costs could be achieved by splitting up the activities of Telia into separate companies, *inter alia* by creating a separate network access company.

Competition is also still weak in a number of other domestic-oriented sectors, such as distribution and construction, owing to a variety of government interventions (*inter alia* zoning regulations and support schemes), and in postal services where the government-owned incumbent on several occasions has been found to act to deter competition. Competitive structures in the construction and the distribution sectors have been reviewed, but further action can only be expected after the on-going review of the 1993 Competition Act, embodying the abuse principle in use in the EU, has been completed. However, the main thrust of this review is to assess the interpretations and demarcations emerging from the decisions of the Competition Authority rather than explicitly strengthening the overall impact of the Act.

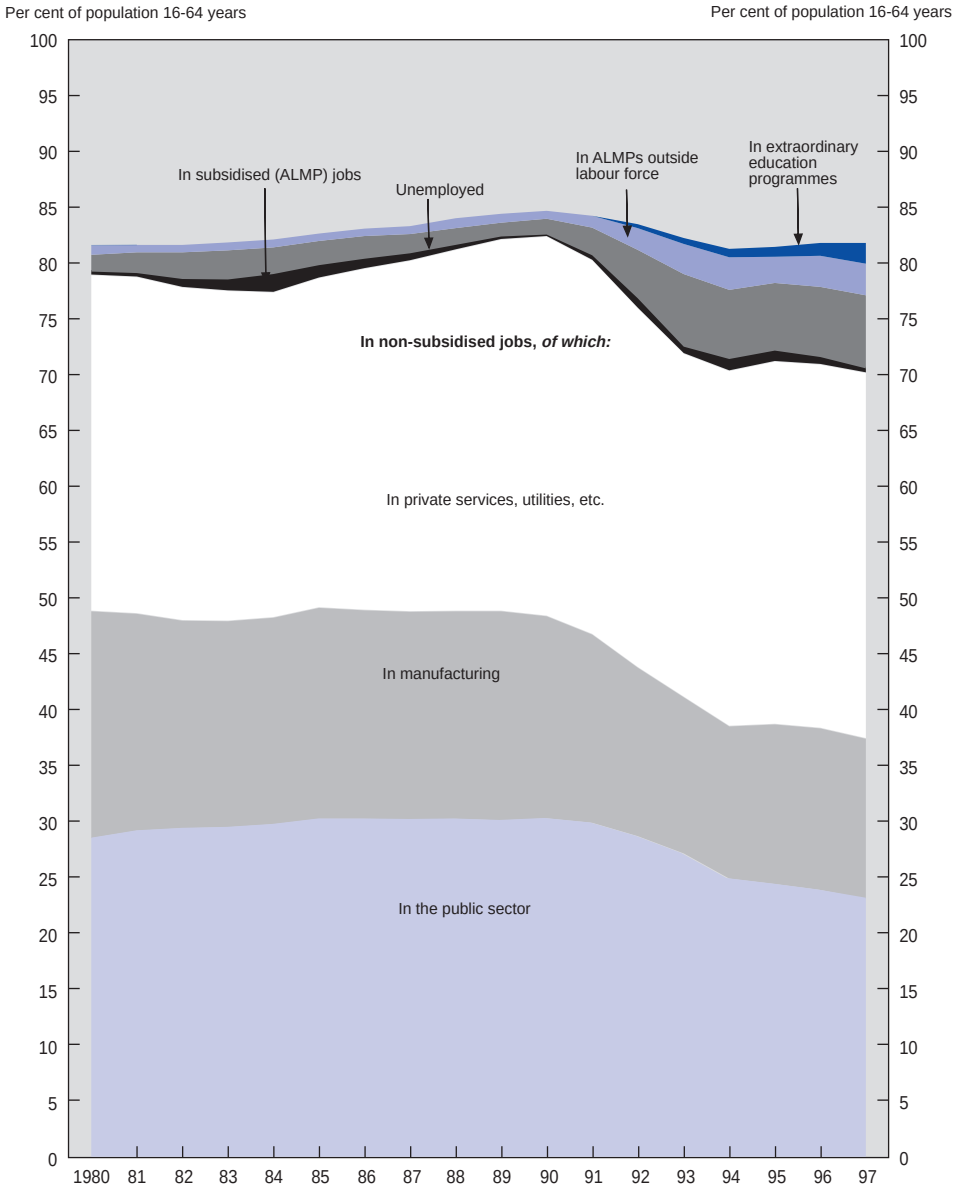
Overview and scope for further action

Employment and labour force participation have remained stagnant since their sharp declines during the recession and are far below their level in the 1980s (Figure 22). Recent underlying employment trends are, however, fairly divergent insofar as employment in the private sector rebounded strongly up to 1995 whereas public sector employment has fallen continuously since 1991. From a longer-term perspective, these trends should be conducive to renewed employment growth. And if wage growth adapts to the levels prevailing in Europe, the labour-shedding and capital-deepening of investment set in train by the 1995-98 wage agreements should gradually slow down. Despite the caveats noted above, competitive forces are strongly felt in most segments of the economy and they can be expected to strengthen and encompass even more sectors as the integration with the European economy proceeds, and this should ensure that price levels converge towards those prevailing elsewhere and price inflation remains low. However, it is evident that the disincentive effects of the tax and benefit systems, as well as of employment regulations, described in the 1997 *Economic Survey of Sweden*, still impose a severe burden on the Swedish labour market. Youths had to bear the immediate brunt of the negative impact on labour demand in the early 1990s, and older workers are already facing an effective retirement age of 58½ years.

The 1996 Employment Bill constituted a concerted effort to improve the flexibility of the Swedish labour market. Most proposals have now been implemented, the major exception being the abandonment of the upper time limit for unemployment benefits. Some of the above measures are being frustrated by the parties affected and slower in having effect than originally envisaged. The legal framework for wage bargaining will be basically unchanged when the social partners enter the 1998 wage-bargaining round, and it remains to be seen to what extent the shift downwards in inflation expectations will feed into lower nominal wage growth and less rigid wage-wage links. The recent modifications to the unemployment insurance scheme will on the whole reduce incentives to effective job search insofar as the higher replacement rate is not offset by a definite upper limit to benefits. Insider-outsider mechanisms are still prevalent and adjustments to the employment protection legislation, although acting to lower hiring and firing costs, have not been of a width and depth to markedly reduce their scope.

Figure 22. **STRUCTURE OF THE LABOUR FORCE**

Per cent



Source: SCB; Ministry of Finance.

In view of the signals and expectations now emerging, Swedish labour market policies are in many respects at a cross-roads. The government is relying heavily on education and training – specifically policies to promote adult and tertiary education – to eventually ensure the compatibility of present levels of labour costs with full employment. Such an approach is to be welcomed, but the education effort needs to be underpinned by economic structures conducive to human capital formation not only in a general way but also to ensure that occupational choice is guided by clear signals. Free product markets help give such signals, but macroeconomic adjustment through product market flexibility can only be partially effective while labour market signals are distorted. Even though the extensive education effort may work to validate the present compression of wages relative to competencies, this will only be fully effective in the longer run and can only belatedly start to offset the forces which are working to keep unemployment high.

The structural reform programme outlined in the Growth Bill of late 1995 and followed up in the Employment Bill thus needs to be revitalised, drawing upon the recommendations following from the *OECD Jobs Strategy*. This would create a foundation for wage formation to be more responsive both to imbalances in the labour market and to structural changes and innovations in product markets, including the diffusion of technology. Such an approach would help to ensure a more reliable allocation of resources to human capital development than one based on artificially compressed wage differentials. The implications following from this are set out in more detail in Box 2 below.

Box 2. **Implementing the OECD Jobs Strategy – an overview of progress**

Macroeconomic prerequisites for economic growth have improved strongly but rigid labour markets contribute to continued high unemployment. The programme to improve labour market structures set out in the 1996 Employment Bill should be implemented in full and without delay

	Job strategy proposal	Action taken	OECD assessment/ recommendations
I.	**Increased wage and labour cost flexibility**		
	• Lower taxes on labour	The 5 per cent surcharge on high incomes to be kept with a more limited coverage after 1998	Lower the highest marginal tax rates to the 50 per cent envisaged in the 1990/91 tax reform and lower the share of tax payers subject to that rate
	• Continue to lower replacement rates in social insurance	Replacement rates increased from 75 to 80 per cent	Lower replacement rates are required to allow the wage structure to adapt to differences in unemployment risk
	• Increase wage flexibility with respect to skills, sector and firms	No action	Proceed in line with proposal
	• Shift financing of UI benefits and ALMPs from government to UI funds themselves	Refunds from UI funds to central government have been doubled	The share refunded is still very small and should be increased
	• Review scope for secondary action	No action	Proceed in line with proposal
II.	**Reform of unemployment insurance and related benefits**		
	• Reduce the generosity of benefits	Subsidised jobs no longer qualify and qualifying period increased from five to six months	An upper limit to the duration of UI benefits should be introduced
	• ALMPs should not be allowed to prolong the effective UI benefit period	ALMPs still available to reinstate benefit eligibility	Proceed in line with proposal
III.	**Increased working-time flexibility and reform employment security provisions**		
	• Review employment protection legislation to improve hiring prospects	Hiring costs of older workers reduced relative to others. Rehiring obligations *vis-à-vis* workers laid off reduced in scope. The scope of fixed-term contracts extended. Stronger employer commitment to part-time and temporary workers.	Hiring and firing costs are still high and further efforts are needed to stimulate enterprises to take on new workers

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	• Decentralise decision-making with respect to terms and conditions	The legal framework has been made neutral with respect to (de)centralisation	Unions should be encouraged to allow a decentralisation in their rules and regulations
IV.	**Enhanced effectiveness of active labour market policies**		
	• Reduce the scope of ALMPs	Expenditure on ALMPs reduced from 5¼ to 4½ per cent of the labour force	A further reduction is warranted
	• Concentrate efforts on the long-term unemployed and others at risk	A reorientation of activities is under implementation	Reorientation to continue
	• Remuneration for subsidised work should give incentives to seek ordinary jobs	Only a few programmes have not been converted to the work-for-benefit principle	Conversion to be completed
V.	**Improve labour force skills and competitiveness**		
	• Sustain the increase in enrolment rates in higher education	Increased capacity in tertiary education and adult education at upper secondary level	*Detailed in Chapter IV*
	• Secure an effective school-to-work transition	Strengthened vocational content of upper secondary education	
	• Lower the share of youths without a vocationally qualifying education	A modernised apprenticeship scheme and vocational education at tertiary level is tried out	
VI.	**Enhance product-market competition**		
	• Secure a level playing-field between private and municipally-owned enterprises	No discretionary action	Should proceed in line with proposal
	• Privatisation to continue with emphasis on separating natural monopolies from competitive activities	No discretionary action	Should proceed in line with proposal
	• Continue to deregulate sheltered sectors	Evaluation of distribution and construction completed and a general review of the Competition Act under way	Further efforts to strengthen competitive pressures should be implemented after the review
VII.	**Nurture an entrepreneurial climate**		
	• Simplify the tax code and the wide array of government support schemes and provisions regulating the activities of SMEs	SME-designated modifications of tax schedules implemented and a broader review of the present tax code for SMEs set up	*Detailed in Chapter V*

IV. Education and training

Introduction

Education has been a central feature of the Swedish economic and social model, embodying the objectives both of creating equal opportunities for all and of stimulating long-term growth through rapid human capital formation. These objectives served to underpin the strong expansion of the formal education system in the post-war decades and the parallel build-up of labour market training and adult education. More recently, the relative decline in Swedish economic performance has focused attention on the role of highly qualified labour in enhancing the capacity to adapt to new technologies, promoting innovation and improving the overall versatility to withstand shocks to economic activity. This focus has been underpinned by a recognition that large differences in unemployment risk can only be avoided if the competencies of low-skilled workers are upgraded to match the strong compression of the wage structure. The formulation of effective policies in these respects raises issues concerning the content and quality of the formal education system, the scope and design of adult education, and of underlying incentives to undertake an education.

The present chapter begins by outlining the institutional set-up of the Swedish formal education system, its claims on resources and the educational achievement of its pupils. The second part summarises the evidence relating to the return to society on the resources invested and the set of economic incentives affecting decisions to defer entrance into work in order to undertake higher education. Publicly-provided adult education in Sweden is an integral part of the formal education system and central issues in this field are also covered in the first two sections; however, the third section is devoted to specific issues emerging with respect to the mix of private and public provision of adult education and training, which are the recent focus of the ''life-long learning'' debate. The concluding part seeks to assess the adequacy of the policy response and trends in underlying structures against the challenges now confronting the system.

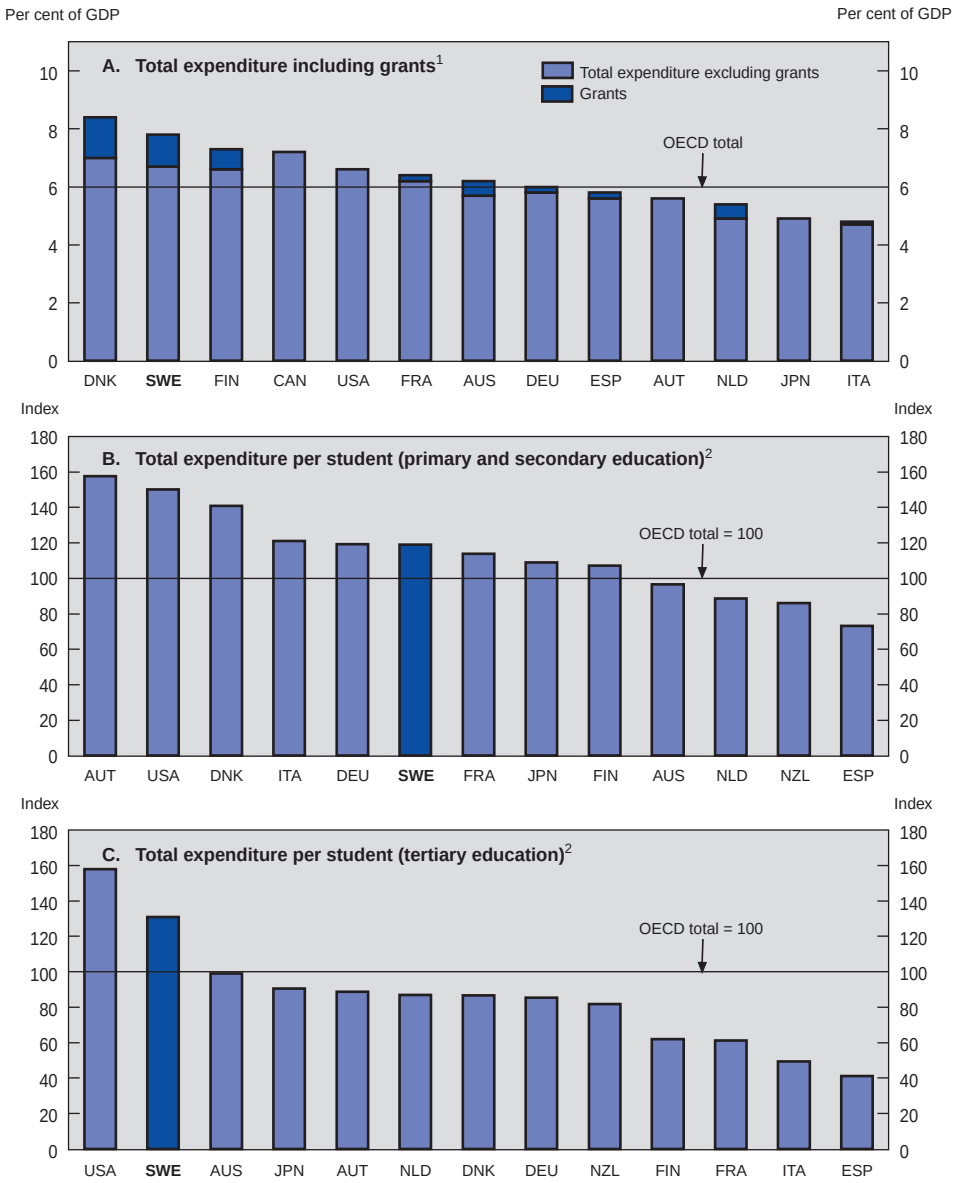
Formal education: developing a new framework

At 6.7 per cent of GDP, the total public and private resources devoted to education in Sweden are among the highest in the OECD area (Figure 23, panel A). When grants to students are included, the share increases to 7.8 per cent, more than one quarter above the OECD average, reflecting both higher spending per student at all levels of education than in most other countries (Figure 23, panels B and C) and a higher overall enrolment rate. As a consequence of the substantial resources committed to education over the past decades, the share of the population with only compulsory (primary and lower secondary) education has fallen from 65 per cent in the early 1970s to below 30 per cent in 1996, while the shares with upper secondary education and tertiary education have increased by close to 20 and 16 percentage points, respectively, to 47 and 24 per cent of the population (Figure 24). The educational effort has succeeded in providing Sweden with a relatively well-educated population compared with other countries: the proportion of the population with only compulsory education is internationally low and that with upper secondary and short (non-university) tertiary education relatively high, while the share with university education is in line with the OECD average. However, the share of each cohort undertaking tertiary education of three years or more, which had been growing well into the first post-war generation, has been declining for some time: the generation born in the first decade after the second world war has the largest share of persons in this educational category (15 per cent), while the corresponding share for those born in the mid 1960s declined to around 10 per cent. The increase in the proportion of higher-educated people thus reflects a propensity to choose education with a relatively short duration.

Institutions, costs and efficiency

The Swedish formal education system has four layers, which become increasingly diversified as pupils move up through the system (Figure 25). Primary and lower-secondary education is compulsory and lasts nine years. Moreover, compulsory education has effectively been extended to include both the pre-school and the upper secondary level: 94 per cent of six-year-olds take part in pre-school education (another 5 per cent enter school directly) and nearly all continue in upper secondary education. Starting in 1992 and fully implemented

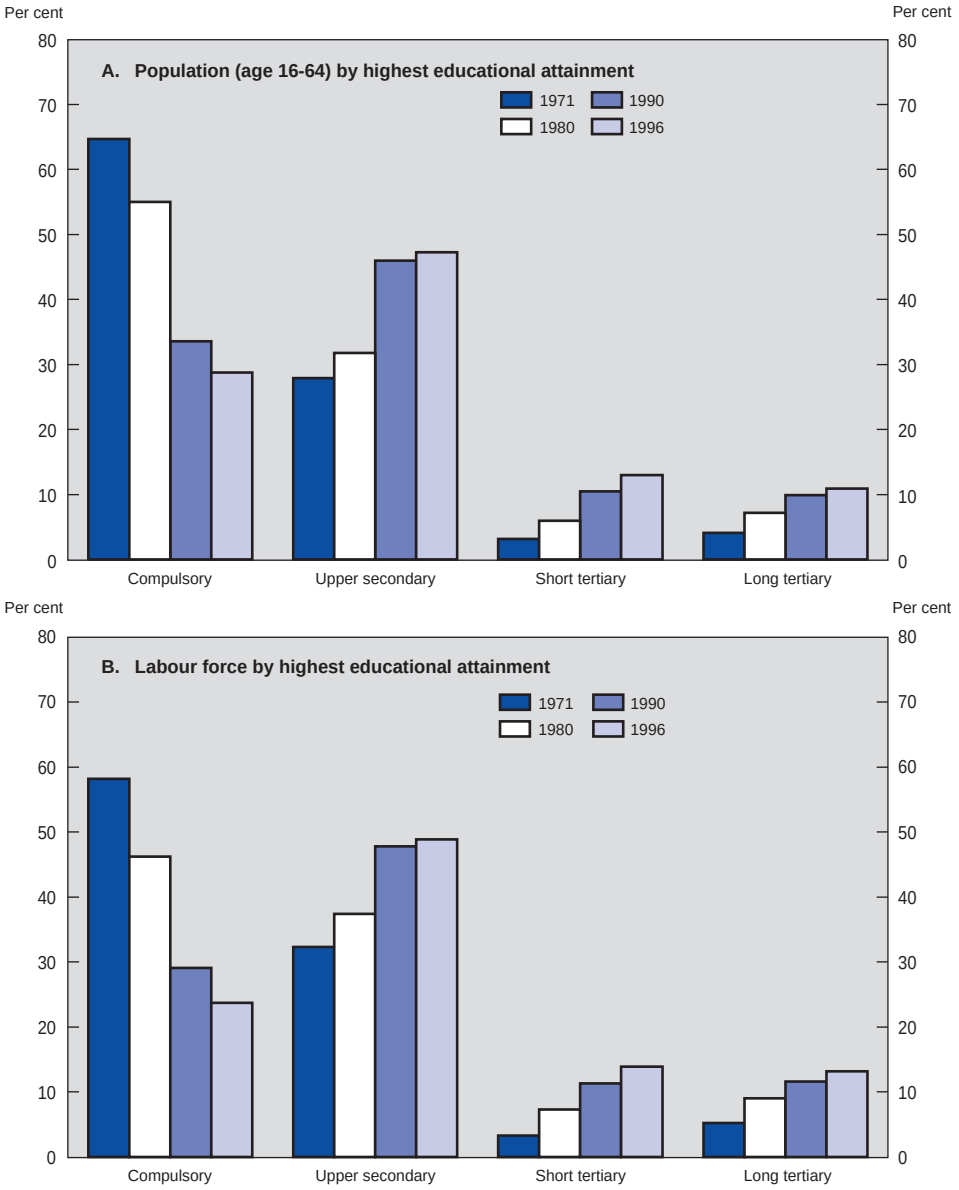
Figure 23. **EDUCATIONAL SPENDING**



1. 1995.
2. 1994. Calculated on the basis of $US converted using PPPs.

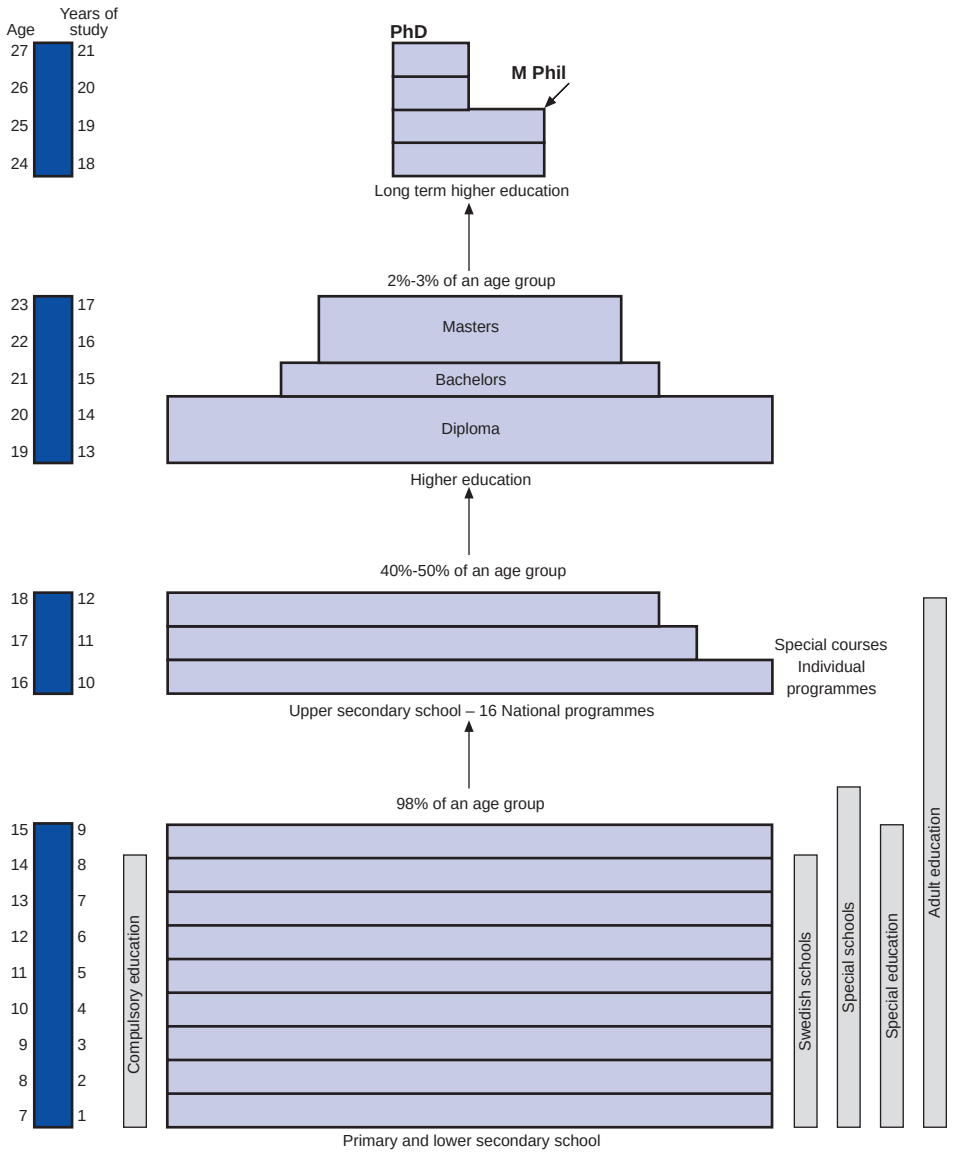
Source: OECD (1997), *Education at a Glance.*

Figure 24. **POPULATION AND LABOUR FORCE BY HIGHEST EDUCATIONAL ATTAINMENT**



Source: SCB.

Figure 25. **THE SWEDISH EDUCATION SYSTEM**[1]



1. 1996.
Source: OECD (1996), *Education at a Glance.*

by 1996, all existing upper-secondary programmes were replaced by sixteen national programmes, all of three years duration. All the programmes qualify for further studies, although fourteen of the programmes – chosen by more than 50 per cent of the students – equip the students with skills aimed at subsequent labour market participation. Tertiary education – pursued by around one-third of a cohort – takes place in either universities or university colleges, leading to a two-year general degree or a three-year bachelor degree. Post-graduate studies can be pursued in a one-year Master programme, a two-year M.Phil. (''*Licenciat*'') or a four-year Ph.D. programme. Formal competencies can also be obtained through the adult education system (*Komvux*), which provides above-age pupils with skills at the compulsory and upper secondary level (see below). A more detailed description of the educational framework is provided in Box 3.

Since the reform of primary and secondary education in 1991, which entailed an extensive decentralisation of decision-making, educational objectives are set out in national legislation, while their implementation, to be detailed in a mandatory education plan, is the responsibility of the municipalities. As a consequence of the reform, schools are financed along with other municipal responsibilities, with local taxes being supplemented by a block grant from the central government to compensate for differences in economic and demographic structure among municipalities. A municipality is not obliged to offer the full range of programmes in its upper secondary schools, but county authorities (which have direct responsibility only for some highly specialised programmes) are supposed to ensure that the full range of programmes is offered within each county. Parents have a free choice of school for their children within a municipality, and they may also choose a private school. Even though these options are in little use, with only 2 per cent of the pupils choosing a public school other than the one recommended by the municipality and 2 per cent of the pupils attending private schools, they have implied a move towards decentralisation even at local level. As a consequence of the right to choose a school, each school is allocated the means to cover its current costs on the basis of its actual number of pupils, and a few municipalities have moved further by explicitly introducing a voucher system. For tertiary education, decentralisation was a cornerstone of a 1993 reform which made universities responsible for planning the education they offer and which – within a system that still restricts access to many tertiary programmes – implied greater freedom of choice and overall easier access for students.

Box 3. **The educational framework: initial education**

Compulsory education (''grundskolan'')

Compulsory education – primary and lower secondary education – lasts nine years. National objectives and curricula are set by Parliament, but since 1991 the municipalities have been responsible for the implementation of these and for the management and financing of the compulsory schools. Municipal schooling is provided free of charge and parents are allowed to select among schools within a municipality. Private (independent) schools are attended by around 2 per cent of the pupils and financed fully by a municipal grant – equivalent to the municipality's average cost. Compulsory schools also offer teaching in more than 100 languages for children with Swedish as a second language. In the school year 1995/96 there were 940 000 pupils enrolled in 5 200 schools and taught by 74 000 teachers at a cost of SKr 46½ billion, equivalent to 2.9 per cent of GDP.

Education at upper secondary level (''gymnasieskolan'')

All qualified pupils, *i.e.* those below 20 years of age with a leaving certificate from compulsory school and a pass in the core subjects, are entitled to an upper secondary education with free choice of school with around 70 per cent entering the programme of first choice. Upper secondary programmes comprise sixteen national programmes with a duration of three years, with most of the programmes branching out in the second and third year. The same division of responsibility applies to central and local authorities for the formulation of curricula and its actual implementation as to compulsory schooling, so that the programmes can also be designed to cater to local needs and conditions. Two of the programmes prepare the students for tertiary education, while the other fourteen programmes, although qualifying for tertiary studies, aim at equipping the pupils with vocational skills – ranging from kindergarten teachers to mechanical engineers – to enter directly into the labour market and include at least fifteen weeks of in-work training. In addition, there are specially designed programmes for students with requirements that fall outside the national programmes and individual programmes for students who are unsure of what studies to pursue. Completion of a programme is dependent both on the number of hours studied and the award of a minimum mark in each subject. In the school year 1995/96, 292 000 students in 640 schools were taught by 20 200 teachers at a cost of SKr 20 billion, equivalent to 1.2 per cent of GDP.

Higher education

Admission to tertiary education is based on *i)* having passed upper secondary education, or *ii)* (for those 25 years or more) having worked at least half time for four years and satisfying upper secondary requirements in Swedish and English (the so-called 25/4 rule). With access restricted to many institutions, a selection is made by institutions themselves, subject to the restrictions that at least one-third of new students should qualify only on the basis of the upper secondary examination, and at least one-third

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should qualify on the basis of a general university aptitude test combined with work experience. Tertiary education programmes vary in length from two to five and a half years, the shorter ones including *inter alia* education for pre-school and child care personnel, qualified technical vocational education, etc. Moreover, Swedish university degrees tend, on average, to be of shorter duration than similar degrees in other countries. Each university or university college decides the organisation of studies and the range of courses on offer – possible combining them into specific educational programmes, leading to specific degrees – while the individual students have substantial freedom in choosing their study route and combining different subject courses into a degree. Each course varies in length from five weeks to eighteen months and gives points according to the number of weeks of full-time study, each full-time term giving 20 points, with a two-year Diploma requiring 80 points, a three-year Bachelor's degree 120 points – including 60 points in major subjects and a 10-point thesis – and the four-year Master's degree 160 points – including 80 points in major subjects and a thesis of 20 points or two of 10 points. In the academic year 1994/95 there were 269 800 students pursuing undergraduate studies and 16 200 pursuing postgraduate studies. The total cost excluding student grants was SKr 26.1 billion in 1994/95 or the equivalent of 1.6 per cent of GDP.

Study financing (''studiestöd'')

Education grants are available to three categories of students: those below 20 years of age in upper secondary education (''*studiehjälp*''); those over 20 years old and in tertiary education (''*studiemedel*''); and for participants in adult education (''*vuxenstudiestöd*'' and ''*utbildningsbidrag*''). With the exception of the first of these, grants are not affected by parental income, but will be scaled down if own income rises above a certain level. For students at university and university colleges, grants are awarded for a maximum of twelve semesters, and after the first year additional grants are dependent on the student maintaining satisfactory progress through the educational programme (75 per cent of full-time study achievement in each semester). The grant amounts to around SKr 1 500 per month on average, and the students may in addition borrow up to about two and a half time this sum. The repayment terms for study loans imply a considerable subsidy element, with repayment beginning formally six months, but effectively two years after completion of the studies and the annual downpayment restricted to a maximum of 4 per cent of earnings and transfers received. When the graduate reaches the age of 65 any outstanding loans are written off. Repayment is therefore slow and the outstanding stock of debt is SKr 94 billion distributed among 1.2 million borrowers.

Control of educational standards and progress

To maintain national standards at compulsory level minimum teaching hours in all specific subjects have been laid down. To ensure a common standard the pupils in compulsory schools are assessed according to national standards at the end of their fifth

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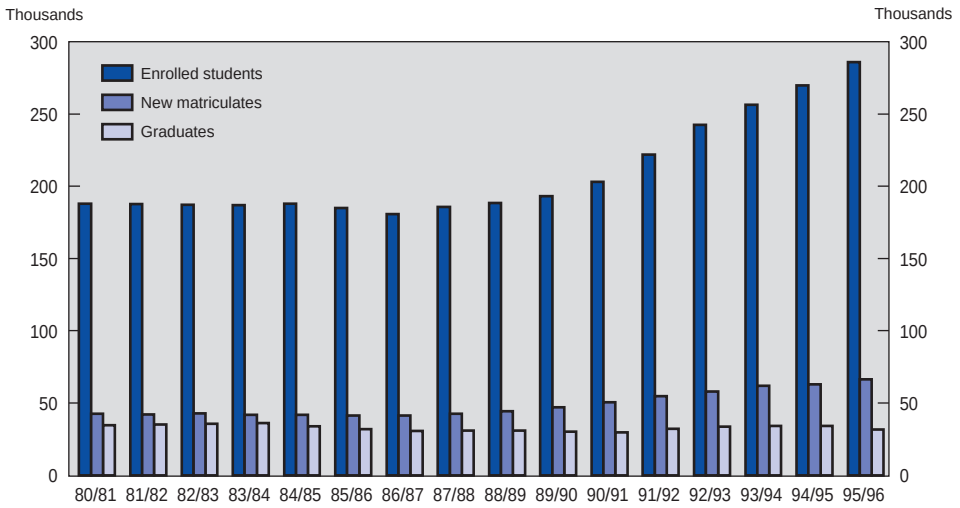
and ninth year. The examination mark system used in primary and secondary education was transformed in 1994 from being based on relative criteria – securing primarily a ranking of graduates – to absolute criteria which measure the knowledge of each individual against some objective standards of achievement set by national authorities. Three marks are used in compulsory school: pass, pass with a distinction; and pass with a very good distinction. At upper secondary level, a fourth mark, fail, is also used. In the upper secondary schools each topic must be passed separately upon completion of the course and according to national standards. However, there are no nationwide examinations to ensure a uniform standard across the country, and there is considerable variation across schools in how the centrally given standards should be interpreted and applied, a case in point is mathematics where some schools never use the grade pass with distinction while others reward the grade to 10 to 15 per cent of their pupils. As a result of the different content of grades across the country a number of university programmes apply their own entry tests to assess the eligibility of applicants. At universities the standards are set by institutions themselves and students must pass an exam at the end of each course. Two national agencies for education, *Skolverket* (for primary and secondary education) and *Högskoleverket* (for tertiary education), are responsible for supervising, monitoring and evaluation to ensure that national objectives are met. Experience gained in the process resulted in 1997 in a revised set of regulations issued by central authorities in order to strengthen quality control systems at local level, underpinned by increased availability of funds. The national control system has also been made subject to further review.

In terms of overall expenditure trends, the education sector benefited during the 1980s from demographic developments which served to lower the number of pupils in primary and lower secondary education, a development that has been only partially reversed in the 1990s. On the other hand, enrolment in upper secondary education has increased somewhat over the period, as the effect of smaller cohorts was more than offset by an increase in the share of pupils continuing in upper secondary schools. Moreover, the annual intake of new students in tertiary education has increased by 55 per cent over the past fifteen years without a concomitant increase in number of graduates, resulting in a considerable expansion in the numbers enrolled at universities and university colleges (Figure 26).

The education system did not realise productivity gains on a par with the private sector during the 1980s and the early 1990s.[20] Real costs per pupil in compulsory education increased during most of the 1980s and, despite a reversal

Figure 26. **ENROLMENT TRENDS IN UNIVERSITIES**



Source: SCB.

of this trend in the 1990s, are still 9 per cent higher than fifteen years ago, and a similar development has taken place in upper secondary education. However, with capacity in tertiary education institutions being utilised more fully, *inter alia* as the expansion of student places has not been matched by a similar increase in the number of teachers, unit costs in tertiary education has fallen (Table 13). The educational reforms in the 1990s, combined with the overall squeeze on local government finances, have brought about a reduction in unit costs which have kept the school sector's share of municipal expenditures fairly stable over the 1990s despite a 6 per cent increase in the number of pupils in primary, secondary and adult education. The cost reduction is concentrated to compulsory schools which have seen real teaching expenditures falling by about 17 per cent as the result of a higher pupil-to-teacher ratio and fewer teaching hours per pupil,[21] whereas the unit cost for pupils' meals has remained unchanged and real rental costs have increased.

Even with the sizeable efficiency gains made, there are still wide differences in cost-effectiveness among municipalities. If adjustments are made for the effects of differences in population density, the share of pupils with parents born

Table 13. **Education expenditure, students and teachers**

	1980/81	1991/92	1995/96	Real growth 1980/81-1991/92	Real growth 1991/92-1995/96
				Per cent	
Compulsory education					
Expenditure (million SKr)	20 831	42 896	47 420	4.2	–4.5
(Share of GDP)	(3.7)	(3.0)	(2.9)		
Students	1 031 964	877 685	958 972	–15.0	6.9
Teachers	86 241	78 695	74 409	–8.7	–5.4
Upper secondary education					
Expenditure (million SKr)	6 588	17 361	19 008	14.6[2]	3.2
(Share of GDP)	(1.2)	(1.2)	(1.2)		
Students	267 827	297 249	312 375	9.0[2]	2.6
Teachers	22 210	21 468	22 016	9.6[2]	2.6
Tertiary education					
Expenditure (million SKr)	4 722	13 421	18 311	19.6	24.7
(Share of GDP)	(0.8)	(0.9)	(1.1)		
Students	187 859	221 901	285 802	18.1	28.8
Teachers	33 000[1]	36 000	41 000	8.2[3]	13.9

1. 1984/1985.
2. 1980/1981 to 1990/1991 due to data break.
3. 1984/1985 to 1991/1992.

Source: SCB.

abroad and the size of schools, differences in expenditure levels of up to 50 per cent can be observed. While expenditure levels also reflect differences in priorities among municipalities, considerable scope for cost reductions at the aggregate level evidently still exists. Cost cuts should materialise as municipalities learn to formulate education plans adapted to decision-making at lower levels, as they develop evaluation and control systems to monitor such plans and ensure that they are adhered to as school managers (headmasters) are given the training necessary to handle administrative responsibilities as well as executing the pedagogical tasks for which they most often are educated.[22]

Student grants and loans have been the fastest-growing component of educational expenditure over the past decade and a half, reflecting primarily a 1989 reform which more than tripled the grant available. Having been stagnant over the 1980s, the number of grant recipients increased by more than 80 per cent

from 1988/89 to 1995/96. The maximum loan extended was, however, kept constant and the share of recipients also borrowing to finance their studies has fallen from almost 90 per cent to just above 70 per cent. The Swedish study-financing system is generous not only in an international context, but also in comparison with alternative sources of income for youths. (Table 14). Students receiving grants and taking out maximum loans – which have a repayment schedule far more favourable than ordinary, secured loans – have a far larger disposable income than out-of-work transfer recipients, the maximum grant and loan amounting to 70 per cent of after-tax income for an unskilled worker. This may be an important factor behind the increased propensity to enter and prolong an education at tertiary level in times of high youth unemployment.

Despite its high resource use, the quantitative output of the Swedish education system has not been above that of other industrial countries. On the most recent comparison (1994), the expected years of schooling for those entering education in Sweden was close to the OECD average. Swedish pupils receive fewer teaching hours than in other countries, and it is doubtful[23] whether this low overall instruction time received is offset by greater teacher attention resulting from the lower student-to-teacher ratio (Table 15). Moreover, homework is less demanding in Sweden than elsewhere, with pupils in seventh grade receiving about 2¼ hours per week, between one third and one half of that used in other countries.

Table 14. **Monthly net income of youths in education, at work and on social benefits**

SKr per month, 1996

	Before tax	After tax
Study grant	1 472	1 472
Study grant plus maximum loan	5 294	5 294
Earnings, unskilled worker	9 765	6 836
Unemployment benefit	7 324	5 127
Social welfare allowance I	2 483	2 483
Social welfare allowance II	3 050	3 050

Source: Gylfason, T. (ed.), T. Andersen, S. Honkapohja, A.J. Isachsen and J. Williamson (1997), ''The Swedish Model under Stress'', *SNS Economic Policy Group, Report 1997.*

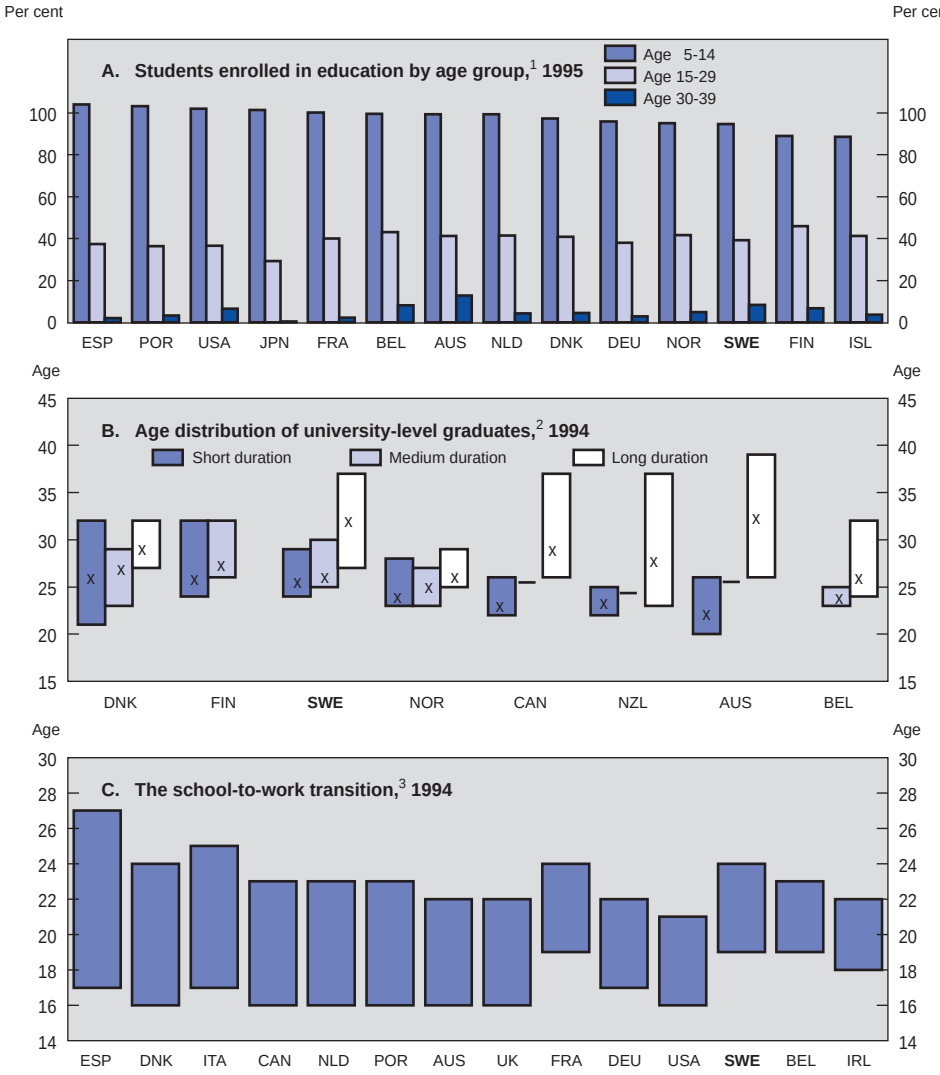
Table 15. **Student-to-teacher ratios and teaching hours by level of education**

	Primary education		Lower secondary education		Upper secondary education		
	ST	TH	ST	TH	ST	TH (general)	TH (vocational)
Austria	12.8	709	9.0	651	8.3	616	629
Belgium	13.0	832	..	720	6.9[1]	660	862
Czech Republic	19.6	687	13.0	657	10.2	627	627
Denmark	11.3	750	9.4	750	9.9	480	750
France	19.4	923	..	660	13.1[1]	660	660
Germany	20.7	760	15.8	712	13.3	650	665
Greece[2]	16.5	696	13.3	569	11.4	569	569
Ireland	23.8	915	..	735	16.2[1]	735	735
Italy	10.6	748	10.0	612	9.7	612	612
Netherlands	22.4[2]	1 000	16.9[2]	954	16.5[2]	954	..
New Zealand	22.7	788	18.6	869	15.0	950	950
Norway	9.5	686	9.4	611	8.2	476	..
Portugal	12.1[2]	828	..	681	13.0[1, 2]	607	607
Spain	16.4	900	16.5	900	13.6	630	630
Sweden	12.3	624	12.2	576	14.9	528	612
Switzerland	15.7	1 085	12.3	1 056	15.4[2]	..	..
Turkey	27.9	830	40.3	996	14.9	996	1 328
United States	17.1	958	17.6	964	15.4	943	..
OECD mean	18.2	818	16.2	760	13.5	688	722

Key: ST = Student to teacher ratio (public education, except for Netherlands: public and private), 1995.
TH = Teaching hours per year (public institutions), 1994.
1. All secondary education.
2. 1994.
Source: OECD (1996 and 1997), *Education at a Glance.*

The pass-through of the system is also rather slow. Pupils start to leave formal education rather late, and this is reflected in the high enrolment in upper secondary education (Figure 27, panel A). A fairly high proportion of pupils undertaking part-time work during upper secondary education (35 per cent) and access to grants for youths above 20 years of age with four years of work experience who enter the *Komvux* system, may also tend to prolong upper secondary education. The strong emphasis given to prior work experience on admission to tertiary studies has served to raise the average age for those entering tertiary education to its current level of 24 years. Students then stay a relatively long time in tertiary education, with the result that for all levels of tertiary education Swedish graduates are among the oldest in the OECD area (panel B).

Figure 27. **ENROLMENT AND TIME USE IN EDUCATION**



1. Based on head counts not accounting for part-time/full time status.
2. "x" represents graduate's median age. A missing bar means no data is available for this university level.
3. The age range starting in the first year of age in which fewer than 75 per cent of the population are in education without working; ending with the first year of age in which 50 per cent of the population are in work, but not in education. For Sweden 1995 data from the National Education Register; for other countries data from labour force surveys.

Source: OECD (1997), *Education at a Glance.*

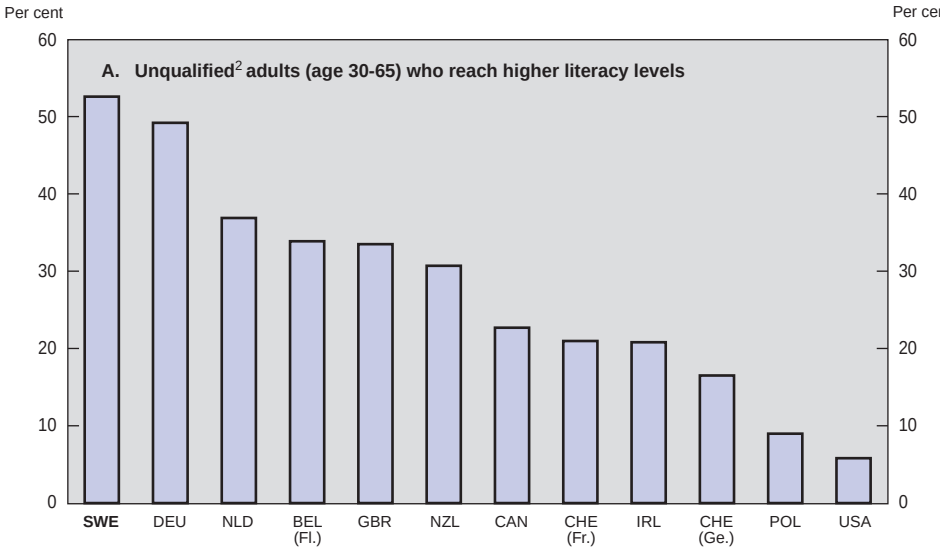
The prolongation of some shorter tertiary programmes, a generous support scheme for students and the weak labour market situation for youths have all combined to prolong the duration of tertiary education in recent years. If trends in labour force participation and continuing education are summarised into one aggregate indicator for the school-to-work transition (panel C), the outstanding feature for Sweden is the late start of the transition (19 years) compared with other countries. The transition is of an average length internationally insofar as by the age of 24 years, more than 50 per cent of youths are in work. This reflects the relatively high share who undertake shorter tertiary studies and the life-long learning approach which tends to spread education out over time.

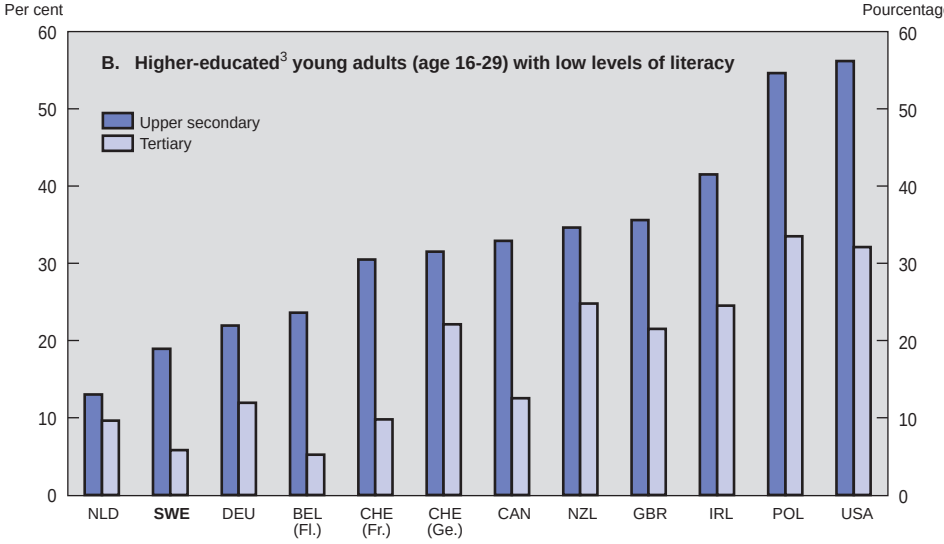
Educational performance

The Swedish educational performance has in many respects compared favourably with other OECD countries. In a survey of the *functional literacy* of the adult workforce in twelve industrial countries carried out over the years 1994-96,[24] Sweden had the lowest proportion of the population with very poor skills, and the highest proportions at or above the skill level regarded as necessary to cope with the demands of modern life and work (75 per cent). While functional literacy by necessity is closely related to educational attainment, in Sweden more than 50 per cent of lower-educated adults – with no formal qualifications beyond lower secondary education – have succeeded in reaching the two highest levels on a five-grade scale. But even in the Swedish case, there was a surprisingly high degree of functional illiteracy among those who had recently completed upper secondary or tertiary education (Figure 28).

This evidence is on the whole consistent with the results from international reading tests which have consistently placed Swedish pupils among the most proficient in their mother tongue. Results in other basic skills such as mathematics and science have, however, not been as favourable; although a tendency for relative scores to improve as pupils progress through compulsory school can be detected, Swedish pupils in general are among the middle achievers in international comparisons (Table 16). Moreover, the gap between the scores of low-and high-performers in all subjects in Sweden is quite as large as in other countries.[25] Arguably, such disparities in educational outcome, with a large proportion of pupils failing to maintain educational progress on a par with others, poses greater challenges for the Swedish system with its emphasis on providing all pupils with

Figure 28. **LITERACY LEVELS OF ADULTS**[1]





1. 1994-1996. Document scale.
2. No formal qualification beyond lower secondary education.
3. Completed upper secondary or tertiary-level education.

Source: OECD.

Table 16. **The achievement of Swedish pupils in an international comparison**[1]

	Reading		Mathematics		Science	
	Mean scores[2]	Inter-quartile range[3]	Mean scores[2]	Inter-quartile range[3]	Mean scores[2]	Inter-quartile range[3]
Sweden	**100**	**100**	**100**	**100**	**100**	**100**
Australia	..	..	102	118	102	118
Austria	..	..	104	113	104	102
Belgium (French Community)	84	84	101	102	88	96
Canada	93[4]	102	102	100	99	99
Czech Republic	..	..	109	115	107	99
Denmark	95	94	97	99	89	97
Finland	103	82	..	..	..	..
France	100	85	104	90	93	88
Germany	94[5]	103	98	104	99	114
Greece	91	81	93	104	93	98
Hungary	..	..	103	110	104	98
Iceland	97	100	94	87	92	92
Ireland	91	95	102	111	101	110
Italy	92	93	..	..	..	..
Japan	..	..	117	117	107	97
Korea	..	..	117	119	106	102
Netherlands	92	88	104	107	105	93
New Zealand	100	127	98	107	98	111
Norway	92	84	97	97	99	96
Portugal	95	69	87	71	90	84
Spain	86	82	94	84	97	87
Switzerland	97	92	105	103	98	104
United Kingdom[6]	..	..	97	108	103	116
United States	97	118	96	108	100	116

1. The comparison is between 14-year old pupils. The underlying surveys were carried out in 1991 (reading) and 1995 (mathematics and science).
2. Sweden = 100.
3. Gap between average scores between upper and lower quartile of students. Sweden = 100.
4. British Columbia.
5. Former Federal Republic of Germany.
6. England.

Source : OECD (1996), *Education At a Glance*.

an upper secondary education that qualifies for university studies than for those countries which rely on different streams more closely adapted to the qualifications of pupils.

International and Swedish evaluations show on the whole a high degree of stability in educational performance in the basic disciplines in the two decades up to around 1990.[26] Recent tests show, however, that basic skills in reading and

mathematics have slipped somewhat over the 1990s. While the overall proficiency in reading and writing is still deemed to be satisfactory, the share of pupils approaching the end of compulsory education without the necessary skills in mathematics increased from 13 to 18 per cent between 1992 and 1995. Basic skills in science were unsatisfactory in 1992 and had not improved up to 1995. The recent changes should be seen in conjunction with the great changes in primary and secondary education in Sweden the early 1990s. The transfer of responsibility for compulsory and upper secondary schools from central to local authorities coincided with a squeeze on public resources following from the severe economic recession and the higher inflow of children to the school system.

In the present economic climate, dropping out of school prior to the normal graduation age or leaving school without obtaining a complete ''exam'' will in most cases seriously compromise labour market careers for the individuals concerned, while also representing a waste of the resources invested. While enrolment in upper secondary education is close to 100 per cent of a cohort, about 10 per cent fail to complete education at this level. This would rank Sweden as one of the best performers by international comparison (Table 17), but a negative feature, most probably reflecting the recent strong rise in enrolment, is the pronounced increase over the past few years in the share of those graduating with incomplete marks, from about 1 per cent to above 4 per cent.[27] At the same time,

Table 17. **Percentage of students leaving secondary education without a certificate**

		Leavers without certificate
Sweden	**1996**	**9**[1]
Canada	1989	34
Denmark	1994	20
Finland	1992	12
France	1986	19
Germany	1988	12
Greece	1987	36
Ireland	1989	9
Italy	1988	11
Netherlands	1986	26
Spain	1988	23
United Kingdom	1989	8

1. Not completed within six years.
Source : OECD (1996), *Lifelong learning for all*; OECD.

about 3½ per cent of graduates have been allowed to weaken the content of their curricula. The impact of these trends on the final skill distribution is, however, alleviated by the extensive adult education system (*Komvux*) which allows above-age pupils to complete their education at compulsory and upper secondary level. For the cohort leaving compulsory school in 1988, more than 20 per cent were found to have passed through the *Komvux* system over the seven years up to 1995, the majority of these as part of an education at upper secondary level. On recent trends, more than 40 per cent of those graduating from upper secondary education can be expected – sooner or later – to enter tertiary education. As noted above, progress through the system has been slow, with only one third completing their tertiary education before reaching 25 years of age. Data from the 1980s and early 1990s indicate that about 20 per cent of those entering higher education fail to complete it.

Economic impact of the education system

Economy-wide effects

The high share of resources devoted to education reflects the assumed effects on long-term economic growth and economy-wide productivity of human capital investment:[28] *i)* it improves the quality of labour as an input to the production process; *ii)* it facilitates the absorption of new technologies and innovations; and *iii)* it improves the versatility of the workforce and thereby the capacity of the economy to respond to economic shocks and structural changes. In an international context, it would appear that Sweden has been unable to profit from its human capital investment to the same degree as other countries over the most recent decades; with a relatively high productivity level in the early 1970s, it has since slipped down the international ranking (Table 18).

For society at large, the return from investing in education is most often expressed by an aggregate measure – the social rate of return – the real discount rate which equates the value of the future income stream generated by such an investment with the costs incurred. Wage differentials can be used to measure the difference in productivity between workers with different levels of educational attainment,[29] and in this case the narrowing of these differentials recorded over past thirty years would indicate that the marginal rate of return from further enhancing the educational attainment of the Swedish labour force has been

Table 18. **Productivity levels in OECD countries**

GDP per man-hour relative to the United States[1]

	1913	1929	1938	1950	1960	1973	1987	1995[2]
United States	100	100	100	100	100	100	100	100
Netherlands	69	74	64	46	54	77	92	99
France	48	48	54	40	49	70	94	99
Belgium	61	55	53	42	45	64	86	93
Italy	37	35	40	31	38	64	79	87
Canada	75	66	58	75	79	83	92	83
United Kingdom	78	67	64	57	56	67	80	81
Austria	48	37	33	27	38	59	74	81
Australia	93	77	75	67	69	70	78	78
Germany[3]	50	42	46	30	46	64	80	78
Sweden	**44**	**38**	**43**	**49**	**54**	**76**	**82**	**77**
Denmark	58	59	54	43	46	63	68	71
Japan	18	22	23	15	20	46	61	70
Switzerland	51	57	55	56	59	67	68	66
Finland	33	32	33	31	36	57	67	63

1. Ranked according to 1995 level.
2. OECD estimates.
3. In 1995 including eastern Lander.

Source: Englander, A.S. and A. Gurney, ''Medium-term determinants of OECD productivity'', *OECD Economic Studies* No. 22.

falling strongly. This would seem to apply in spite of the rationed access to tertiary education which *per se* would imply an upward bias in wage premia as they incorporate a ''scarcity rent''. While the market valuation of the human capital produced by the Swedish education sector kept pace with the increase in resources allocated to it up to 1980, it fell while educational inputs were constant over the 1980s (Table 19). Incomes generated seem hardly to cover the costs entailed, indicating a marginal rate of return on educational investment below that of other types of investment.[30] On the other hand, if wage differentials are institutionally determined, implying that they reflect neither direct productivity differentials nor indirect effects as one worker's education may raise the productivity of his fellow workers, the market evaluation of educational output would understate the return on such investment. But in such a case, there would also be a one-off gain in aggregate productivity and production – at the cost of a wider dispersion in incomes – from previously undertaken educational investment if wages were allowed to reflect underlying productivity differentials and indirect effects, and if qualified labour were reallocated in correspondence with these differentials.

Table 19. **Output from and input into the education system**

Fixed prices, 1990 = 100

	Inputs (expenditure)	Market evaluation of output at 5½ per cent discount rate[1]		
		Wages after tax	Wages pre-tax	Wage costs
1967	80[2]	138	142	119
1973	88[3]	148	144	123
1980	101	158	140	139
1990	100	100	100	100
Memorandum item:				
1990, SKr million	66 042	22 520	37 630	51 700

1. The original analysis is based upon a framework where individuals determine their supply of labour on the basis of after-tax wage rates. In the Swedish case, it may be more appropriate to assume that enterprises determine employment on the basis of gross labour cost, including both wages paid and social security contributions.
2. 1970.
3. 1975.

Source: Ahlroth, S. *et al.* (1994), ''The Output of the Swedish Education Sector'', *Working Paper No. 43*, NIER.

The capacity of the Swedish economy to utilise its manpower resources to the full and to reallocate labour swiftly in response to changes in demand and external shocks was discussed in the *1997 OECD Economic Survey of Sweden*. A general labour market rigidity was documented, which can be expected to impede the allocation of qualified labour. It was also noted that the public sector has been prominent in defining both the content and the capacity of the tertiary education sector. At the macro level, the large public sector in Sweden may have depressed measured productivity growth relative to other countries somewhat.[31] More importantly, the large share of the output from the education system going to the public sector may have diverted the supply of highly qualified university graduates from other sectors, *inter alia* in the scientific and engineering field. However, smaller enterprises appear to be able to attract qualified labour on a par with others insofar as data at sector level show no clear tendency for the share of higher-educated employees to increase with enterprise size and small enterprises also have wage levels comparable with other enterprises.

Individual incentives to human capital investment

The efficient allocation of the public funds depends on individual choices, which may reflect a variety of influences, including expected life-time income taking account of unemployment risk and the effect of tax and transfer schedules.

The rationing of access to tertiary institutions may have delayed entrance for some pupils and made others enter a programme of second priority, so that choices may at times have been distorted. But the large variations in enrolment and annual intake at tertiary education institutions from the late 1960s, through to the early 1990s can generally be explained by changes in economic variables, such as the university wage premia, marginal tax rates, generosity of student grants and unemployment rates.[32] Increased enrolment at the end of the period was attributable to a host of factors working in the same direction: changes in wage premia, the 1990-91 tax reform and unemployment movements.

From a longer-term perspective, wage premia for education have fallen from around to 10 per cent in the late 1960s to around 5 per cent in the mid-1980s, a level from which they increased only slightly to the early 1990s (Table 20, panel A). Trends over the 1990s do not on the whole indicate substantial changes to this picture. Wage differentials between the low-skilled and higher-educated have in the aggregate remained fairly stable, but wider differences in unemployment risk have worked to increase the private return from education.[33] In a comparison with wage premia in other Scandinavian countries, which are on the lower side in an international context, the effect on hourly wages of an additional year of schooling beyond compulsory education has been estimated at 5 per cent for Sweden compared with 4½ per cent for Denmark and, respectively, 5 and 7 per cent for Norway and Finland (panel B). Market-based incentives to undertake a higher education should still be characterised as weak, the individual incentives depend to a large degree on the cost of undertaking an education not being brought home to the student.

A major effect of undertaking higher education is to move the income stream over the life cycle, with income foregone at the start of working life being offset by higher incomes at the end. A progressive tax system will have a disproportionate impact on such an income stream compared with a flat income profile and acts to lower the return of undertaking additional education. The increase in taxes implied an increase in the effective ‘‘tax rate on education’’ from about 25 per cent in the late 1960s to close to 90 per cent in the early 1980s, but the 1990-91 tax reform brought about a return to the 25 per cent level. Later modifications of the tax system have again increased tax levels and the highest marginal tax rate is approaching 60 per cent, but with tax increases taking place at all income levels, the overall progressivity has remained fairly stable

Table 20. **The effect on hourly wages of post-compulsory schooling**

Per cent

A Estimates for Sweden

		1968	1974	1981	1984	1986	1988	1991
Effect of additional year of schooling[1]				4.4		3.8	3.2	3.6
Effect relative to nine-year compulsory education[2]	12 years	10.2	5.6	4.8	5.6	4.5	5.7	5.2
	15 years	11.2	6.9	4.7	4.8	5.7	5.8	5.9
	16 years	13.4	6.6	5.1	5.2	5.4	5.6	6.0
Effect from an additional year of education[3] when starting with								
• Compulsory education		7.6	4.4	4.3				..
• Secondary education		7.5	3.9	3.2				2.9
• Short tertiary		7.3	3.4	1.8				2.5
• Long tertiary		9.6	4.5	1.1				3.7

B. Nordic comparison (an additional year of schooling)

	Sweden	Denmark	Norway	Finland
Total	**5**	4½	5	7
Men	**5**	5¼	5	7¾
Private sector	**6¼**	6	6½	7¾
Public sector	**5**	5¼	4½	8¼
Women	**4½**	3½	5	6½
Private sector	**5½**	4	5½	5
Public sector	**4**	3½	5	7½

1. Mellander (1994).
2. Edin and Holmlund (1993).
3. Björklund and Kjellström (1994).

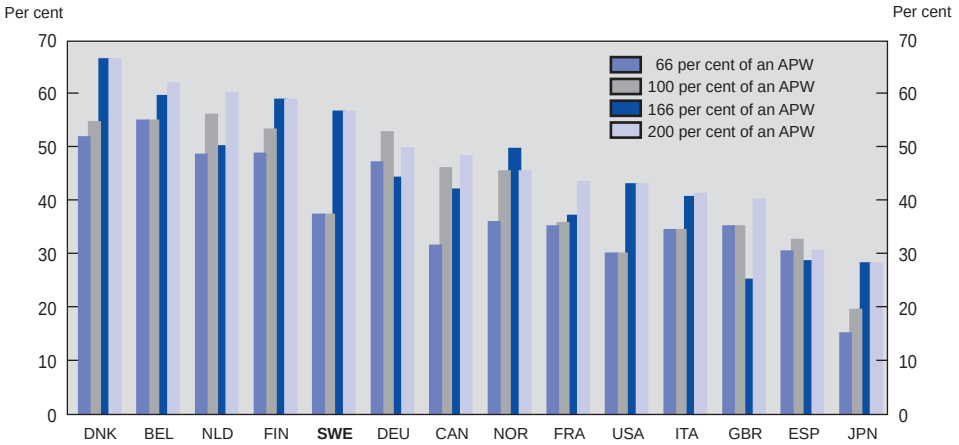
Source: Panel A: Industriförbundet (1996) *Lönar sig utbildning?*, Björklund A. and C. Kjellström (1994): ''Avkastningen på utbildning i Sverige 1968 till 1991'' in Erikson, R. and J. Jonsson (eds.), *Skola och sortering.*
Panel B: Asplund, R. *et al.* (1994) ''Human Capital and Earnings in the Nordic countries'' in Asplund, R. (ed.) 1994, *Human Capital Creation in an Economic Perspective.*

(Figure 29). Maintaining private incentives to education has been dependent on the rather generous student finance system which has been reformed on several occasions, most recently in 1989, and which would now appear to more than neutralise the effect of the tax system on the return of undertaking tertiary education (Table 21).

The increasing number of people entering and staying longer in higher education may be seen as a natural response to the overall rise in youth unemployment in the early 1990s and the opening up of a difference of about 5 to

Figure 29. **MARGINAL TAX RATES**[1]

Per cent



1. 1995. Income levels are given relative to the earnings of an Average Production Worker (APW).
Source: OECD.

6 percentage points in unemployment rates between graduates from tertiary education and workers with only compulsory education. The weakening of the labour market also made it necessary to extend the use of ALMPs as an inroad to the labour market, with several targeted youth schemes being introduced over the 1990s. For the cohort leaving compulsory school in 1988, the employment rate fell below 40 per cent in 1993 and was still below 50 per cent seven years after

Table 21. **Effects of taxes and grants on the return from undertaking university education**[1]

Percentage points

	1968	1974	1981	1991
Taxes	–3.8	–3.3	–3.8	–1.5
Grants	+3.2	+2.0	+1.2	+3.1

1. Effects on private rates of return of an additional year of university study. For males.
Source: Björklund, A. and C. Kjellström (1994) *op. cit.*; Edin, P.A. and B. Holmlund (1993), ''Avkastning och efterfrågan på högre utbildning'', *Ekonomisk debatt*, 1993 No. 1.

graduation, with another 6 per cent on ALMPs and 10 per cent unemployed. A decade earlier, the same cohort would have enjoyed an employment rate above 70 per cent with only 7 per cent unemployed or in ALMPs. However, while lower unemployment rates and a higher effective retirement age for the higher-educated boost their expected life-time income relative to other groups, the high graduation age documented above reduces their remaining effective working life, lowering the private as well as the social rate of return.[34] In sum, market-based incentives to undertake higher education are on the whole weak and while the effect on the return of individuals has been addressed by educational grants, the rate of return to society appears to be low, viewed from the perspective both of present wage premia and long-term productivity trends.

Life-long learning: a pioneering effort

It has been recognised for some time that recurrent training and education of the adult workforce is essential to promote structural change and sustain long-term economic growth while securing high utilisation of resources. Such an approach gained broad acceptance in Sweden from the early 1950s insofar as it seemed to reconcile these objectives with the Swedish emphasis on securing an equitable distribution of earnings. This framework served to underpin both the strong expansion of the formal education system through the following decades and the parallel build-up of labour market training and the adult education system (*Komvux*) providing competencies at lower and upper secondary level for above-age pupils. As in the other Nordic countries, skill acquisition in *Folkhögskolan* (''folk high school'') and *Studieförbunden* (''study groups'') emanate from a century-long tradition of participation in voluntary organisations, but this system can now be seen as highly integrated with the publicly-provided adult education (Table 22). However, overall developments in adult education do not only reflect choices made in the public domain: with an enterprise structure dominated by large manufacturing plants with the capacity to define the need for and carry out human capital investments on their own – and with a large public sector committed to implement broad policy guidelines – on-the-job and other forms of enterprise-financed training would appear to be of greater importance than publicly-provided training and education of the workforce.

Table 22. **Traditional and alternative educational avenues**[1]

Per cent of population 18 to 75 years

Avenue	Total	Year of birth		
		1916-29	1930-49	1950-73
Traditional				
University	16.4	5.1	18.4	19.4
Upper secondary	24.9	1.4	5.7	48.3
Vocational schools	15.3	13.5	23.2	10.3
Alternative[2]	13.6	16.2	16.7	10.3
of which:				
Folk high schools	2.3	3.6	2.6	1.6
Study circles	1.0	2.4	1.0	0.4
Labour market training	1.6	1.1	1.5	1.9
Enterprise training	4.7	5.8	6.2	3.2
Private (other)	2.5	3.2	3.3	1.6
Formal adult education *(Komvux)*	1.5	0.1	2.0	1.6
Compulsory only				
Lower secondary	8.3	4.9	8.0	9.9
Primary	21.5	58.6	28.1	1.9
Total	100.0	100.0	100.0	100.0

1. 1991.
2. Courses lasting at least six months; for *Komvux* only with examinations passed.

Source: *SOU* 1993:85, *Ursprung och utbildning. Social snedrekrytering till högre studier.*

The resources devoted to adult education increased throughout the 1980s and early 1990s, reaching 4.5 per cent of GDP in 1995, reflecting both the overall emphasis outlined above and the heavy reliance on labour market training to restrain the rise in unemployment. While inter-country comparisons are difficult in this area, they would seem to put Sweden on top among European countries, a result confirmed by surveys of training of the workforce. Compared with Denmark,[35] which has a similar emphasis on training of the workforce, Sweden has retained a far larger role for private provision of adult educational services. Apart from the *Komvux* system, which was set up to provide formal competencies for adults with government support to participants available both as grants and as loans, enterprises are on the whole responsible for upgrading the skills of their workforce, purchasing training services both from private and public producers. Public intervention is concentrated on the unemployed who have free access to labour market training, with educational grants equivalent to their unemployment

benefit. Overall, transfers from the public sector cover about 30 per cent of the expenses for adult education, to be compared with 70 per cent in Denmark, and about 20 to 25 per cent in Great Britain and Germany, while the provision of such services has a 70/30 per cent split between private and public suppliers. The main features of the Swedish adult education system are set out in Box 4.

Labour market training: adapting to new policy priorities

In terms of claims on resources, labour market training has historically been the most important part of adult education, the average cost being higher than for *Komvux*, although participation is lower. The evidence available[36] indicates that the expansion of labour market training in the early 1990s went too far, as the positive interaction between a strict interpretation of job availability criteria, intensive job placement efforts from labour market offices and participation in labour market programmes was lost. Increasing average unit costs and weaker motivation on the part of participants combined to produce strongly decreasing returns to scale. The effect of labour market training on labour market outcomes was further diluted by the opening up of this programme to be used to reinstate unemployment benefit eligibility.

The effect of labour market training on subsequent labour market outcomes depends closely on the overall labour market situation when training is completed. During the 1980s, about 60 to 70 per cent of participants would be in work six months later, but in the early 1990s this share fell below 30 per cent (Figure 30, panel A). The majority of evaluations relating to periods of strong labour demand have found that the net discounted value of future earnings resulting from such training does not match the costs incurred.[37] While some of the effects of labour market training may be difficult to capture, it would seem safe to conclude that labour market training needs to be closely integrated with the day-to-day work of labour offices in placing job seekers. This implies a still further downsizing of ALMPs in general to levels commensurate with the overall capacity of labour market offices, if labour-market training is to have the desired impact.[38] Moreover, in terms of direct effects on open unemployment, labour market training would appear to be a rather costly instrument. In terms of both current cost and educational grants, it is far more expensive than both formal education and *Komvux* (panel B).

Box 4. **The educational framework: adult education and training**

Formal competencies at compulsory and upper secondary level for above-age students (20 years or more) are provided through the *Komvux* system which is a municipal responsibility, subject to the same educational standards and surveillance as applying to the formal education system. By 1996, the share of the population (20 to 64 years) participating in *Komvux* had reached 3.1 per cent, with 21 per cent of participants pursuing courses leading to a complete compulsory education (courses that municipalities are legally obliged to supply to all who demands them), 76 per cent taking part in courses at upper secondary level and 4 per cent in additional vocationally-oriented courses. The average age of students is 32 years, with 25 per cent in the 20 to 24 years interval and another 20 per cent in the 25 to 29 years interval.

Additional school-based education – with curricula set by institutions themselves under the supervision by a separate body – is provided through the *folk high school* system, with schools either owned by local government (covering 34 per cent of students) or by non-profit organisations (66 per cent) and with as much as 4 per cent of the population (20 to 64 years) participating in courses lasting from a few days to a full year. Of those participating in courses lasting more than half a month, 60 per cent were below 30 years of age and 40 per cent had not completed education at upper secondary level. In the mid 1990s, about 35 per cent of activity was financed by additional public grants as part of a programme to reduce unemployment. Moreover, about 30 per cent of the entire population take part in further education through correspondence courses and study groups organised by non-profit organisations (*Studieförbunden*); activities also benefiting from public support.

Labour market training was reorganised in 1986 with a split of the provider and production functions. Training services for the registered unemployed are provided by the local labour market boards and produced by a government-owned corporation, *Amu-gruppen*, in competition with private suppliers and public producers such as *Komvux*, upper secondary schools and folk high schools. However, ordinary formal education cannot in general be completed through labour market training. Courses are in general short and formulated to equip participants with specific vocational and practical skills in order to enable them to return to ordinary jobs as fast as possible. Labour market training is an integral part of active labour policy programmes and adjusted to the current state of labour demand. At its peak in 1992, labour market training comprised on average 86 000 persons, (close to 2 per cent of the labour force), but it had been scaled back to 46 000 by the second half of 1996. A majority of participants (57 per cent) has only compulsory or short (two-year) upper secondary education, and close to 80 per cent of participants are above 25 years of age.

Government support to *enterprise-organised training* is limited, covering only cases which can be motivated for active labour market policy purposes. Presently, support is provided to enterprises who temporarily take on unemployed when their ordinary

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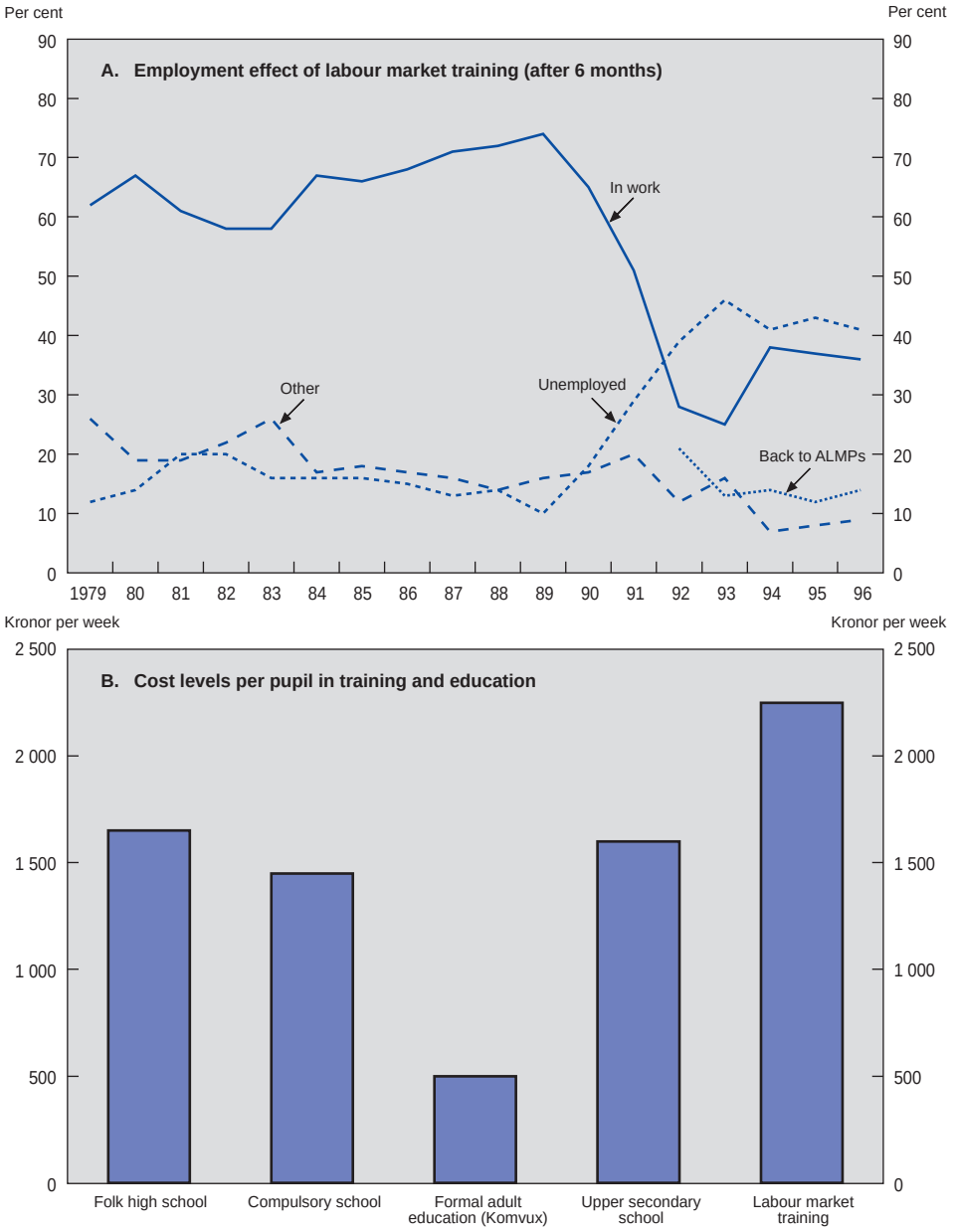
(continued)

employees are granted education leave and to enterprises who have to carry out a restructuring, in order to soften the impact on the local labour market. In general, enterprise-organised training – on the premises or purchased from outside suppliers – covers about 40 per cent of employees in a single year, but compared with other countries the duration of such training at six to seven days per employee per year is fairly short. Employees with an education at tertiary level receive a more than proportionate share of such education, both in terms of participation (58 per cent) and length of courses.

Participants in adult education can access the *support and borrowing schemes* available for students in the formal education system, but some additional schemes have been formulated to cater to their particular needs. After four years of work, adults can obtain educational grants – taxable but also giving pensions rights – equalling 65 per cent of their potential unemployment benefit upon entering *Komvux*, a compulsory or an upper secondary school. Unemployed with three years of work and above 21 years of age can obtain a similar grant; and in both cases the grant can be topped up to UI-benefit level with a loan. With effect from 1997, a new support scheme is available for unemployed people without a full (three-year) upper secondary education undertaking studies up to one year, with grants at UI-benefit level. The same option is given to employees with five years of work provided that a long-term unemployed person is hired as a replacement. Participants in labour market training also receive an educational grant at UI-benefit level.

Underlying the reassessment of labour market policies in 1995-96 was the recognition that the character of labour market imbalances had shifted sharply, with unemployment in danger of becoming structural unless labour market institutions and labour market policies were altered substantially. The widening gap in unemployment rates between low-skilled workers and other categories ensured that efforts were redirected towards the formal education sector, with labour-market training sharply reduced in scope. Training programmes are one of the most expensive forms of labour market intervention. In order to be effective they must be carefully targeted to both individuals' and labour market needs. This precludes large and indiscriminate programmes and access should be reserved to the most urgent cases. This is still not the case in Sweden, those with a compulsory or two-year upper secondary education having a lower representation in labour market training (below 60 per cent) than in overall or long-term registered unemployment (70 to 80 per cent). With broad policy guidelines being modified

Figure 30. **FEATURES OF LABOUR MARKET TRAINING**



Source: Arbetsmarknadsstyrelsen, Riksdagens Revisorer.

to make long-term unemployed the focus of active labour market programmes, the return on labour market training should increase even if benefit eligibility can still be obtained through labour market training.

The reduction in labour market training since 1992 has been associated with falling real costs per participant per week. Competition would seem to have improved insofar as the market share of the major publicly-owned producer of training services, *Amu-gruppen*, has fallen from 85 per cent in the late 1980s to under a half, while the share of private suppliers, who are most often active both in this market and in enterprise-organised training, has increased to a third. The revenues and profits of the major government-owned supplier have been falling strongly, *inter alia* because of an unsuitable organisational structure, costs brought forward from its former role as a government agency and a passive ownership,[39] leading to a further injection of equity capital in 1996. In the longer run, enhanced efficiency would require a full privatisation of the company, all the more pertinent insofar as there would seem to be no particular economic reason for *Amu-gruppen* to have a presence in all regions and in all types of training services.

Increasing reliance on formal adult education

Following the decentralisation of responsibility for all education at primary and secondary level to the municipalities, these activities were to be financed out of their tax revenues and block grants, on a par with all other local government activities. However, targeted grants to municipalities for adult education of the unemployed became available in 1993, and helped reverse a downward trend which started in 1991, resulting in a volume increase in total resources allocated to *Komvux* by 11 per cent from 1990 to 1995. The Employment Bill from the spring of 1996 extended this programme to cover the five school years 1997/98 to 2001/02, with the aim of increasing the coverage of adult education by an additional 60 000 full-time persons.[40] Underlying this major effort – *Kunskapslyftet* – is the perception of an educational mismatch among the population, arising partly from the existing number of the low-educated, and partly from additions to their ranks, as pupils fail to complete upper secondary education. It would appear that close to 1 million persons between 16 and 65 years of age have no qualifications beyond compulsory education, and this number is rising by 20 000 to 25 000 a year. Drawing upon the results from the

functional literacy study, it can also be estimated that – in spite of the overall favourable Swedish performance – about 1.4 million have a literacy level below that corresponding to a complete education at compulsory level.

Evidence from the adult literacy survey and elsewhere indicates that a complete upper secondary education is a minimum requirement for an efficient school-to-work transition and to secure a lasting foothold in the labour market. The target groups of the adult education effort are the unemployed and employees without a complete education at the upper secondary level, and the subsidies from the central government to municipalities are targeted on increasing capacity at upper secondary level, while – in accordance with their legal obligation – municipalities themselves will have to fund in full increased demand for education at the compulsory level.[41] The effort is underpinned by more generous student allowances, available at UI benefit level for the first year of education.

This ambitious strategy raises a number of issues. On a practical level, the rapid increase in capacity may easily lead to administrative problems, with respect both to space constraints and the supply of teachers. In the introductory stage, these problems seem so far to have been overcome. The municipalities have relied on their in-house provider, *Komvux*, to supply most of the places but 17 per cent of the places have been purchased from other suppliers of educational services. Even if quantitative targets may be achieved, the rapid expansion requires close monitoring of the quality of education offered, a concern amplified by the slow progress at municipal level in establishing and evaluating school plans evident after the decentralisation of educational responsibility in 1991. Overall, weaknesses are now being exposed in the formal education system as the relative abundance of resources that characterised the 1980s gives way to a relative scarcity of resources which affects most public sectors. Dealing with these weaknesses while implementing a strong expansion of capacity over a short time may end up with less than satisfactory outcomes for both.

The educational output of the *Komvux* system has not been subject to thorough evaluation. Participants coming back to complete a formal education in *Komvux* can on average be expected to be more motivated than youths proceeding directly from compulsory to upper secondary school. On the other hand, as many as one third prefer to finance their studies through full-time or part-time

work. A few years ago, as many as 14 per cent of entrants into the system were reported to drop out completely, while another 10 per cent did not complete all courses they registered for, not just because of a wider choice of studies and job offers, but also through educational failure. The labour-market impact of *Komvux* may also be weaker than desired, as can be seen from a small sample of those participating in 1993/94.[42] The proportion unemployed, which fell immediately after the education spell, seems to have rebounded markedly over time, while the employment rate remained fairly stable (Table 23). In terms of immediate labour market prospects, *Komvux* is less effective than labour market training; its main effect may be seen in its impact on the propensity to enter tertiary education.

As a labour market strategy, upgrading the qualifications of the less-skilled part of the workforce to match the present compression of wage differentials may turn out to be a rather inefficient and time-consuming way of stimulating labour demand. The short-term effect will be a reduction of open unemployment since participants in adult education are registered as out of the labour force, thereby contributing to the official objective of lowering open unemployment to 4 per cent by 2000. While the labour market prospects for individuals undertaking such an education will undoubtedly improve, the fact that workers with upper secondary qualifications still face above-average unemployment rates, and wage differentials for the lower-educated segments of the labour force do not, in general, respond significantly to changes in demand and supply,[43] indicates that the

Table 23. **The effects of formal adult education *(Komvux)***

1993/1994, per cent of total

	Status ahead				Status after		
	3 years	2 years	1 year	Immediately	Immediately	1½ years	2½ years
Employed	60	55	39	20	20	28	24
Tertiary education	0	0	0	0	17	29	32
Upper secondary education	13	14	9	8	–	–	–
Komvux	2	4	13	15	36	11	6
Unemployed	7	14	28	49	22	27	36
Others	19	13	11	8	5	5	2

Source: Paldanius, S. *et al.*, ''Olika betydelser av kommunal vuxenutbildning för arbetslösa'', *Rapport No. 1997:03*, Institutionen för pedagogik, University of Gothenburg.

capacity to absorb a larger supply of such workers will still be insufficient when they return to the labour market. The education strategy implies a return to society above what can be inferred from a purely labour market determined outcome, insofar as where wages do not fully reflect productivity levels, government initiatives to promote government initiatives to promote investment in human capital can partly compensate. Nevertheless, if wages were more strongly differentiated to reflect relative scarcity of qualifications demanded – although income inequality may increase in the short run – incentives to acquire higher education would automatically improve, thereby securing a more stable foundation for a long-term strategy of upgrading the competencies of the workforce.

Overall, human capital investment will be fully efficient when the allocative function of the labour market is exploited to the full. In the short run, both differences in wage levels and in unemployment levels could be used to signal what skills are in demand, but in the long run emphasis needs to be on the former, both to assign workers with different competencies to the tasks they are best qualified to perform and to signal to those seeking education which talents are in demand. If this allocative function is not in place, education will increasingly function as a screening device for employers, increasing the risk for a substantial excess investment in education in a labour market now characterised by overall excess supply of labour. The overall return on the investment in human capital is also likely to suffer from the distortions to incentives introduced by more generous educational grants for above-age formal education and the strong increase in capacity in this system. The informational feedback from labour market conditions to individuals and labour market organisations may be obstructed if the government consistently intervenes to offset the consequences of not seeking a qualifying education. Moreover, as demonstrated above, the return to society of providing higher education is highly dependent on the number of years that investment is utilised.

An appropriate setting for enterprise-based training?

Enterprise-based training is one of the major parameters by which enterprises adapt to changes in their economic environment. In the Swedish case, this parameter is of larger significance than in most other countries insofar as enterprises are subject to a rather strict employment protection and industrial relations legislation, limiting their possibility to adjust their workforce to changes in the

overall level and composition of demand for their products. Wage bargaining, mainly taking place at the industry level between organisations of employees and employers, would appear to give comparatively little leeway to enterprises to adjust wage rates to adverse developments in revenues. Indeed, this framework was deliberately set up to provide enterprises with incentives to upgrade the qualifications of their workers, so as to improve versatility over the longer run and to occupy them with in-plant training in times of slack.[44]

General work experience has a stronger influence on lifetime income in Sweden than in the other Nordic countries, with job tenure, which may be taken as indicating the firm-specific human capital investment, having a far smaller role. The wage effect of spending ten years in the same job is estimated at 3 per cent, while general experience of a similar length raises wages by 16 per cent.[45] Undertaking enterprise-based training has immediate wage effects, but they do not increase over time because the emphasis is on providing general training rather than firm-specific knowledge. In fact, training seems to serve as a (non-taxable) reward to well-performing employees (especially those with an education at tertiary level)[46] and to compensate high-skilled employees for the strong compression of wages. The fact that turnover among employees is comparatively high in Sweden may discourage enterprises from implementing firm-specific human capital investment even if they were given the opportunity; but the high turnover itself may be traced to the compressed wage structure within enterprises which has made job mobility the easier way of obtaining higher earnings.[47]

For some parts of the manufacturing sector, available evidence points to a positive return to enterprises from carrying out training themselves,[48] and this issue has been receiving attention for some time in wage bargaining agreements in manufacturing sectors such as pulp and paper, engineering and printing and publishing, more recently in the municipal sector. These agreements provide a framework for local initiatives rather than numerical prescriptions for allocations to training at enterprise level. On the whole, this framework, combined with the long-standing focus on recurrent training, has ensured that training of the workforce has taken hold at most levels of the economy. Small enterprises (fewer than 100 employees) involve just as many of their staff in enterprise-based training as do medium-sized and larger enterprises,[49] indicating the absence of a ‘‘training gap’’ often found in other countries (Table 24).

Table 24. **Enterprise-organised training in the private sector by enterprise size and length of training**[1]

Per cent

Enterprise size	Less than 1 week	1 to 4 weeks	More than 1 month	Total	Share of employed receiving training	Share of employment
0-9[2]	82	14	4	100	18	27
10-49	85	13	2	100	25	19
50-99	79	16	5	100	10	7
100-249	76	21	3	100	12	} 16
250-499	78	19	3	100	8	
500-999	82	15	3	100	7	7
1 000+	77	20	3	100	21	23
All[3]	79	17	4	100	100	100

1. 1996.
2. Including self-employed.
3. Including non-allocated by enterprise size.

Source: SCB.

In conclusion, it should be emphasised that adult education and training in Sweden embodies a division of responsibilities that leaves the public sector as the main provider of training for the unemployed and enterprises as the provider of training services for their employees. This should be the most efficient way of organising this market, relying on private incentives for the broad segment of the workforce while focusing public resources on those affected by labour market imperfections. Such an approach is supported by the findings of the functional literacy survey, which indicate that formal schooling and continuous labour-market attachment are the main factors contributing to a high literacy level, with a much weaker link to adult vocational education and training.[50] The division of responsibility has been strengthened over the 1990s as tax expenditures for training have been phased out and government funds to stimulate the development of industrial organisation and labour market structures abolished.[51] On recent evidence, the allocation of training resources within enterprises may still be rather inefficient, and would benefit from a decentralisation of wage bargaining to allow for wage contracts that reward long-term employer-employee commitments. This would appear to be a more efficient way of promoting such commitments than through employment protection and industrial relations legislation which only serve to prolong imbalances in the labour market.

Assessment and scope for further action

Reform of the Swedish education system has been on the agenda continuously in the post-war period. The result was a fully integrated and comprehensive nine-year compulsory school system, implemented in 1969, a fully-integrated upper secondary school system with both a general branch and a vocational branch, implemented in 1970, and a gradual introduction of formal adult education (*Komvux*) from 1968. This framework has continued to develop, with particularly extensive reforms in the 1990s, many of which should act to improve educational performance in the years ahead (see Box 5). However, the output of the educational system has to be seen in terms of the ample resources allocated to it over the past few decades, the years of schooling received, the pass-through of the system and the performance of Swedish pupils in international comparisons. More importantly, the Swedish option of using the education and training system as the instrument for reconciling the requirements of the labour market with equity objectives means that average, or even ''good'' performance in the domain of education and training may not be sufficient given the burden imposed on the instrument of securing full employment. Priorities for further action to achieve such ends are offered in the following paragraphs.

The overall direction and the full effect of the new institutional framework for primary and secondary education and tighter resource constraints facing all public sectors in recent years will only be evident in some years' time, as pupils start to graduate from the new institutional set-up. But on the most recent evidence, educational performance would appear to be declining. Large inter-municipal differences in educational costs, combined with a low dispersion of educational quality between schools, bear witness to a substantial potential for increasing output within present resource disbursements. Giving priority to education spending within local government budgets and relying on user fees to finance supporting expenditures would serve to improve educational standards. Moreover, an important potential remains to be exploited in the still low pupil-to-teacher ratio, the low teaching hours effected in the course of a year and the scope for redirecting pupils time to school and homework. Time use for teaching, preparation etc. has so far been determined by detailed wage bargaining agreements between the organisations of teachers and municipalities. These agreements are now being reformed, with only total working hours (school presence) specified, which could lead to more extensive use of school facilities and pupils'

Box 5. **Educational reform in the 1990s**

Revisions to objectives and curricula and extensive trials of alternatives within upper secondary schools in the 1980s paved the way for more extensive reforms in the 1990s:

- The structure of upper secondary education was reformed over the years 1992-96, with all programmes qualifying for tertiary education;
- The responsibility of primary and secondary education was decentralised to municipalities which allocate resources to schools in direct competition with their other responsibilities;
- The line system in the universities was replaced by an exam system, with each exam yielding a number of points and the award of a degree depending on the accumulation of points;
- To promote decentralisation and increased focus on quality and performance, the financing of the universities has become dependent on the students' progress;
- An absolute mark system has been implemented throughout the formal education system;
- The scope for private schools has been extended;
- Pre-school will be closely integrated with compulsory school, with municipalities being obliged to offer such education to all six-year olds;
- Vocational training is being reassessed with trials of a modernised apprenticeship programme (two years in school, two years in work-based training) and two-year vocational education at tertiary level.

On the quantitative side, participation in upper secondary education is approaching 100 per cent, while the target of 50 per cent of each cohort entering tertiary education is underpinned by an increase in the coverage of tertiary education by 45 000 places from 1997 to 2000. Moreover, a five-year programme with 100 000 central government-financed places in adult education, designed to increase capacity by 50 per cent, has been set in train to cover those with qualifications below upper secondary level.

time during working hours. More flexibility at local level would also provide more scope for alternating between class-room and targeted-group teaching than now practised. While the recent wage agreement has allocated an additional 10 per cent of the annual wage sum to be used over five years to reward educational innovation at school level, progress will be hampered by a still extreme compression between starting and top wages by OECD standards, seniority determining the progress between these two levels and the relatively high average age of Swedish teachers.

Implementing a fully decentralised system for school management is proceeding at a different pace among municipalities. Some have brought it to its logical conclusion and have formally introduced vouchers which make the revenues of schools dependent on their capacity to attract pupils. Others rely on budgeting principles which *de facto* adjust appropriations to the actual number of pupils. On the whole, such systems are not fully developed, leaving school administrations with little discretion to adjust inputs in line with enrolment or to retain a surplus of revenues over outlays as a result of savings stemming from local initiatives. There are few specifications as to the results required and no adjustment of appropriations or vouchers to intra-municipal differences in the composition of pupils, *inter alia* with respect to the need for targeted teaching. Rectifying such omissions would enhance the demand orientation of the system and, if combined with the broader parental influence on school boards now being tried out, make primary and secondary schools adjust more rapidly to inefficiencies and sub-standard educational results both at school and class level.

Administrative efforts would, however, have to be accompanied by stronger feed-back from nationwide pedagogical evaluations. National tests are at present obligatory only from the end of compulsory school and then for municipal schools only. Below that level, nationwide and nationally-formulated tests are only recommended, both with local evaluation only. A consistent use of nationwide exams with external censors from an early stage and for all schools would ensure more uniform standards both across schools and over time. Individual teachers and school administrations would be more aware of the requirements at national level and of their own relative performance if such results were made regularly available; moreover, if school averages were to be publicised, parents' choice of school could be based on objective facts rather than the present word-of-mouth method. Nation-wide tests underpinned by a minimum requirement of educational proficiency would serve to uphold standards at a time when the upper secondary level is under pressure both from the requirement to admit pupils with incomplete marks at the lower secondary level and to make all their pupils qualify for studies at tertiary level. In the end, the transparency and consistency of such a system would allow universities to base entrance primarily on the exams from upper secondary school, rather than maintaining a dual set of entrance criteria system which allows equal weight to applicants assessed on the basis of an aptitude test. Indeed, steps towards such a system have been taken

recently, introducing minimum requirements in the core subjects (Swedish, English and mathematics) for entering national programmes in upper secondary education, a new final examination from upper secondary education, and more extensive use of results from nationwide tests at primary and secondary level for evaluation purposes.

Up to 1991 all vocational education was provided in a school-based setting with the traditional apprenticeship system having been abandoned for practically all occupations. The 1992 reform of the upper secondary school incorporated a longer period of training in enterprises, and the recent initiatives to reintroduce a modernised apprenticeship system and provide vocational studies at tertiary level can be interpreted as a recognition of the neglect of the needs of business during a period when the expansion of the public sector resulted in a constant excess demand for labour. At a practical level, this has resulted in a stock of teachers and equipment which has not kept up with modern business practices. This is a concern insofar as for the majority of youths upper secondary school will still be the place to obtain the skills necessary to start their working life. A few large and established enterprises, primarily in manufacturing, have offset these weak links by setting up their own enterprise-run independent schools within the framework of upper secondary education, providing general qualifications in line with national requirements but integrating the vocational content closely with the operation of their local plants. However, for rapidly expanding enterprises such an option hardly exists in the short run when it matters. On the whole, reform of the vocational education system would need to be substantial if an interaction between schools and enterprises on a par with Denmark, Germany and Austria is to be achieved – if it can be achieved at all.[52]

The time-consuming pass-through of the Swedish education system reflects to a large extent the incentives embedded in the system. The restricted access to certain tertiary programmes may have prolonged pass-through as students sought to re-qualify, but with the strong increase in capacity this effect should be of declining significance compared with the effects of: *i)* more favourable grants for completing upper secondary education after 20 years of age than below; *ii)* the possibility to qualify for tertiary studies through general work experience; *iii)* the generosity of student grants and loans compared with transfer schemes available for unemployed youths; *iv)* the formulation of the student grants and loans scheme which – by relating repayment to income rather than the principal of the

loan – implies a higher effective interest rate for smaller than for larger loans. On the other hand, with the compressed wage distribution and small wage premia, market-based signals to further education are weak, and this effect has dominated in times of readily available jobs. With little improvement in the private return from undertaking education during the 1990s, it may well dominate again when labour market imbalances recede.

Improvements in the student grant and loan scheme have served to raise the private return to education, and a further increase in generosity has been advocated by a parliamentary-based commission. However, expanding such schemes in the absence of sufficient wage premia to education and in the face of the progressivity of the tax system which further reduces the existing premia, is a rather inefficient way of addressing the lack of incentives. Not only does it imply both higher taxes and higher expenditure, but as long as student grants are not combined with cost-related tuition fees to bring differences in costs home to students, they can only enhance the general incentive to pursue an education, but give little guidance which studies to pursue. The present formulation of the scheme thus not only serves to prolong tertiary studies once they have been embarked upon, but also reinforces a tendency to study for a long time without graduating with any particular skill or qualification. This can only serve to strengthen the use of academic degrees as a general screening device, signalling to employers the general competence level of applicants rather than their specific qualifications. Overall, educational choice would improve if wage premia were allowed to open up and marginal tax rates on labour income brought down, reducing the need for a generous student grant system. Within the latter, the present emphasis on loans in relation to grants enhances the investment element of education and should be retained; however, grants for above-age students undertaking upper secondary education and the rule linking repayment of loans to income serve to delay and prolong the time spent in education, and both features should be reconsidered during the planned review of the scheme. Indeed, signals have been given that the emphasis on repayment will be strengthened.

In the sphere of adult education, the division of responsibilities is organised so as to rely on private initiatives for the broad segment of the workforce while focusing public resource on those affected by labour market imperfections. Public efforts are now being concentrated on the provision of formal education for above-age students, which is already more extensive than in most other countries.

The strategy relies on supplementing freely-provided education services with more generous grants and borrowing facilities. Insofar as this approach is not supplemented by increased reliance on price signals from the labour market to guide the allocation of labour and educational choice, there is a clear risk that inefficiencies will remain, lowering the overall returns of investment undertaken. While an effort concentrated on those left behind by the Swedish education system is warranted for a host of reasons, future efforts should focus on maintaining and improving the output of the initial education system and modifying labour market structures. Recommendations to improve educational performance, both within the formal education system and in the different fields of adult education, are set out in Box 6.

Box 6. **Recommendations to improve effectiveness and efficiency in the education sector**

Formal education

Sweden's educational performance is still impressive in an international comparison, but underlying weaknesses are being exposed at a time when ambitions in the educational sphere are being raised further. It is imperative that corrective actions be made to ensure full use of the substantial allocation of resources to formal education, underpinned in particular by a wage structure that gives incentives for youths to seek out education in ready demand by enterprises and to move efficiently through the education system. More specifically, efforts should concentrate upon:

- *Ensuring acquisition of basic skills at compulsory level*. Educational standards in mathematics and science are below those achieved in reading, and there are large differences in skill acquisition within each youth cohort. The time both of teachers and pupils could be utilised more fully to this end.
- *Stimulate educational innovation by completing in full the decentralisation of primary and secondary schooling embarked upon in 1991*. This implies a more explicit use of educational vouchers at municipal level, more explicit independence for school administrations combined with more parental influence, and a continuation of the trend towards decentralised wage setting embarked upon in the recent wage agreements for teachers, options which are all available within the existing legal framework.

(continued on next page)

(continued)

- *Ensuring nationwide educational standards*. Following on the first steps taken in upper secondary education in 1998, pedagogical feed-back to pupils, teachers and schools would improve if nationwide test using external censors were made obligatory at most levels and in more subjects, while the option to progress with incomplete marks should be tightened.
- *Reviewing the formulation of the framework for vocational training*. The evaluation of vocational education under progress is praiseworthy, but the resource implications of rebuilding a fully-fledged structure for vocational training within formal education should be assessed against a strengthening of basic skills combined with more scope for enterprises to formulate training of youths, more fully integrating the current demands of industry and business.
- *Reviewing the structure of student grants and loans*. While the loan element serves to underline the investment dimension of education, both generous grants and the repayment schedule of loans serve to prolong both upper secondary and tertiary education.

Adult education and training

Adult education and training embodies a division of responsibilities that leaves the public sector as the provider of training for the unemployed and enterprises as the provider of training services for their employees. The transparency of this regime may be somewhat reduced with the expansion of formal adult education now embarked upon, because the consequences to individuals of not seeking qualifying education at the outset and to labour market organisations of not adapting structures to ensure that low-skilled workers can compete for and retain ordinary jobs, are alleviated by government intervention. The overall outcome would improve if:

- *The adult education effort combined more strongly than at present a close investigation of the needs of each applicant with a general adaptation of the adult education curricula to the needs of industry and business*; moreover, the effort is dependent upon increased labour market flexibility to ensure that the higher outflow from adult education can be absorbed in due time.
- *The social partners modified wage structures to allow for local adaptations to improve incentives for enterprises to supply firm-specific training*. With the dismantling of tax credit schemes and government funds to promote industrial organisation, the setting of enterprise-financed training has gained in clarity and incentives improved. This process should be brought to its logical conclusion.
- *The process of adapting labour market training to serve the long-term unemployed and to underpin the job placement process at Employment Offices continued, at the same time the scope of the publicly-owned producer in a more competitive market should be reconsidered.*

V. Entrepreneurship and job creation

The macroeconomic crisis from 1990 to 1992 brought massive labour-shedding in the large industrial conglomerates, which have been the traditional focus of Swedish business sector policies, and even though this sector is now growing, this stimulates productivity growth rather than employment growth. The policy focus has therefore shifted toward entrepreneurship and small and medium-sized enterprises, both as a source of future jobs and as way to ensure a more dynamic and quickly-adjusting enterprise sector. One of the conclusions of the *OECD Jobs Study* was that the economies best equipped to exploit future opportunities will be characterised by dynamic entrepreneurship.[53] As a follow-up to the *Jobs Study*, this chapter attempts to present some of the factors influencing the entrepreneurial climate in the Swedish economy, with special regard to those affecting the establishment and growth of small and new enterprises.

Entrepreneurship and enterprise growth

At its most general, entrepreneurship can be defined as the dynamic process of identifying economic opportunities and acting upon them by developing, producing and selling goods and services. Dynamism is not necessarily related to size, since the force of competition should ensure that firms are growing and shrinking throughout the distribution scale. But from a historical perspective they appear to be related in Sweden where an unusually large share of the business sector's employment is attributable to large enterprises. In the Swedish case the business sector is dominated by large industrial conglomerates with roots in the resource-based industries, most of which were major players well before the second world war (one outstanding exception is described in Box 7). And whereas some of these conglomerates have shown considerable ability to renew and innovate – Ericsson and Electrolux being frequently quoted examples – it is

Table 25. **Share of employment in selected sectors, by enterprise size**

	Size of enterprise	Share of sector employment						
		Sweden	Germany	France	United Kingdom	Netherlands	Denmark	EU average
Manufacturing	< 20	14	10	11	15	15	20	18
	20 to 99	16	14	20	17	27	26	17
	100 to 499	20	19	21	16	26	30	20
	500 +	50	57	48	52	32	24	45
Construction	< 20	37	50	41	64	33	59	57
	20 to 99	19	28	28	12	38	21	21
	100 to 499	10	13	13	11	20	10	12
	500 +	34	9	18	13	9	10	10
Trade, restaurants and hotels	< 20	43	52	44	40	45	54	59
	20 to 99	21	19	25	20	21	19	17
	100 to 499	14	11	12	27	13	12	12
	500 +	22	18	19	13	21	15	12
Business services	< 20	48	44	24	31	26	49	41
	20 to 99	19	15	19	14	18	21	13
	100 to 499	17	15	17	11	24	12	14
	500 +	16	26	40	44	32	18	32

Source: Submission from the European Observatory for Small and Medium-sized Enterprises.

not obvious that their success can be emulated by others. Rather, the perceived inability of small enterprises to grow beyond medium-size has at times been highlighted as an indicator of serious impediments to entrepreneurship in the Swedish economy, lowering the growth potential of the economy.

Actually, the share of manufacturing employment found in large enterprises is only slightly above average in Sweden (Table 25), but Sweden has an unusually large share of manufacturing in the business sector. Only in a few non-manufacturing sectors – construction and retail sales in particular – is there a significant tendency toward comparatively few small enterprises. The overall importance of large companies in the Swedish labour market and job creation thus relates more to its relatively large manufacturing sector in international comparison than to concentration within individual sectors. However, the share of self-employment in the Swedish labour force is internationally low; only during the last recession has it approached the levels prevailing in other North European countries.

Box 7. **A recent example of Swedish entrepreneurship: IKEA**

One of the relatively few outstanding examples of entrepreneurship after the second world war is the success of IKEA, which at the same time drives home the point that a company does not have to be in the ''high-tech'' category to be both fast-growing and profitable.

IKEA was established during the war by 17 year old Ingvar Kamprad in rural Småland. The region has no industrial tradition and very little contact with the traditional clusters of Swedish industry, but it is characterised by a high share of craftsmen and traders in the population and by a large propensity toward self-employment. The company started mail-order sales shortly after the war, but did not enter into its later business selling self-designed wooden furniture through its own outlets until 1955. The company's business idea has remained broadly unchanged ever since: make quality furniture available to a larger group of people through the lowering of prices. The prices have been kept low by a policy of: *i)* sale in large-surface outlets; *ii)* an element of do-it-yourself on the part of the purchaser; and *iii)* sub-contracting most of the production, more recently to low-cost countries.

The initial expansion focused on setting up outlets in the neighbouring countries, but the Danish EU entry in 1973 led IKEA to move its administrative headquarters to Copenhagen. This became the start of a period of rapid international expansion (Table 26). By 1996, the company's annual turnover came close to 3 per cent of Sweden's GDP.

Table 26. **Company characteristics – IKEA**

	Outlets	Countries	Employment	Turnover ($ million)
1954	1	1	15	0.5
1964	2	2	250	15
1974	10	5	1 500	139
1984	66	17	8 300	837
1996	136	28	33 400	5 709

Source: *Facts and Figures 95/96* – IKEA.

The founder still controls the company, but formal ownership has passed to a Netherlands-based family foundation in order to reduce some of the effects of Swedish taxation. However, the tax treatment of personally-owned companies has reportedly induced the family to consider leaving Sweden altogether.

Job creation and destruction: does size matter?

In Sweden, as elsewhere, net enterprise creation has contributed almost as much to net job creation as the expansion of existing enterprises,[54] particularly in the years before the recession. The considerable increase in labour shedding in the early 1990s seems primarily to have been related to the contraction of enterprises, whereas net job creation from new enterprises held up during the early stage of the recovery. Jobs associated with new enterprises also tend to last for a comparatively long time, as almost 70 per cent of new enterprises still exist after three years, and an internationally high 60 per cent survive after five years.[55] However, the average employment in these enterprises after ten years is fewer than three persons (including the owner), with very few employing more than ten persons and around half employing no additional personnel. It should also be noted that around one-third of these start-ups are unemployed persons benefiting from the government's support schemes for self-employment.

Despite the limited size of new enterprises, job creation in existing companies is usually related to their expansion. The 1995 *OECD Economic Survey of Sweden* showed that during the past decade, the net contribution of large establishments to Swedish net job creation was negative, the largest positive contribution coming from establishments with fewer than 20 employees. Studies based on data for the period 1985 to 1989, in which overall job creation was high, find that small single-establishment firms account for more than 60 per cent of net private sector job creation, while accounting for around 30 per cent of private sector employment.[56] Moreover, gross job creation and losses were both much higher in the small and medium-sized segment than in larger companies, indicating a more dynamic process. Finally, confirming an international trend toward increasing globalisation, small and medium-sized enterprises currently account for around one third of total exports – and around 20 per cent when the exports by subsidiaries of larger companies are excluded.

The role of the service sector

One possible explanation for the importance of new and small companies in job creation would be the increasing importance of the service sector: around 60 per cent of the net employment contribution of small firms occurred in trade,

Table 27. **New enterprises and bankruptcies by sector**

	1994		1995		1996		Average 1992 to 1996	
	New enterprises	Bank-ruptcies	New enterprises	Bank-ruptcies	New enterprises	Bank-ruptcies	New enterprises	Bank-ruptcies
	Per cent of all enterprises within group							
Industries	5.9	3.8	5.6	2.9	5.2	2.6	4.9	4.2
of which:								
Manufacturing	5.5	3.2	5.2	2.6	5.0	2.4	4.5	3.9
Construction	6.3	4.4	6.0	3.2	5.3	2.8	5.1	4.5
Services	8.4	3.4	7.9	2.8	7.7	2.9	7.2	3.7
of which:								
Trade, restaurants and hotels	6.7	4.4	6.2	3.6	6.0	3.6	5.8	4.8
Transportation	5.6	3.0	5.3	2.3	4.4	2.4	4.7	3.2
Finance and consultancy	10.7	2.8	10.3	2.4	9.9	2.8	9.7	3.2
Personal and social services	12.2	1.5	10.4	1.3	10.3	1.0	9.2	2.0
Total	7.8	3.6	7.4	2.8	7.1	2.6	6.7	3.8

Source: SCB.

household services and construction. Data on net new enterprise creation, in terms of the number of companies, seem to confirm this trend. The total number of service companies remained broadly unchanged even at the depth of the recession in 1992, whereas the industrial sector shrank significantly. Over the past four years taken as a whole, the number of enterprises in manufacturing and construction has been broadly unchanged, whereas the service sector expanded considerably. The growth was particularly pronounced in such parts of the service sector as business services – which may, partly, reflect outsourcing of certain services from industrial enterprises to smaller subcontractors – and, most significantly, personal and social services (Table 27).

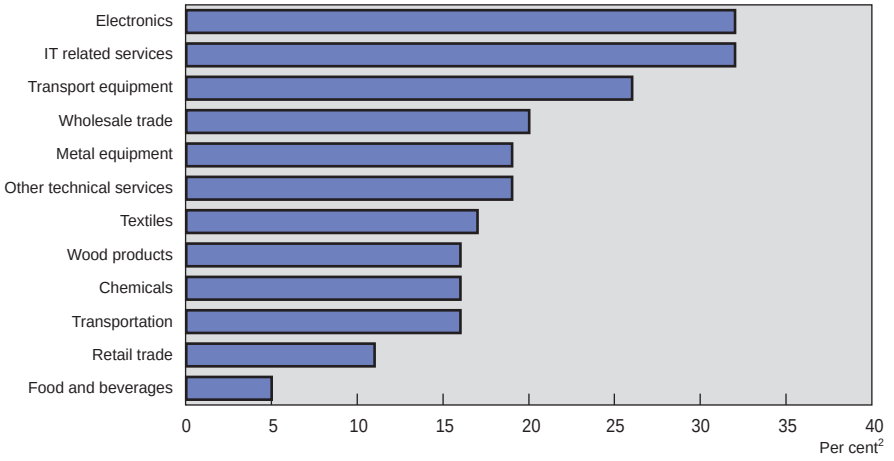
Fast-growing enterprises

A recent Swedish study demonstrates that fast-growing companies (FGCs), defined as limited companies which have more than doubled their turnover over the period 1992 to 1996, tend to be concentrated among small and medium-sized firms.[57] Half of all fast-growers were found to be enterprises with fewer than

20 employees, and small enterprises with particularly high turnover were the fastest growers. However, reflecting the dominant share of very small companies with a low turnover among Swedish enterprises, only one half per cent of all enterprises with fewer than 20 employees were fast-growing. Only 7 per cent of all larger enterprises (more than 200 employees) were fast-growing, but these accounted for the largest absolute job creation among the fast-growers.

Particularly fast-growing European enterprises – generally concentrating on developing new products and technologies – are usually dependent on exports as a source of growth. Swedish enterprises are no exception. As for the sectoral distribution of fast-growing companies, two broadly-defined sectors stand out. Owing to the export boost over the reference period – but also reflecting rising export prices due to currency depreciation – the export-oriented parts of the manufacturing sector (electronics, mechanical and transport equipment) recorded a large share of fast growers (Figure 31). Also the service sector, and particularly business-related services, recorded a share of fast-growing firms significantly above average. While clearly related to the rapid growth in industry, this also

Figure 31. **FAST-GROWING COMPANIES IN SELECTED SECTORS**[1]



1. Companies with an average annual growth in turnover exceeding 25 per cent from 1992 to 1995.
2. Per cent of all companies in selected sector.

Source: NUTEK (1996), *Tillväxtforetag i Sverige.*

seems to reflect the increasing importance of services in the Swedish economy. A final observation to be drawn from this study is that, irrespective of size classes, fast-growing enterprises tend to be somewhat younger than average.

The regional dimension

Swedish public debate often tends to focus on differences in business performance across regions, a common perception being that geographically-disadvantaged counties in the north and the middle of the country are performing worse than the south and the large urban centres. In recent years there has been a certain tendency for the latter parts of the country to have a higher gross enterprise creation, but the differences in net enterprise and job creation in recent years are not compelling. A recent study finds that gross enterprise creation relative to working-age population from 1990 to 1993 was somewhat above the national average in Stockholm and the south west of the country, but several of the regions presumed to be less well-performing held up strongly in this comparison. There is, however, a tendency for enterprise creation to cement traditional differences in the structure of the business sector: gross enterprise creation in the manufacturing sectors was particularly pronounced in the counties with a strong reliance on traditional industries (north and centre), where regional clusters in forestry, metal processing and machinery and electronics industry persist. Moreover, since these industries taken as a whole have reduced their employment, it is reasonable to assume that this enterprise creation was, at least partly, connected with the downscaling and outsourcing of existing activities. Most of the creation of service-sector enterprises, on the other hand, took place in the larger urban areas.

The regional distribution of fast-growing enterprises has been remarkably even. Although on the whole the large urban centres recorded a slightly higher share of fast growing enterprises, the difference is not significant, and only a couple of geographically-isolated counties seem to be significantly below the national average.

Implications

Summing up, new enterprises play a crucial role in net job creation, and fast-growing companies tend to be concentrated in the SME sector. Rapid employment growth in a subset of new enterprises combined with downsizing in

larger and maturing companies should be seen as an inherent feature in a well-functioning market economy. However, the large majority of SMEs show no tendency to grow and can consequently not be considered as particularly entrepreneurial, nor important for new employment creation. Given the importance of both net and gross enterprise creation for net and gross job creation as well as for the dynamism of the product and service markets, it is safe to conclude that reducing the number of obstacles to enterprise creation and growth is a way to promote entrepreneurship and thereby increasing overall incomes and, most likely, employment.

Start-up of enterprises: incentives and impediments

The first question to arise is whether incentives to create enterprises – and for existing enterprises to grow – are sufficiently strong in Sweden. According to surveys of enterprises started in 1995, around one-fourth of the entrepreneurs were motivated by a desire to avoid unemployment, one fourth by a wish to work independently and another fourth by a desire for self-realisation. Only 16 per cent perceived entrepreneurship primarily as a way of earning more money. Moreover, more than 40 per cent of the existing small and medium-sized companies reported that they either did not see any possibility to grow, or that they saw a possibility but did not wish to seize it. Of those who did not want to grow, more than half cited either a preference for leisure over possible gains, or too low an expected return on expansion relative to risk.

The welfare state and entrepreneurship

The egalitarian principles embedded in the Swedish social model have resulted in high marginal personal tax rates, which have almost certainly restrained the development of the business sector. Notwithstanding significant lowering of the rates in the 1990-91 tax reform, Sweden has retained one of the higher marginal tax rates in the OECD area. While income taxation is broadly neutral *vis-à-vis* the growth decisions of a going concern, the decision to start an enterprise or undertake specific risky projects is influenced by weighting the default risk (ultimately bankruptcy) against expected after-tax gains, encompassing not only the effect of tax rates but also of provisions allowing losses to be carried forward or set against other types of income. In 1996, loss-offset

provisions for Swedish unincorporated enterprises were improved as capital losses from new establishments were allowed to be offset against labour income. In the absence of complete loss-offset provisions, a country maintaining high marginal tax rates is likely to have less business start-ups and a lower share of self-employed persons.

The ''solidaristic wage policy'' is also likely to have disadvantaged new enterprises insofar as it has led to a compressed wage structure. According to recent cross-country studies, larger and older companies pay significantly higher wages. In most European countries, a wage level of around 15 to 20 per cent below national averages is normal in small enterprises, and in the United States the difference is around 40 per cent.[58] In Sweden, a recent study found a differential of 7 per cent, almost all of which could be attributed to differences in skill levels and seniority of employees in small and larger enterprises.[59] While an equalisation of wages across skills and seniority levels may contribute to a more efficient static resource allocation, there may be a dynamic loss of efficiency stemming from the compressed wage structure in Sweden to the extent that lower wages are necessary to retain profitability during the start-up phase of enterprises.

Moreover, high taxes and an equal wage distribution, together with an extensive public pension system, have contributed to internationally low household savings, and shifted savings away from financial wealth toward tax-favoured schemes such as housing investment and private pension schemes.[60] Recent figures show that Swedish household net financial assets (including private pension savings) are an estimated 80 per cent of GDP, compared with 140 per cent in Germany, 200 per cent in the United Kingdom and 275 per cent in the United States. Since most capital in the earlier stages of an investment is provided either by the entrepreneur himself or persons close to him, low household wealth may reduce the capital available for enterprise start-ups. The possibility of quantitative rationing in the demand for start-up capital is underpinned by a recent survey-based study which found that households reaping large windfall gains have a significantly higher probability of starting an own enterprise.[61] Another recent empirical study concludes that the compression of wealth differentials has had a negative impact on the rate of enterprise starts.[62]

The development of the private service sector has been particularly disadvantaged because wage compression, high income taxes and high indirect taxes have raised the difference between the cost of personal services and the

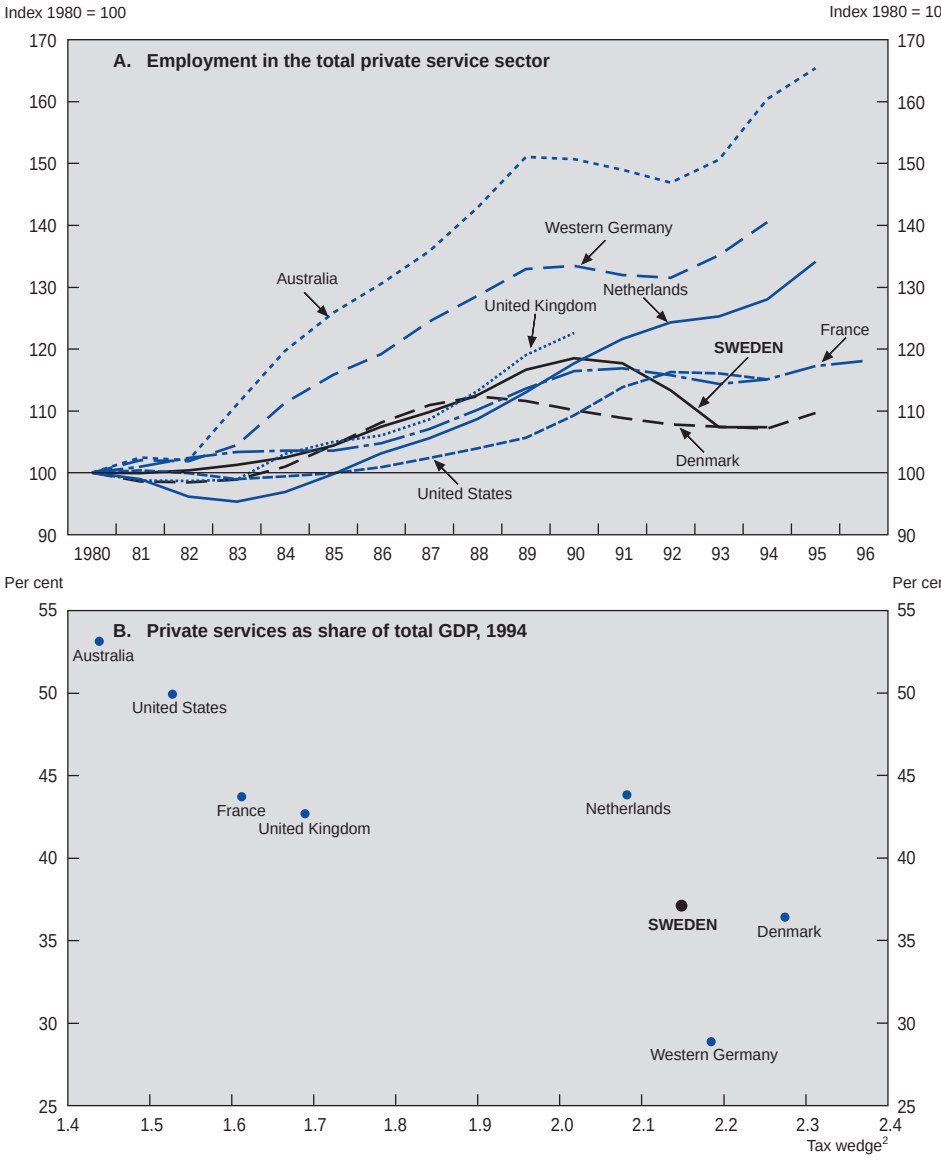
take-home pay of the service provider (the so-called ''tax wedge'') to one of the highest in the OECD area (Figure 32). This has almost certainly shifted household demand away from personal services and toward other kinds of consumption. It has also discouraged the development of economic activities in the service sectors which are close substitutes to the households' own domestic effort (cooking, cleaning, gardening, etc.), where new enterprises can be established at a low entry cost and where few formal qualifications are required. Indeed, this sector has recorded considerable net enterprise creation in many other OECD countries.

Start-up and closure: rules and administrative burdens

Slow or expensive procedures in connection with the establishment of a new enterprise can also lower the start-up rate. In Sweden, however, there seem to be only limited problems of this kind. The number of administrative requirements and the delays appear to be below the international averages, and the costs involved are negligible (Table 28).

On the other hand, the rules for bankruptcy and otherwise closing an enterprise are restrictive, and this is arguably a factor which significantly discourages entrepreneurship. Concerns about limiting abuse and outright fraud are indeed legitimate ones, since if left unchecked, they distort competitive conditions facing entrepreneurs and undermine the tax base. The challenge is to combat abuse and fraud, if necessary through the relevant provisions in the criminal code, while not restricting genuine risk-taking, which can often result in failure.[63] Moreover, there is a trade-off between protecting creditors and encouraging entrepreneurs. In the Swedish case, the absence of a discharge clause (Table 29) implies that an unsuccessful attempt at setting up an own enterprise can, in the worst case, lead to financial obligations for the rest of one's life. (This may be contrasted with the United States experience where ''a good try'' is encouraged, and where many successful entrepreneurs have one or two bankruptcies behind them before they succeed.) A further tightening of the bankruptcy rules is currently under consideration. According to a proposal from an expert commission, persons involved in three corporate bankruptcies within ten years and persons still liable for tax debt exceeding SKr 217 000 (approximately $27 000) from bankrupt companies will be legally barred from starting a new enterprise during a period of three to ten years. In seeking to strike a proper balance between controlling abuse and fraud and avoiding the creation of barriers to entrepreneurship, it would seem important to ensure that the criteria used do not discriminate against genuine entrepreneurial dynamism.

Figure 32. **TAX WEDGES AND THE SERVICE SECTOR**[1]



1. The private service sector includes wholesale and retail trade, restaurants, hotels, transport, communication, finance, insurance, real estate, business services.
2. Total labour cost relative to purchasing power of take-home wage for an average production worker.`

Source: OECD, *National Accounts*; OECD, *The Tax/Benefit Position of Production Workers*; OECD, *Revenue Statistics.*

Table 28. **Registration requirements and costs**

Unlimited liability (limited), private limited company and public limited company

	Legal form[1]	Pre-registration requirements[2]	Registration offices[3]	Post-registration requirements	Time (weeks)[4]	Minimum charter capital (ECU)[5]	Cost (ECU)[6]
Australia	Proprietary	3	1	1-5	1	0	200-480
	Public	4	1	1-5	1	0	200-480
France	Artisan	6	1	4	1-7	0	1 100-2 700
	SARL/EURL	10	1	5	4-8	8 000	1 900-4 600
	SA à conseil	14	1	7	7-15	40 000/250 000	2 200-6 100
Germany	KGT	1	1	2	1 day	0	10-25
	GmbH	6	2	2	8-24	25 000	750-2 000
	AG	6	2	2	8-24	50 000	750-2 000
Italy	Artigiana	7	1	4	4-16	0	1 150
	SRL/SuRL	17	5	3	4-16	10 000	2 200
	SPA	18	4	3	22	100 000	7 700
Netherlands	Eenmanszaak	1	1	5	3-7	0	0
	BV	3	1	5	12	19 000	1 000-
	NV	2	1	5	12	0	0
Spain	EI	0	3	5	1-4	0	0
	SRL	7	5	5	19-28	3	330+
	SA	7	5	5	19-28	62	330+
Sweden	Enskild Firma	0	1	2	0-4	0	90
	AB	3	1	3	2-4	12 000	1 130
	AB (public)	3	1	3	2-4	60 000	1 130
United Kingdom	Sole Trader	0	0	3	0	0	300
	Private	1	1	3	1	2	420
	Public Limited	1	1	3	1	70 000	900
United States	Sole Proprietor	0	2-6	2-5	1-2	0	200-800+
	LLC	0	2-6	2-5	1-2	0	200-800+
	Corporation	0	2-6	2-5	1-2	0	200-800+

1. Legal form. Three different forms have been selected among each country's many different forms: unlimited liability forms, limited liability corporate form and public corporations.
2. Registration requirements. The number of procedures which must be completed before registering.
3. Registration offices. The number of offices where the business entity must be registered.
4. Time. The number of weeks required before the registration has been processed by the authorities.
5. Minimum charter capital requirements. The value of assets which a business entity must have and maintain.
6. Cost. The direct (fees paid to the registering authorities) and indirect (fees paid to lawyers, agents and consultants) costs of registration.

Source: OECD (1998), *Economic Survey of Australia.*

Table 29. **Bankruptcy proceedings**

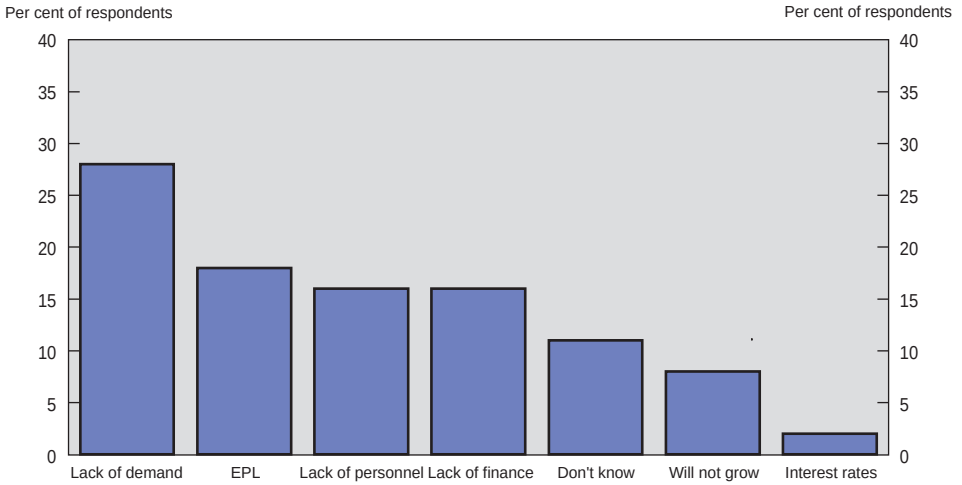
Country	Filing	Discharge clause
Australia	Entrepreneurs can voluntarily file for bankruptcy. For limited companies a number of procedures for liquidation and voluntary administration are in place.	Discharged after three years. For negligible amounts discharge can be granted after three months. Managers are not liable unless involved in improper dealing.
Germany		Currently, no discharge. Company managers incur civil liability and may also be liable for criminal penalties. Reforms to be enacted will discharge liability companies after seven years.
Sweden	Entrepreneurs can voluntarily file for bankruptcy. However, high-priority debt must be settled, and some additional costs are involved.	No discharge. Managers and owners of limited companies can be made personally liability for tax debts.
United Kingdom		Discharged after two years if the debt is lower than £20 000; three years if the debt is greater.
United States	Entrepreneurs can voluntarily file for bankruptcy. Many bankruptcies are settled outside the courtroom.	Discharge effective immediately.

Source: Submissions from national authorities.

Obstacles to expansion

Among those companies which wanted to grow, the major obstacles have varied over time, factors such as weak demand and scarcity of labour showing strong cyclical variation. Small companies (up to 20 employees) list lack of venture capital and labour market legislation as equally important obstacles to growth, with a tendency for high-tech sectors to focus somewhat more strongly on capital (Figure 33). On the other hand, surveys including somewhat larger enterprises (up to 50 employees) conclude that labour market restrictions are by far the most important obstacle.[64]

Figure 33. **IMPEDIMENTS TO ENTERPRISE GROWTH**[1]



1. Based on a survey of 1 807 enterprises with less than 20 employees.
Source: Föreningsbanken, Företagarnas Riksorganisation, LRF Konsult (1997), *Småföretagsbarometern,* Fall 1997.

Disadvantages of small scale? Financing, training and R&D

Small enterprises may be disadvantaged in comparison with larger ones in three ways (the three so-called ''gaps''): availability of finance; training of staff (''human capital investment'') and development and use of new technology.[65] As for finance, it is often claimed that small businesses are disadvantaged in the ability to raise loan capital relative to larger competitors due to information asymmetry and larger risk. Banks, unable to correctly assess the credit risk involved, charge excessive interest rates or decline credit altogether. In this case, small and new enterprises could obtain cheaper financing via injections of equity capital which allow investors to cover the high default risk against the expected share of the profits of successful enterprises.

There is some past evidence of a financing gap, where the subset of small companies which intended to undertake innovative activity perceived a lack of venture capital as a hindrance.[66] Indeed, following the banking crisis, which particularly affected the financial system around 1992, bank lending was scarce

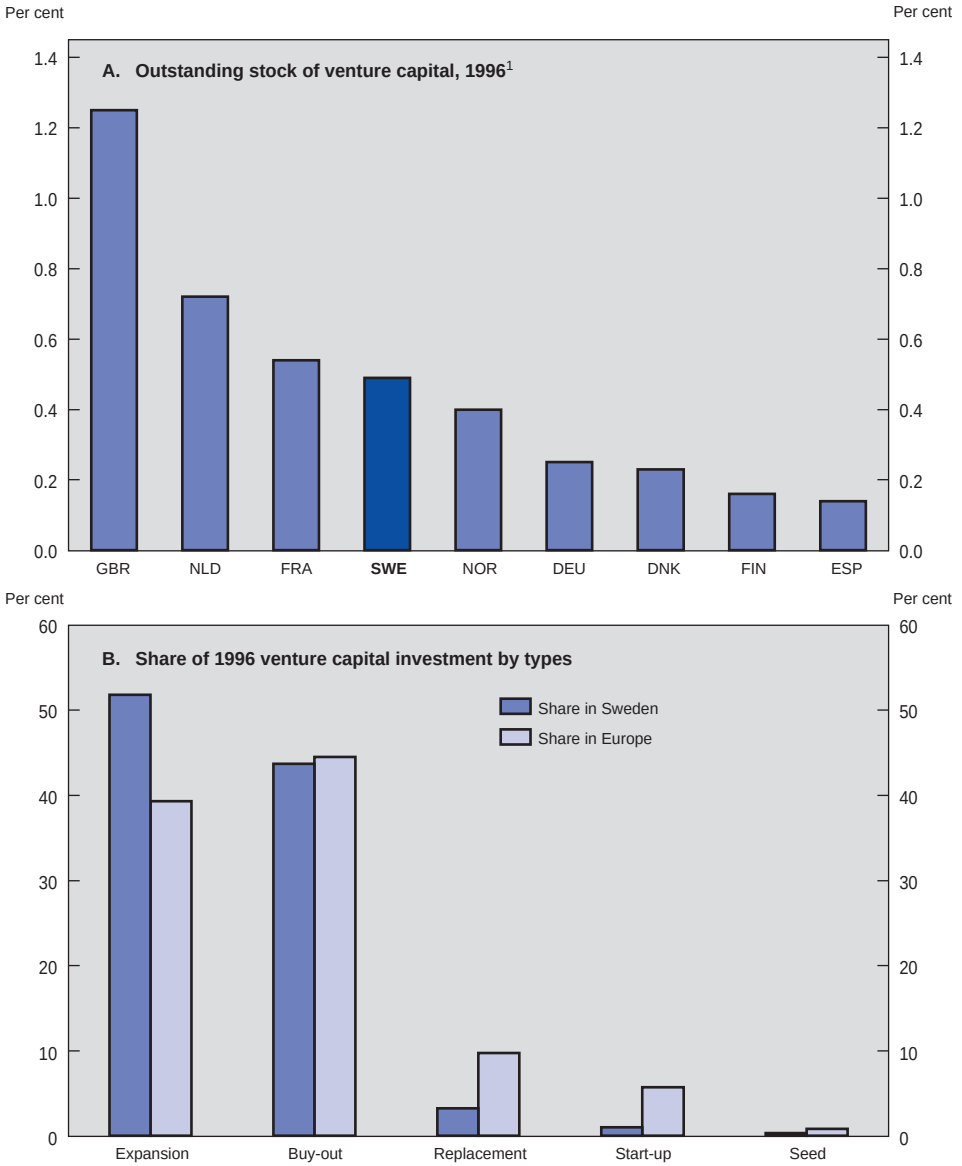
in the Swedish economy and this disproportionally affected companies which were either weakly capitalised or attempting to grow fast. More recently, surveys have not pointed to any widespread lack of finance, and the *1997 Economic Survey of Sweden* found no evidence of bank lending rates to small businesses which were higher than justified by default risks.

However, high-risk enterprise start-ups and expansion may more appropriately be financed through equity capital than borrowed funds. As for venture capital there may have been a problem in the past, but it would appear that the supply of venture capital is now rapidly increasing. Indeed, a recent cross-country comparison shows that Sweden has the fourth largest venture capital stock (relative to GDP) in Europe (Figure 34, panel A). Furthermore, the annual investment of new venture capital quadrupled to more than SKr 3 billion from 1995 to 1996 and the increasing trend continued into 1997. The venture capital market has developed rapidly since the abolishment of the Stockholm Stock Exchange monopoly on listing of equity six years ago. First, development markets – the so-called OTC and O-listings, broadly equivalent to the United States NASDAQ – were established. Second, private listings (*e.g. Stockholm Börsinformation* and *Innovationsmarknaden*) targeting small and new companies developed. Third, individual stockbrokers started listing equity issues by small and unlisted clients on their electronic trading systems. It would appear from recent experience that companies reaching a size corresponding to some fifteen employees can seek venture capital within the framework of this system.

Venture capital for the early stages of an investment continues to be in relatively scarce supply, insofar as the share of venture capital is below the European average, although the proportion allocated to expansion exceeds the average (Figure 34, panel B). The need for early-stage venture capital is in some countries to a large extent filled by informal investors – known as ‘‘business angels’’ – who closely monitor, or even work with, the entrepreneur. While there are no formal studies of such investors in Sweden, the presumption used to be that they are rather limited in number, reflecting *inter alia* the scarcity of household wealth mentioned above. Recently, however, there has been anecdotal evidence that large portions of newly-floated non-listed equity have been taken up by individuals, so that the degree of risk-willingness among Swedish households may be increasing.

Figure 34. **VENTURE CAPITAL**

Per cent



1. Stock as a per cent of GDP.

Source: OECD (1996).

Both in Sweden and in the industrial economies in general there is little concrete evidence of new and small businesses being disadvantaged in the acquisition and use of new technology. There does seem to be a tendency for small businesses to invest comparatively less in R&D than large companies, but such comparisons fail to take account of the fact that small businesses are often concentrated in sectors with little overall R&D, and that there is a confirmed tendency for small companies without their own R&D department to under-report their R&D effort. Indeed, while comparable data are not available, studies of the R&D activities of small enterprises in Sweden and other Scandinavian countries seem to show that they are considerably more innovative than usually assumed.[67] Moreover, formal R&D is not the only source of technology acquisition. According to a recent study, small businesses acquire a significantly higher share of their ''innovation'' from non-R&D sources – purchase of technology and consultancy, hiring of outside specialists – than larger enterprises.

The question of training in enterprises is treated in detail in Chapter IV of this *Survey*. Recent investigations into training efforts elicit no conclusive evidence of small companies using less economic resources on training their staff than larger ones. Moreover, those enterprises which do engage in training activities allocate broadly the same number of working hours per employee irrespective of company size.

Taxation of enterprises

The structure of the Swedish business sector has been significantly influenced by the business taxation regime prevailing up to the 1990/91 tax reform. This system combined high statutory tax rates with considerable leeway for individual enterprises to reduce the tax base through accelerated depreciation of capital stock and inventory and allocations to in-house ''investment funds''. The system, motivated by a wish to smooth investment over the cycle and to encourage reinvestment of corporate earnings, in fact discriminated against newly-established enterprises and enterprises with few tangible assets while reducing capital mobility across enterprises, thus favouring large and capital-intensive industries.[68] The tax reform redressed this, and introduced a universal corporate tax rate of 30 per cent which was lowered to 28 per cent in 1994. This general taxation of business earnings applies equally to incorporated and

non-incorporated enterprises, thereby implying a more equal treatment of small and large enterprises – albeit at the cost of a considerably more burdensome tax compliance.

The tax reform also diminished the tax-distortions between different forms of enterprise financing, most notably by reducing the prohibitively high marginal tax rate on new share issues in companies owned by households. A temporary abolition of double taxation of dividends in 1994 contributed further to this (Table 30). The reintroduction of double taxation of dividends in 1995 can be considered as impeding the supply of risk capital, affecting not the larger enterprises with internationally-traded shares but listed and unlisted enterprises which rely on domestic equity finance. While a partial exemption for unlisted businesses was re-introduced with effect from 1997, the present rules still imply disincentives to grow and to become listed. More generally, it may be noted that the high effective taxes facing households as investors, especially as purveyors of equity capital, have contributed to a long-term decline in households' participation in the equity market. For small and growing enterprises, this may historically have had particularly adverse effects on the supply of risk capital as a direct placement market failed to emerge.

Table 30. **Effective marginal tax rates**

Real pre-tax of return 10 per cent at actual inflation rates

	Debt	New share issues	Retained earnings
1980			
Households	58.2	136.6	51.9
Tax exempt institutions	–83.4	–11.6	11.2
Insurance companies	–54.9	38.4	28.7
1994			
Households	32.0/27.0 [1]	28.3/18.3 [1]	36.5/26.5 [1]
Tax exempt institutions	–14.9	21.8	21.8
Insurance companies	0.7	32.3	33.8
1995			
Households	32.0/27.0 [1]	67.7/57.7 [1]	48.0/38.0 [1]
Tax exempt institutions	–3.5	25.7	25.7
Insurance companies	21.0	53.3	50.4

Note: All calculations are based on the actual asset composition in manufacturing.
1. Excluding wealth tax. Wealth tax on unlisted shares was abolished in 1993.
Source: Henrekson, M. (1996), *Företagandets villkor*.

Further adverse effects of the corporate tax code on enterprise growth may emanate, paradoxically, from the desire to provide some relief for particularly small companies. Enterprise wage bills below SKr 850 000 (the wage costs of three to four employees) face a lower social security contribution than those above that level, increasing the costs for a further expansion. The complexity of tax rules stem also from efforts to prevent tax arbitrage between wage and capital income, resulting from the large discrepancy between internationally-competitive capital income taxes at 28 to 30 per cent and wage income taxes close to 60 per cent. For active owners of closely-held companies,[69] earnings less than 5 per cent of a benchmark – calculated as the capital invested corrected for certain wage payments[70] – are exempt from the tax on capital income and only taxed at the corporate level (28 per cent); income from 5 to 12 per cent of the benchmark is taxed at both the corporate and the personal level (giving an overall tax burden of approximately 50 per cent). However, income exceeding 12 per cent of the benchmark is taxed at the high marginal tax rate applying to wage income which together with the corporate tax on the underlying income results in an overall tax burden approaching 70 per cent. While tax arbitrage may be prevented, the complexity of the system may discourage entrepreneurs and involve relatively high administrative costs for small enterprises.

The personal wealth taxation may create some advantages for small businesses, compared with larger enterprises. According to recent changes in the tax code, all equity listed on the OTC and the O lists of the stock market, along with equity unlisted altogether, is considered as working capital. Such working capital is untaxed, whereas ownership of publicly-listed equity (equity appearing on the so-called A-list) is taxed. In order to avoid disincentives to public listing in the future, main owners of companies moving from the OTC and O lists to the A-list will retain their exemption from wealth taxes.

The labour market

Labour market legislation is listed by many employers as one of the main impediments to enterprise growth. Regulations which raise the costs of laying off labour inevitably discourage companies from hiring and thereby affect job creation in general, but there are reasons to believe that small and medium-sized businesses are particularly affected. For example, to the extent employment protection legislation (EPL) discourages lay-offs of staff ill-suited for their

current work assignments, companies with a limited number of workplaces are in a disadvantageous position compared with larger enterprises, who can redeploy and use natural wastage to adapt. The *Jobs Study Follow Up* in the *1997 OECD Economic Survey of Sweden* concluded that the EPL in Sweden was not just more restrictive than the average in Europe, and considerably more so than in the United States, but that the rules guiding the order of dismissal and re-hiring of employees discriminated against small and – particularly – medium-sized employers.[71]

Finally, it has been argued that the payroll taxes are excessive and that they are one major reason for sluggish growth of Swedish enterprises.[72] However, while high total labour costs definitely discourage hiring, most studies indicate that the long-run incidence of payroll taxes falls on the wage earner rather than on the employer. The implication of this is that lower payroll taxes – while they might have some positive short-term impact – would eventually result in higher wages rather than more jobs. On the other hand, improving labour market flexibility to allow smoother adjustment of wages to demand and supply in the labour market would indeed help improve the entrepreneurial climate.

Market factors

Weak competition impedes the creation and growth of new enterprises, insofar as it gives incumbent companies an advantageous market position. Therefore, a strict and rigorously enforced competition law is favourable to entrepreneurship. Swedish competition laws were tightened in 1993, notably to encompass the ''prohibition principles'' prevailing in EU area, which must be considered an improvement in the framework for small and new enterprises. Some problem areas remain, notably in the service sector (see chapter III), which has performed less well as a creator of new jobs than in most other countries. Cartelisation remains a problem, *inter alia* in transport services, and the market for professional services remains strongly regulated.

As a particularly important example, the retailing sector has recorded no increase in employment over the last two decades, compared with a yearly growth of around 1 per cent in most European countries, and 1½ to 2 per cent in North America. While this is partly linked to slower growth in private consumption and restrictive opening-hour legislation which was lifted only in 1989, it also reflects an internationally very concentrated retail sector. For example, three food

retailers continue to cover close to 70 per cent of the total market (Table 31). While this oligopolistic power is not actively encouraged by the authorities, there is some evidence of local authorities using their control of the zoning rules to discourage changes to the traditional pattern of retail outlets.

Table 31. **Characteristics of the distribution sector in some European countries**

	Retail outlets per 100 000 inhabitants			Concentration	
	Total	Supermarkets[1]	Hypermarkets[2]	1995 market share of the three top food retailers	Share of sole proprietorship
Sweden	**940**	**23.5**	**0.8**	**68**	**29.8**
Germany	850	12.1	1.4	40	81.5
France	970	12.6	1.9	38	71.8
Italy	1 710	7.4	0.3	17	90.4
United Kingdom	810	3.4	1.3	49	n.a.
Spain	1 340	19.1	0.6	26	92.0
Netherlands	800	13.8	0.3	57	70.1
Belgium	1 410	19.8	1.0	61	74.5
Denmark	1 000	17.3	0.3	50[3]	76.6
Norway	920	31.0	0.7	86	18.0
Finland	770	20.6	1.5	83	41.9

1. Supermarkets are defined as covering between 400 and 2 499 square metres.
2. Hypermarkets cover 2 500 square metres and more.
3. Top two retailers only.

Source: OECD.

Whereas the central government's involvement and ownership in the business sector remains modest, local authorities' business activities can be seen as an impediment to the development of the private business sector in some areas. At the beginning of 1994 local governments had control (a minimum 50 per cent equity stake) of close to 1 500 companies with nearly 50 000 employees and an annual turnover of SKr 115 billion (8 per cent of GDP), the strongest involvement being in housing and utilities. Swedish rules allow government providers to compete with private providers, and although the competition authorities have the right to intervene in the case of dumping and predatory pricing practices the burden of proof still falls on the private sector operator. But this creates problems of ensuring that prices offered by government entities actually cover all costs,

especially capital and rental costs. Overall, the provision of a wide range of free or subsidised services which are operated by private companies in most OECD countries – the municipal laundry services are a frequently quoted example – have discouraged the development of those private service sectors which have been among the most rapidly-growing elements in the US and UK labour markets. As a result of the focus on the actual and potential distortions of competitive conditions in this field, the government has decided to establish a council which will review and monitor the competitive inter-relationship between public and private producers, with a mandate to formulate general rules as well as analysing specific complaints.

Policies to promote entrepreneurship

Swedish economic policy has traditionally worked to consolidate and promote the expansion of large enterprises. In the past, competition policy focused on promoting economies of scale, while the tax system, capital market regulations and the foreign direct investment regime combined to favour large capital-intensive enterprises. In addition, a tendency for secondary and tertiary education systems to focus on meeting the needs of large industry and public administration helped to create a ''wage-earner culture'' where risk-taking and entrepreneurship were not actively encouraged. Starting in the early 1990s, financial markets and tax policy have become more neutral with respect to enterprise size, but small enterprises rarely expand into larger ones. Over the years several policies and targeted programmes aimed at nurturing small enterprise growth have therefore been put in place. The main areas covered relate to investment, exports, technology, R&D, management and education, consultancies and environment issues – albeit at a comparatively limited budgetary cost. In all there exist 140 types of subsidies with an additional 110 available from the EU. Some of the most important of these are listed in Box 8.

In addition to policies pursued at the national level, regional authorities and, particularly, municipalities have been active in support of the local business community – and have given this area increasing attention in the face of growing labour market imbalances. The kind of assistance offered varies significantly across regions. In areas dominated by ''knowledge intensive'' parts of the business sector (the three main urban centres and certain areas around major

Box 8. **Government programmes for assisting new, small and medium-sized enterprises**

Swedish enterprises have access to EU schemes aimed at promoting business sector growth, of which several can be said to be of particular relevance to smaller firms. For example, the *BRE* and *BC-Net* programmes and the *Enterprise* initiative facilitate the search for business partners in other European countries; the *Europartenariat* aims at promoting business links with less favoured regions; the *SME Initiative* supports SME investment in improvement of production and organisation; the *Framework Programme for Research and Technical Development* aims at co-ordinating and supporting research in different areas; and the *CRAFT* programme encourages networking between enterprises facing common technical need.

Furthermore, the following national programmes are in force:

- ***Start-up and information.*** The *ALMI Business Partner* company, which is owned by the government, and 22 regional development companies owned by county councils, offer consultation, financing and information to persons planning to start a new business. The initial consultation is free of charge, later services are offered in return for a fee. The *National Board for Industrial and Technical Development (NUTEK)* operates a special telephone service with information and advice for new-starters. Moreover, these two entities provide SMEs in general with legal, technical and commercial advice related to product development, financial solutions, marketing and patent issues.
- ***Financing*** ALMI Business Partner provides new firms with soft loans with a maturity of six to twelve years, covering up to 30 per cent of the total capital requirements. Generally these loans are interest-free and not amortised during the first two years. Moreover, credit guarantees can be granted. For existing enterprises investing in equipment or engaging in product and process development, loans and credit guarantees for a period up to eight years are given. *NUTEK* finances technical innovation at an early stage, before any product has been introduced on the market. Support is provided as loans, capital against royalty or project guarantees to a maximum of 50 per cent of the cost of the project. The *Swedish Industrial Development Fund* manages the following programmes in favour of smaller enterprises: *i)* loans for specific projects (maximum 50 per cent of total costs); *ii)* capital against royalty (maximum 50 per cent); *iii)* credit guarantees (maximum 80 per cent of the loan); and *iv)* venture capital in exchange for shares or convertible loans. The *Innovation Centre* supports the early stages of the innovation process, such as technical and commercial licenses.
- ***Trade promotion.*** The *Swedish Trade Council* (joint initiative of the government and private enterprises) promotes exports through information, advice, arranging trade fairs and joint marketing. Special priority is given to SMEs. The *Swedish Export Credits Guarantee Board* provides credit insurance to enterprises and

(continued on next page)

(continued)

banks in relation to exports and investment in other countries. The premium for the guarantee varies with the estimated risk, but an element of subsidy is included. The *Swedish Export Credit Corporation* (partly government-owned) provides medium and long-term export credits. Market-based interest rates are charged throughout the credit period.

- ***Women and minority groups.*** ALMI Business Partner has a special scheme for loans to enterprises wholly owned by women. These loans, which can be granted for both start-ups and existing enterprises, generally have a term of ten years including a grace period of one to three years, and a one to two year interest-free period. *NUTEK* finances several programmes in favour of female-owned enterprises: *i)* female business advisors in many regions; *ii)* training, information and networks for women entrepreneurs; *iii)* support to R&D on female entrepreneurship; *iv)* conferences for female entrepreneurs; *v)* scholarships.

education centres) the municipalities' efforts are concentrated on fostering contacts and facilitating the exchange of information among existing enterprises. In areas dominated by ''basic industries'' (the northern parts of the country and the rural south-east) half of all municipalities are involved in developing new business activities together with local enterprises. In addition to the formal programmes for business development, local authorities reportedly use their influence on labour market programmes and their control over primary and secondary education – where far larger budgetary amounts are involved – to create a favourable climate for local enterprises.

A new policy orientation?

Since the recession of the early 1990s, which particularly hit regions and segments of the labour markets which had become dependent on a few dominant enterprises, some additional reorientation of Swedish industrial policies has been under consideration. The government's programme for promoting growth and reducing unemployment by half by year 2000 singles out policies to promote entrepreneurship and SMEs as a field of action. A catalogue of measures intended to exploit entrepreneurial potential and further level the playing field between small and large enterprises has been adopted, easing the tax burden and the

labour market rigidities for new and small businesses. In particular, the government has taken the position that small and medium-sized knowledge-intensive enterprises are likely to be the main source of private sector growth and employment creation in the near future. Policies relying on the diffusion of technology and networking in local areas thus play a prominent role. The main elements relating to the business sector are summarised in Box 9.

One element of the action plan which is still under consideration is the tentative devolution of some business-sector related policies to the regional and local authorities. The political debate has focused on whether local authorities should have a freer hand in reallocating the state-financed unemployment compensation toward job creation in the enterprises and the public sector. The 1997

Box 9. **The Swedish government's programme for growth and employment**

Development, renewal and growth for enterprises: (three year programme)

- Increase the diffusion of knowledge and technology by *i)* improving contacts between locally based tertiary education institutions and small and medium-sized enterprises; *ii)* encouraging the transfer of technology from large to small companies through the, largely self-financed, Industrial Development Centres; *iii)* increasing the supply of technologically skilled labour through the establishment of Science Centres; and *iv)* supporting the development of environmentally oriented technologies and industries.
- Make enterprise start-up and growth easier through *i)* a 5 per cent reduction in employers' social contributions up to a (low) limit; *ii)* some easing of EPL, in particular improved access to temporary employment and a right to deviate from certain dismissal rules by local agreement; *iii)* an exemption from double taxation of parts of the dividends from unlisted equity; and iv) some increase in the amount self-employed persons can earn subject only to corporate tax rates.
- A strengthened role for the regions, including in particular increased co-operation of local authorities among themselves and with the business sector, and the devolution of some central government tasks to the regional and local level. Concrete measures include SKr 500 million (in addition to existing programmes) to locally administered labour-market related programmes for self employment and a programme to promote tourism.

and 1998 experiment with extending the subsidised employment in the local authorities (the ''Kalmar model''), is seen by some discussants as a test model for ''active'' use of the labour market funds in local areas.

Assessment and recommendations

To the extent that entrepreneurship is a function of the cultural and educational features of Swedish society and that are embedded in the nature of the Swedish welfare state, the shortcomings analysed above cannot be easily redressed without conflicting with other policy objectives. However, economic factors seem to have exerted a negative impact on the entrepreneurial climate in general, and on the start-up and growth of small enterprises in particular, and these could be addressed at a low direct cost. Progress has already been made in removing some of the biases in favour of large enterprises stemming from the taxation, corporate finance and regulatory systems. However, a recent OECD study found that policies to encourage the growth of smaller enterprises tend to be focused more toward investment, R&D and goods exports and are thus more effective within the manufacturing sector than in the service sector.[73] More fundamentally, while the proliferation of schemes to support small businesses may be seen as a way of correcting the negative effects of previous policies which favoured large Swedish enterprises, such support can only play a limited role with respect to the impediments arising from other sources, in particular restrictive labour market regulations, a tax code which may still discourage investment in both human capital and technological diffusion and a public procurement process which is weighted against private-service provision.

To deal with these problems, new initiatives are needed in the following areas:

- Labour market legislation, and particularly employment protection legislation, limits enterprise growth and job creation and should be critically reviewed. To the extent that it creates special problems for new and small businesses, such companies could, as a second best, be granted statutory exemptions from parts of such legislation;
- If the marginal tax on labour income could be lowered towards that of capital income, less emphasis would have to be put on closing off windows for tax arbitrage; as a consequence, the tax code could be

simplified. The tax code could be made more neutral, *inter alia* with respect to enterprises' decisions to expand and innovate and to seek a stock-exchange listing;

- While entry costs are modest, exit costs seem prohibitively high. The concern to limit the abuse of corporate bankruptcy may be a valid one, but the penalties involved in a personal bankruptcy are such that they discourage entrepreneurial activity;
- The efforts to improve companies' market access should be continued. The strictest possible enforcement of competition laws on the domestic market environment will, in particular, benefit small and new enterprises. This should include fair and equal competition with government entities providing market-produced goods and services.

With respect to state support for diffusion of information and technology, it is important that these be promoted through generic programmes:

- Where support services for newly-established enterprises are provided it is important to ensure that such activities do not conflict with work which is done – or could equally well be done – by private business consultants, lawyers, accountants and venture capitalists;
- Public authorities can facilitate the diffusion of information among enterprises: for example small enterprises faced with a particular problem are often unaware of the solutions at hand. Also, networking toward technological development or marketing can sometimes be facilitated though outside participation.

In addition to the national policy measures and support schemes, a case could be made for increased local authority involvement in nurturing an entrepreneurial climate. Due to their closeness to the local markets, regional authorities and municipalities may be well placed to diagnose and solve specific problems. However, local authority involvement is necessarily circumscribed by their own budget constraints and should be subject to mechanisms which assure transparency, and a standard of quality in design which ensures that competitive forces are not inhibited.

This *Survey* has found little evidence that small and new enterprises suffer any systematic disadvantage of scale – except possibly in the financing of the earliest stage of an investment, where it is not obvious that the government should play any direct role. The introduction of further support schemes for

selected parts of the business sector thus does not seem advisable. The number of specific programmes is already large, and it should be borne in mind that the proliferation of current schemes risks creating bureaucratic conflicts and information problems and diverting enterprises' attention from entrepreneurial activity toward rent-seeking. Broader structural policy reform is necessary for the improvement of the institutional and regulatory environment. The most important role for public authorities remains the provision of a stable macroeconomic situation, a neutral system of taxation, a well-functioning labour market and a competitive business sector environment.

Notes

1. Households own directly 15 per cent of Swedish shares and even more indirectly through mutual funds. Direct ownership is highly concentrated among a relatively small section of the population which may account for the weak short-term link with household demand.
2. The Institute for Housing Research has estimated that the ratio between second-hand prices and construction costs for houses of comparable quality (the so-called Tobin's Q) was on average close to 0.8 in 1996. Price and cost developments since then have only eliminated about 5 percentage points of the gap. However, in the faster-growing parts of the country, investing in a new house is now more profitable than buying a house on the secondary market.
3. Inflation targeting requires a high degree of transparency. In the Swedish case, the main instrument to effectively communicate to the general public and financial markets the continued commitment to the overriding objective of price stability is the Inflation Report of the central bank which sets out economic trends over the policy horizon.
4. The Bill also contains proposals for amendments to the Swedish constitution which require acceptance from two consecutive parliaments, implying a first vote in spring 1998 and a second vote after the general election in September 1998.
5. At the inception of EMU from 1 January 1999, a separate mechanism for exchange rate co-operation between the European Central Bank and national banks of non-participating countries will be set up, in most respects emulating the present ERM but based on central rates against the euro instead of the parity grid of the latter. Sweden has indicated that it will not join the new mechanism, also called ERM2, at its inception, but maintain its present regime. Annex I provides an overview of the main features of inflation targeting in Sweden and other countries applying such a framework.
6. The Taylor rule and assumptions applied are documented in Annex I.
7. See SOU (1996), ''Sverige och EMU'', *SOU 1996:158*.
8. The adjustments of the Government Debt Office comprise items which do not fundamentally affect the net wealth of the central government, the adjustments of the National Audit Office also incorporated in Table 10, comprise also temporary items such as the membership rebate obtained by Sweden during its first three years of EU membership.
9. See further details in 1997 *OECD Economic Survey of Sweden*, pp. 63-65.
10. Tax expenditures are calculated at the margin and adding them up, although not entirely meaningful, is done to assess their importance, in total and in different areas.
11. See Hussénius, J. and J. Selén (1994), ''Skatter och socialforsäkringar över livscykeln – en simuleringsmodel'', *ESO Report Ds 1994:135*. Björklund, A. and M. Palme (1997), ''Income redistribution within the life cycle versus between individuals: Empirical evidence using

Swedish panel data'', *Working Paper Series in Economics and Finance No. 197, Stockholm School of Economics* find the same effect for child and housing allowances, using a somewhat different methodology.

12. Tendencies in the retirement propensity and its sensitivity to economic factors and the administration of social insurance schemes are discussed in Palme, M. and I. Svensson (1997), ''Social Security, Occupational Pensions and Retirement in Sweden'', *Working Paper Series in Economics and Finance no. 184, Stockholm School of Economics*.

13. See OECD (1997), *Economic Survey of Sweden*, pp. 75-122.

14. The first expert report, delivered in late 1997, proposed establishing a new public body by January 1999 with a broader mandate than the present government-appointed mediator.

15. The reduction in the replacement rate from 90 to 80 per cent in 1993 was associated with a reduction of the length of unemployment spells by 8 per cent. This reflects basically lower reservation wages while the effect of more intensive job search was of lesser importance, see Harkman, A. *et al*, ''Arbetslöshetsersättningen och arbetsmarknadens funktionssätt'', *Report Ura 1997:1 from The Labour Market Board*.

16. By mid-1997 the take-up under the first of these schemes was 6 000 persons, while 9 500 persons had taken up the benefit.

17. A recent committee evaluation of the Working Time Act – yet to be assessed by the Government – has proposed that the legal text concerning agreements on working time should not specify at which level, be it at industrial or local level, they are to be finally authorised. At present, the central organisations have to give their assent to local agreements. It also proposes that the legal restriction of 200 hours per annum of overtime work should be lowered to 100 hours.

18. Temporary work contracts have been a significant source of job creation over the past few years. This was, however, also the case during the late-1980s and the higher level in the mid-1990s may reflect the changing pattern of employment growth with private services (*inter alia* information technology) now being the main source of new jobs.

19. See SOU (1996), ''Kompetens och kapital – om statligt stöd til företag'', *SOU 1996:69*.

20. See Murray, R. (1996), *Productivity Trends in the Public Sector in Sweden*, which reports an 11 per cent productivity decline over the period from 1980 to 1992 for the primary and secondary schools as a whole, while productivity in tertiary education is estimated to have increased by 31 per cent. For the education sector as a whole productivity would thus seem to have stagnated.

21. The cost increase in the 1980s reflected partly the expansion of special and second language teaching, and the recent reduction in costs reflects a (partial) reversion of this expansion. Documentation is provided in Wahlström, S. (1997), *Resurs- och resultatutvecklingen i skolsektorn efter avregleringen 1991 – en nationell beskrivning*, Skolverket.

22. See documentation and discussions in Skolverket (1996), *Varför kostar elever olika?* and Lundahl, C. (1997), *Resurs- och resultatutvecklingen i skolsektorn efter avregleringen 1991 – en översikt av utvärderingsresultat*, Skolverket.

23. The studies cited in Murray, R. *op. cit.* indicate that the teaching process is fairly independent of changes in class sizes between 15 to 30 pupils, it is only when they fall below or increase above this range that results are perceptibly affected. Class sizes are typically higher than the pupil to teacher ratios reported in Table 15.

24. The survey defined literacy in three dimensions to indicate the capacity of workers to independently acquire and upgrade the skills required for their jobs: how to extract and use information from standard texts (prose literacy), from other formats (document literacy) and how to transform textual information into numerical operations (quantitative literacy). See OECD/Statistics Canada (1995), *Literacy, economy and society*, and OECD/Human Resources Development Canada (1997), *Literacy Skills for the Knowledge Society*.

25. The difference in performance between the upper and lower quartile of pupils corresponds to a difference in schooling of three to four years in all subjects, and this difference increases through compulsory education.

26. See studies cited in Fägerlind, I. (1993), ''Utbildningen i Sverige och det mänskliga kapitalet''. *Annex 9 to SOU 1993:16*; and Taube, K. (1993), ''Reading comprehension among Swedish students: a comparative analysis of IEA studies from 1970 and 1991'', *Scandinavian Journal of Educational Research*, Vol. 37, No. 1. Developments over the 1990s are covered in Skolverket (1997), *En kunskapsmätning i grundskolan 1995*.

27. At the end of compulsory education, close to 6 per cent of pupils fails to achieve ''passable'' marks. However,from 1998 pupils will not be accepted in the standard national programmes unless they have at least ''pass'' in Swedish, English and mathematics.

28. Hansson, P. and M. Henrekson (1994), ''Catching up, Social Capability, Government Size and Economic Growth'', *Working Paper No. 123, Trade Union Institute for Economic Research* find that whereas public expenditure in the aggregate has a negative impact on economic growth, educational expenditures have for a sample of fourteen OECD countries had a positive effect on growth.

29. This concept is discussed in more detail in OECD (1995), *Education and employment*. While the compressed wage differentials in Sweden may understate the productivity effects of higher education, the estimated return to education may be upward biased by ascribing effects of natural talents (intelligence) to education. See Kjellström, C. (1997), ''Omitted Ability Bias and the Wage Premium of Schooling: New Swedish Evidence'', *Working Paper, Swedish Institute for Social Research, Stockholm University.* Ottersten, E.K., E. Mellander, E.M. Meyerson and J. Nilsson (1994), ''Pitfalls in the Measurement of the Return to Education'', *Working Paper No. 414, The Industrial Institute for Economic and Social Research* find that selection, omitted variable bias and measurement errors broadly cancel out for the years investigated (1984 and 1986).

30. See Ahlroth, S., A. Björklund and A. Forslund (1994), The Output of the Swedish Education Sector'', *Working Paper No. 43, The National Institute of Economic Research*. The analysis compares the market valuation of an additional year of education, given by the wage premia observed, with the annual input into the education system in terms of teachers' wages, equipment, depreciation of investment etc. as recorded in the national accounts. By necessity, the analysis applies only at the margin for providing education above the compulsory level.

31. On national accounts conventions, output in the government sector is measured by its inputs, thus implicitly assuming zero productivity growth in this sector.

32. See Fredriksson, P. (1997), ''Economic Incentives and the Demand for Higher Education'', *Scandinavian Journal of Economics* Vol. 99, No. 1 and Edin, P.-A. and B. Holmlund (1995), ''The Swedish Wage Structure: The Rise and Fall of Solidarity Wage Policy?'' in Freeman, R.B. and L.F. Katz (1995), *''Differences and Changes in Wage Structures''*.

33. Using more stylised assumptions with respect to wage relativities, but institutional data for 1994, Löfgren, A.K. (1997), *''För vem lönar sig utbildning?''* En delrapport från LO-projektet Ökad Sysselsättning, LO finds a wage premia of 3.9 per cent, somewhat below available results for 1991.

34. It has been estimated that the effect of increasing the starting age for a three-year university education from 19/20 years (the earliest possible) to 28 years (about the average at the time) would lower the already low returns in 1981 by 1 to 3 percentage points, depending on wage measure used. See Björklund, A. and C. Kjellström, *op. cit*. Löfgren, A.-K. (1997), *op. cit.* indicates that entering a tertiary education is profitable until an age of 37 years.

35. See *1997 OECD Economic Survey of Denmark*, for an overview and assessment of the Danish adult education system.

36. See the works cited in Zetterberg, J. (1996), *''Effekter av arbetsmarknadspolitik – en översikt av svensk empirisk forskning''*, Annex to SOU 1996:34 and the overall assessment in the *1997 OECD Economic Survey of Sweden*.

37. See Forslund, A. and A. Kreuger (1994),''An evaluation of the Swedish Active Labour Market Policy – New and Received Wisdom'', *SNS Occasional Paper* No 65.

38. In 1989/90, resources to labour market training, youth measures and subsidised employment were matched by administrative resources in the proportion 3:1. The proportion peaked at 8:1 in 1992/93 before falling to 5:1 in 1995/96.

39. The evidence is surveyed in a report from the Auditors of the Parliament (1997), ''Riksdagens revisorers förslag angående *Amu-gruppen AB'', 1996/97: RR8.*

40. The central government will finance about 105 000 places in adult education, 45 000 to 50 000 of these serve to carry on activities hitherto financed by central government, close to 60 000 to expand capacity by approximately 50 per cent. Out of the 105 000 places, 10 000 places are allocated to the Folk High schools and 3 600 places to the recently established programme for advanced (tertiary) vocational training, while the remainder is allocated to the municipalities. The adult education effort should also be seen in conjunction with the further increase in the permanent capacity in tertiary education, with 15 000 new places coming on stream in each of the years 1998-2000.

41. Displacement effects are to be avoided by the establishment of a baseline number of upper secondary places in *Komvux* corresponding to the average of municipally-financed places over the three years preceding the initiative. Only places above that baseline will receive central government subsidies.

42. See Paldanius, S. *et al.* (1997), ''Olika betydelser av kommunal vuxenutbildning för arbetslösa'', *Rapport nr. 1997:03, Institutionen för pedagogik, University of Gothenburg.* On the whole, *Komvux* has been far less extensively analysed and evaluated than labour market training.

43. Machin, S. *et al.* (1996), ''Technology and Changes in Skill Structure: Evidence from an International Panel of Industries'', *CEPR Discussion Paper* No. 1434 report a fairly small

elasticity of substitution between unskilled and skilled workers in Swedish manufacturing, indicating a need for rather large movements in relative wages to offset changes in relative demand for workers with different skills. On the whole, wage differentials by educational attainment have been stable in Sweden through the 1990s despite the large movements in relative unemployment levels.

44. To this end, unemployment benefits were not accessible for workers on temporary layoffs as in many other countries but were financed partly by the enterprise itself, partly by contributions from a separate fund which enterprises themselves had to contribute to.

45. See Regnér, H. (1997), *Training at the Job and Training for a New Job: Two Swedish studies*. Ph.D. Thesis, Swedish Institute for Social Research, University of Stockholm. A Nordic comparison is provided in Asplund, R. *et al.* (1994), ''Human Capital and Earnings in the Nordic countries'', in Asplund, R. (ed.) (1994), *Human Capital Creation in an Economic Perspective.*

46. See Eliasson, G. and E.K. Ottersten (1994), *Om forlängd skolgång*, The Industrial Institute for Economic and Social Research, which also interpret the concentration of training expenditures on high-wage employees as a hidden wage drift.

47. It has been argued that employees will prefer training to be general insofar as it provides them with ''observable'' qualifications that may justify wage rises. See studies cited in Ottersten, E.K. *et al.* (1996), ''Cost and productivity effects of firm financed training'', *Working Paper No 455 from The Industrial Institute for Economic and Social Research.*

48. See Ottersten, E.K. *et al., op. cit*. The study covers the period 1989-95, and there is a risk that the overall efficiency drive in Swedish manufacturing over that period may have been ascribed to changes in the quality of labour brought about by enterprise-financed training.

49. It may also be argued that with a very low difference in average pay between enterprises according to size, even compared with other Nordic countries, small and medium-sized enterprises in Sweden have in general been able to obtain workers of the same quality as larger enterprises. The extensive unionisation may also play a part in explaining low wage differentials, but this factor is not of great influence in those countries for which it has been quantified. See Albæk, K. *et al.* (1996), ''Employer Size-Wage Effects in the Nordic countries'', *Working paper 96-03 from Centre for Labour Market and Social Research*, Aarhus.

50. See OECD (1996), *Education at a glance – Analyses.*

51. An overview of government initiatives and policy perceptions in this area at the end of the 1980s is provided in OECD (1993), *Industrial training in Australia, Sweden and The United States*.

52. The different avenues chosen by other countries is discussed in Soskice, D. (1993), ''Social skills from mass higher education: Rethinking the company-based initial training paradigm'', *Oxford Review of Economic Policy, Vol. 9, No. 3.* An underlying theme is that some countries may have gone too far in dismantling the apprenticeship system, leaving them with no options but to concentrate human capital formation in the formal education system. The Swedish case is discussed in detail in Eliasson, G. and E.K. Ottersten (1994), *op. cit*.

53. More recently, The Denver Summit of the Eight listed an effort for small and medium-sized enterprises as a priority area to ensure growth and employment.

54. OECD (1995), *Economic Survey of Sweden*.

55. EFER (1996), *Europe's 500 Dynamic Entrepreneurs. The Job Creators*, Brussels. For an overview, see OECD (1994), *Employment Outlook* and ENSR (1995).

56. Davidsson P. (1995), ''SMEs and Job Creation in Sweden'', paper presented at the OECD High-Level Workshop ''SMEs: Employment, Innovation and Growth'', in Washington DC, United States. OECD (1997), *Economic Survey of Sweden.*

57. Blixt, L. (1997), ''Tillväxtföretag i Sverige – stora som små och från norr till söder'', Teknikpolitiska analyser, *Arbetsrapport*, NUTEK. This study includes only companies active during the whole period 1991/92 to 1994/95 with a turnover in the last year of at least SKr 25 million. Companies growing through mergers and acquisitions are disregarded.

58. These figures are, however, not corrected for sectoral influences.

59. See Albæk, K. *et al., op. cit*.

60. Of these, pension savings cannot be used for setting up an own enterprise, whereas housing capital may be used as collateral for loans.

61. Lindh, T. and H. Ohlsson (1996), ''Self-Employment and Windfall Gains: Evidence from the Swedish Lottery'', *The Economic Journal*, Vol. 106. Lindh and Ohlsson, using data from the early 1980s, study the effect of winning large sums in the national lottery on the propensity to start an own enterprise.

62. Lindh, T. and H. Ohlsson (1998), ''Self-Employment and Wealth Inequality'', *Review of Income and Wealth*, March 1998 (forthcoming).

63. It should be noted that government-run schemes which guarantee wage claims outstanding in case of bankruptcy also enter into this legislative sphere insofar as they may affect employees' surveillance of financial developments of their enterprises.

64. Industriförbundet (1996), ''Tillväxthinder för småföretag. Södra Norrland'', *Rapport No. 2, Nyindustrialiseringsprojektet*. Older surveys quoted in OECD (1995), *Economic Survey of Sweden* also concluded that lack of capital was not among the major obstacles to enterprise growth.

65. Swedish Government (1996), ''Vissa åtgärder för att halvera arbetslösheten till år 2000, ändrade anslag för budgetåret 1995/96, finansiering m.m.'', Regeringens proposition 1995/96:222 explicitly refers to these gaps.

66. Deiaco, E. (1992), ''New Views on Innovative Activity and Technological Performance. The Swedish Innovation Survey'', *STI Review*, No. 11, OECD.

67. NUTEK (1996), *Forsknings- och utvecklingsverksamhet inom forskningsnära och tekniker-intensiva småföretag 1993* and Nordic Council of Ministers (1995), ''Forskningsnära företag i Norden – forskning och utveckling 1993'', *TemaNord 1995:575.*

68. The possibility of transferring the ownership of enterprises to tax-exempt institutions continues to favour large enterprises over new and small enterprises, insofar as a larger share of the latters' earnings are necessary for the subsistence of the owner(s).

69. A closely held company is defined as a company owned by ''few'' persons (usually fewer than ten), and where persons employed by the company hold more than 70 per cent of the equity.

70. The formula is: invested capital plus the total wage cost minus salaries to active partners minus SKr 363 000.

71. The Swedish EPL imposes rules on: notice periods at dismissal; the reason for dismissal; trial employment; and the order in which employees may be dismissed and re-hired. The law stipulates that in the case of labour shedding the last person hired must be the first person laid off, and that in the case of re-hiring any person laid off within the last nine months must be the first person re-hired. Some easing has been put in place recently. Moreover, Sweden has one of the shortest legislated maximum trial periods in Europe.
72. See Swedish Employers' Confederation and Federation of Swedish Industries (1996), *Abolish government aid to companies in exchange for lower payroll taxes*.
73. OECD (1997) *Economic Survey of Sweden*.

Annex I

Supporting material to Chapter II

Table A1. **Inflation targeting regimes in OECD countries**

	Date	Variable[1]	Target range[2]	Target horizon	Caveats[3]	Operational target	Policy instrument	Published projections	Response to a breech of target
Australia	1993	CPI (underlying)	2-3 per cent on average	Over the business cycle	None	Not explicit	Overnight cash rate	No	Not explicit
Canada	February 1991	CPI (underlying)	1-3 per cent	End-1998	Explicit	Monetary conditions index	Overnight rate operating band	No	Not explicit
Finland	February 1993 to October 1996[4]	CPI (underlying)	2 per cent	1995 onwards	None	Not explicit	Tender rate	2 years ahead	Not explicit
New Zealand	March 1990	CPI (underlying)	0-3 per cent	August 1998[5]	Explicit	Monetary conditions index	Settlement cash balances	Up to 2-3 years ahead	Reserve Bank Governor can be dismissed
Sweden	**January 1993**	**CPI**	**2±1 per cent**	**1995 onwards**	**None**	**Not explicit**	**Repo rate**	**Up to 1-2 years ahead**	**Not explicit**
United Kingdom	October 1992	RPIX	2.5 per cent[6]	Up to 2 years ahead	None	Commercial bank's base rate	Money market dealings rate	Up to 2 years ahead	Letter from the Bank of England indicating appropriate response

1. The price indices that these variables refer to are as follows: Australia: consumer price index excluding mortgage interest and credit debt charges, goods and services subject to government regulation or influence and goods and services whose prices are seasonal; Canada: consumer price index excluding food, energy and temporary effects of indirect taxes; Finland: CPI excluding indirect taxes, subsidies and housing capital costs; New Zealand: CPI excluding significant changes resulting from changes to indirect taxes, government charges, terms of trade, direct effects from interest rates and natural disasters; United Kingdom: retail price index excluding mortgage interest payments.
2. The range refers to the current target. Previous intermediate targets have been as follows: Canada: 3 per cent, ±1 per cent between February 1991 and the end of 1992; 2.5 per cent, ±1 per cent from 1993 to mid-1994; and 2 per cent, ±1 per cent until end-1995; New Zealand: 3 to 5 per cent between March 1990 and end-1992; and 0-2 per cent between 1993 and end-1996; and United Kingdom: 2.5 per cent or less between October 1992 and June 1997.
3. Caveats for Canada include large supply shocks and the effects of indirect taxes; for New Zealand: significant changes in the terms of trade; changes in indirect taxes or the GST, natural disasters, a significant price level impact arising from changes to government levies, and movements in interest rates that causes a significant divergence between the change in the CPI and the change in CPI excluding the interest cost component.
4. Finland entered the Exchange Rate Mechanism (ERM) of the European Union on 14 October 1996.
5. The Policy Target Agreement which sets out the inflation targeting framework is an agreement between the Minister of Finance and the Governor of the Reserve Bank and expires at the end of the Governor's term of office, after which it is renegotiated.
6. If inflation deviates more than 1 percentage point on either side of the target, a letter from the Bank of England to the Chancellor of the Exchequer must be forthcoming.

Source: OECD.

The Taylor rule and assumptions applied

The Taylor rule* was developed to assess in a mechanistic way the short-term interest rates set by central banks, incorporating two considerations: controlling inflation as the overriding objective but seeking to stabilise output around its long-term trend. The rule is calculated as

$$r = r^* + ep + a(p - tp) + b(GDP - GDP^*),$$

where

r = the short rate implied by the rule

r^* = the neutral (long-run) short rate

ep = expected inflation

p = actual inflation

tp = the central bank inflation objective

GDP^* = potential output and

GDP = actual output

and a and b are parameters which indicate the strength of the monetary policy response when actual inflation deviates from the objective and actual output deviates from the potential.

The neutral rate is set to 4.0 per cent for Sweden, somewhat below the average over the 1986-96 period of 4.8 per cent; the inflation objective is 2.0 per cent annual increase in the official consumer price index from 1995 and has been assumed to be approached gradually from the instigation of inflation targeting in early 1993. Expected inflation is based on survey-based expectations of household and business. For larger economies, the weights a and b are normally set at 0.5. In the present case, two set of weights have been used, *i)* the standard ones used for other countries; and *ii)* a lower set at 0.25 which may be interpreted as less emphasis on short-term stabilisation of both prices and output, but also to reflect the absence in the Taylor rule of the direct price impact of interest rate changes through the exchange rate channel which is of particular importance in small open economies.

* See Taylor, J. (1993), *Macroeconomic Policy in a World Economy* and Taylor, J. (1996), ''How should monetary policy respond to shocks while maintaining long-run price stability – conceptual issues'' in *Achieving Price Stability*, The 1996 Federal Reserve Bank of Kansas Symposium.

Annex II

Calendar of main economic events

1996

April

The Government tables its Spring Bill implementing a system of expenditure ceilings for 27 major expenditure categories and covering the years 1997-99. Central government expenditures are set to fall from 41.0 per cent of GDP in 1997 to 37.8 per cent in 1999. On the projections presented, the budget deficit falls somewhat short of reaching 3 per cent of GDP in 1997 and balance by 1998, and offsetting measures are proposed. The permanent effect of the measures proposed is estimated to be around SKr 8 billion with additional temporary reinforcements of SKr 6 billion for 1997-98.

The proposal incorporates a programme to increase the capacity in adult and tertiary education, to be covered by lower spending on ALMPs and by higher energy taxes, and an agreement with municipalities (later converted into legal requirements) which obliges them to keep tax levels unchanged during 1997-98 while block grants from the government are kept unchanged in nominal terms.

The repo rate is lowered to 6.90 per cent at the end of the month. The central bank's deposit and lending rates are lowered to 6.00 and 7.50 per cent respectively.

May

The repo rate is lowered to 6.50 per cent at the end of the month.

June

The government puts forward an Employment Bill, aiming at bringing down open unemployment to 4 per cent by 2000. The length of spells on unemployment insurance is to be limited, the replacement rates in social insurance schemes are to be increased to 80 per cent and to be financed with less generosity in other aspects, and employment protection legislation is to be modified. Payroll taxes for small enterprises are to be lowered, and they also benefit from a targeted relief from the double taxation of dividends. The obligation for employers to pay wages during employees' sickness is extended from the first fourteen to the first 28 days of an absence.

The repo rate is lowered to 6.10 per cent at the end of the month. The central bank's deposit and lending rates are lowered to 5.25 and 6.75 per cent respectively.

July

The elements of the employment package are approved by Parliament. They include an additional 130 000 places in adult and tertiary education over the period 1997-2001 and work-for-benefit for long-term unemployed persons over 55 years of age.

The repo rate is lowered to 5.55 per cent.

August

The repo rate is lowered to 5.25 per cent. The central bank's deposit and lending rates are lowered to 4.75 and 6.25 per cent respectively.

September

The Finance Bill for 1997, embodying the detailed appropriations following from the adoption of the expenditure ceilings, is presented together with the third update of the Convergence Programme. No additional savings are required to comply with the objective of budget balance in 1998.

The repo rate is lowered to 5.05 per cent.

October

The repo rate is lowered to 4.60 per cent. The central bank's deposit and lending rates are lowered to 4.25 and 5.75 per cent respectively.

November

The repo rate is lowered to 4.30 per cent.

The proposal for modifications of Unemployment Insurance is withdrawn.

December

The repo rate is lowered to 4.10 per cent. The central bank's deposit and lending rates are lowered to 3.75 per cent and 5.25 per cent respectively.

1997

January

The revised Employment Protection legislation enters into force, embodying modifications in *i)* the criteria determining the length of notice periods; *ii)* enterprises' rehiring obligation *vis-à-vis* laid-off workers; *iii)* a wider scope for fixed-term contracts; and *iv)* a strengthened position for part-time workers and workers on replacement contracts.

February

The Government reaches an agreement with two opposition parties on a programme to dismantle Sweden's nuclear power generators, beginning with the decommissioning of two privately-owned reactors in the south of Sweden in 1998 and 2001.

March

The Government submits a new proposal for the Unemployment Insurance Scheme, incorporating *i)* a higher replacement rate; *ii)* stricter eligibility rules; *iii)* higher refunds from the UI funds to the central government; and *iv)* an integration of the cash benefit (KAS) for those not members of UI funds into the overall UI scheme. These modifications will have been entered into force by 1 January 1998.

April

The Spring Bill proposes a long-term goal of achieving a surplus on the government balance of 2 per cent over the economic cycle, to be reached via surpluses of 0.5 per cent in 1999, 1.5 per cent in 2000 and 2.0 per cent in 2001. The overall ceilings for central government expenditures already adopted for 1998 and 1999 are retained, while the three-year horizon for the ceilings is extended to 2000. Expenditures are to fall from 39.1 per cent of GDP in 1998 to 36.7 per cent in 2000. Fiscal consolidation is progressing faster than expected, creating scope for measures to lower unemployment, summarised in a five-point programme amounting to SKr 16 billion in 1998 and reaching SKr 23 billion in 2000. The programme increases spending on: *i)* education, health and welfare services by increasing transfers to municipalities; *ii)* tertiary and adult education by bringing on stream in total 45 000 new places in tertiary education 1998-2000 and 140 000 new places in adult education; *iii)* environmental infrastructure projects; *iv)* the framework for small and medium-sized enterprises; and *v)* active labour market policies.

May

The Government and four centre-right parties in *Riksdagen* (the Parliament) reach an accord on the revision of the Central Bank Act, enshrining the commitment to price stability and the independence of the central bank.

June

The Government announces that it will not recommend Swedish participation in the European Economic and Monetary Union (EMU) at its inception 1 January 1999.

July

A new telecommunications act lays down the principle that the Government will not use its position as the sole owner to instruct the major telecommunications company,

Telia, but that all economic and social considerations will be embodied in provisions in the act. Price regulation for the main telephony products as well as clearer rules for interconnectivity are introduced.

September

The Finance Bill for 1998 retains the overall fiscal policy framework of the Spring Bill. Projection of higher economic growth is utilised to expand the five-point programme by SKr 6½ billion in 1998, increasing to SKr 15½ billion in 2000. The scope for older workers to go on early retirement is increased, and the wage obligation of employers during employees' sickness is reduced to fourteen days.

December

The repo rate is increased to 4.35 per cent.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
A. Percentage change from previous year at constant 1991 prices										
Private consumption	4.6	2.4	1.2	–0.4	0.9	–1.4	–3.1	1.8	0.8	1.3
Gross fixed capital formation	8.2	6.6	11.3	1.3	–8.9	–10.8	–17.2	2.0	12.4	3.7
Residential	8.8	8.4	4.8	7.2	–2.4	–7.3	–32.8	–35.9	–28.0	14.7
Business	9.3	5.8	15.0	–1.1	–15.2	–15.5	–13.0	18.6	29.0	4.6
Public sector	0.9	6.2	10.5	–2.0	8.3	0.6	6.8	11.1	–5.7	–8.0
GDP	3.1	2.3	2.4	1.4	–1.1	–1.4	–2.2	3.3	3.9	1.3
GDP price deflator	4.8	6.5	8.0	8.8	7.6	1.0	2.6	2.5	3.7	1.0
Industrial production	2.8	2.9	2.9	0.2	–5.2	–1.1	0.5	11.9	9.1	2.1
Employment (persons)	1.0	1.1	1.8	1.0	–2.0	–4.5	–5.5	–1.0	1.6	–0.9
Compensation of employees (current prices)	8.3	9.2	12.8	12.5	5.1	–1.3	–1.8	3.5	4.3	6.1
Productivity (GDP/employment)	2.2	1.1	0.6	0.4	0.9	3.3	3.5	4.4	2.3	2.2
Unit labour costs (compensation/GDP) (current prices)	5.0	6.8	10.2	11.0	6.3	0.2	0.4	0.1	0.3	4.8
B. Percentage ratios										
Gross fixed capital formation as % GDP at constant prices	18.6	19.3	21.0	21.0	19.4	17.5	14.8	14.6	15.8	16.2
Stockbuilding as % GDP at constant prices	–1.0	–0.8	–0.7	–0.3	–1.5	–0.4	–0.9	0.4	0.9	–0.2
Foreign trade as % GDP at constant prices	2.4	1.7	0.6	0.8	1.5	1.9	4.9	5.7	6.9	8.0
Compensation of employees as % GDP at current prices	59.9	60.1	61.3	62.5	61.8	61.2	59.9	58.6	56.7	58.8
Direct taxes as % of household income	21.7	22.2	22.7	21.4	18.6	18.1	17.9	18.2	18.9	19.4
Household saving as % of disposable income	–2.8	–4.8	–4.9	–0.6	3.1	7.7	8.3	8.0	6.5	4.7
Unemployment as % of civilian labour force	2.1	1.7	1.5	1.6	3.0	5.3	8.2	7.9	7.7	8.1
C. Other indicator										
Current balance ($ billion)	0.0	–0.6	–3.1	–6.2	–4.7	–8.7	–4.1	0.7	4.9	5.9

Source: Statistics Sweden, *National Accounts*; OECD.

Table B. **Supply and use of resources**

SKr billion, current prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Consumers' expenditure on goods and services	537.9	584.4	632.7	692.7	771.3	777.3	796.4	834.5	863.5	885.1
General government expenditure on goods and services	273.4	290.3	322.6	372.1	394.4	402.5	406.1	416.2	425.7	443.1
Gross fixed capital formation	197.9	225.1	271.0	292.5	280.4	244.6	205.7	209.2	240.4	249.6
Change in stocks	–4.8	–3.6	–0.5	–2.5	–21.2	–6.7	–13.7	7.2	14.2	–3.3
Domestic expenditure	1 004.5	1 096.2	1 225.9	1 354.8	1 424.9	1 417.8	1 394.4	1 467.1	1 543.8	1 574.5
Exports of goods and services (non-factor)	332.4	359.7	394.5	406.8	404.2	401.6	473.3	557.8	675.4	675.1
Imports of goods and services (non-factor)	313.3	341.4	387.8	401.8	381.8	377.6	421.5	493.7	569.2	561.4
Gross domestic product in purchasers' values	1 023.6	1 114.5	1 232.6	1 359.9	1 447.3	1 441.7	1 446.2	1 531.1	1 649.9	1 688.2
Indirect taxes	176.9	182.8	199.8	233.4	256.8	235.1	226.0	228.8	237.1	279.3
Subsidies	48.4	48.9	55.7	63.3	72.7	78.6	84.9	80.8	83.7	79.4
Gross domestic product at factor cost	895.1	980.7	1 088.5	1 189.8	1 263.2	1 285.3	1 305.2	1 383.1	1 496.5	1 488.3
Depreciation and other operating provisions	128.1	142.1	160.2	178.9	191.6	193.3	203.5	205.6	207.6	207.6
Net domestic product at factor cost	767.0	838.5	928.4	1 010.9	1 071.7	1 092.0	1 101.7	1 177.6	1 288.9	1 280.7
Memorandum item:										
Gross domestic product in purchasers' values, Maastricht definition	..	..	..	..	..	..	..	..	1 657.0	1 696.7

Source: Statistics Sweden, *National Accounts.*

Table C. **Supply and use of resources**

SKr billion, 1991 prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
A. Gross domestic product by type of expenditure										
Consumers' expenditure on goods and services	740.5	758.5	767.3	764.3	771.3	760.5	737.2	750.4	756.2	766.0
General government expenditure on goods and services	364.1	366.2	374.0	383.8	394.4	394.2	395.0	392.4	388.7	388.0
Gross fixed capital formation	255.9	272.9	303.8	307.7	280.4	250.0	207.0	211.0	237.1	245.9
Change in stocks	–5.8	–4.8	–1.9	–0.7	–21.2	–5.6	–13.0	6.4	13.0	–3.4
Domestic expenditure	1 354.7	1 392.8	1 443.3	1 455.1	1 424.9	1 399.2	1 326.2	1 360.1	1 395.0	1 396.5
Exports of goods and services (non-factor)	385.3	394.9	407.1	413.8	404.2	413.7	445.3	507.4	572.9	607.8
Imports of goods and services (non-factor)	352.5	371.3	398.8	401.5	381.8	386.1	376.4	425.9	469.5	486.9
Gross domestic product in purchasers' values[1]	1 379.4	1 410.5	1 444.0	1 463.7	1 447.3	1 426.8	1 395.1	1 441.6	1 498.4	1 517.4
B. Gross domestic product by kind of activity										
Primary sector	30.0	29.4	32.6	34.0	31.2	30.3	30.8	31.4	33.2	32.2
Mining, manufacturing, gas, etc.	312.5	318.7	320.2	318.7	301.5	292.0	294.8	330.5	368.0	374.9
Construction	90.0	91.4	98.4	98.7	98.5	94.1	84.8	81.2	80.8	81.4
Services	810.5	831.6	852.4	872.4	876.3	874.7	854.8	878.6	901.6	915.7
Adjustment[2]	144.1	146.6	147.7	144.4	139.5	132.1	123.9	117.7	113.5	112.8
Statistical discrepancy	–7.6	–7.3	–7.5	–4.6	0.2	3.6	5.9	2.2	1.4	0.4
Gross domestic product	1 379.4	1 410.5	1 444.0	1 463.7	1 447.3	1 426.8	1 395.1	1 441.6	1 498.4	1 517.4

1. On the basis of GDP by production.
2. Import duties, value added tax and other taxes net of subsidies, on sales and production of commodities less imputed bank service charge.

Source: Statistics Sweden, *National Accounts.*

Table D. **General government revenue and expenditure**[1]

SKr million, current prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Current revenue	635 099	686 050	785 348	860 747	870 804	857 210	849 736	887 179	950 182	1 040 729
Direct taxes	243 679	270 319	310 960	318 612	287 962	294 921	301 012	324 697	355 634	377 122
Indirect taxes	176 860	182 766	199 786	233 362	256 807	235 067	225 955	228 755	237 106	279 292
Social security contributions	140 836	156 757	186 883	211 889	223 768	213 145	207 560	220 701	242 825	268 169
Other	73 724	76 208	87 719	96 884	102 267	114 077	115 209	113 026	114 617	116 146
Current expenditure	600 358	640 887	710 565	797 372	876 511	939 213	985 485	1 023 272	1 058 659	1 069 886
Consumption	273 408	290 266	322 630	372 130	394 394	402 508	406 071	416 167	425 668	443 064
Property income payable	66 747	63 175	67 377	69 293	75 896	79 903	90 959	105 443	117 293	121 339
Subsidies	48 395	48 919	55 719	63 268	72 710	78 634	84 948	80 781	83 685	79 356
Current transfers	211 808	238 527	264 839	292 681	333 511	378 168	403 507	420 881	432 013	426 127
Net saving	34 741	45 163	74 783	63 375	–5 707	–82 003	–135 749	–136 093	–108 477	–29 157
Capital formation	26 321	26 195	30 277	32 787	32 683	39 144	15 481	45 706	47 659	34 097
Capital transfers net	16 284	115	–2 217	0	–3 337	–15 831	–51 828	–2 326	45	36
Consumption of fixed capital	18 734	19 988	24 054	26 376	25 404	24 993	25 566	25 977	27 422	26 735
Net lending	43 438	39 071	66 343	56 964	–16 323	–111 985	–177 492	–158 148	–128 669	–36 483
Per cent of GDP	4.2	3.5	5.4	4.2	–1.1	–7.8	–12.3	–10.3	–7.8	–2.2
Memorandum items:										
Net lending, Maastricht definition	..	..	..	..	..	..	..	..	–115 714	–60 644
Per cent of GDP, Maastricht definition	..	..	..	..	..	..	..	..	–7.0	–3.6

1. According to OECD definition.

Source: Statistics Sweden, *National Accounts.*

Table E. **Balance of payments**

SKr billion

		1989	1990	1991	1992	1993	1994	1995	1996
A.	Trade balance	15.0	14.8	30.6	34.2	53.1	67.1	108.5	118.6
	Exports of goods	329.0	336.0	328.8	322.2	384.2	463.1	571.2	569.8
	Imports of goods	314.0	321.2	298.2	288.0	331.1	396.1	462.7	451.2
B.	Services, net	–8.7	–14.0	–8.2	–11.6	–0.4	–1.0	–3.6	–5.1
	Transportation	11.3	11.2	12.1	9.4	10.0	10.2	9.1	9.2
	Travel	–16.3	–20.0	–21.7	–22.8	–14.2	–15.8	–14.1	–17.2
	Other services	–3.8	–5.2	1.4	1.7	3.9	4.6	1.3	2.9
C.	**Trade in goods and services (A + B)**	6.3	0.8	22.4	22.6	52.7	66.1	104.9	113.5
D.	Wages and salaries	–1.0	–1.0	–1.3	–1.1	–1.1	–0.8	–1.2	–1.7
E.	Investment income	–14.0	–25.4	–37.5	–57.3	–67.0	–45.1	–45.0	–48.6
F.	Unrequited transfers	–13.2	–13.5	–12.4	–15.2	–13.9	–14.3	–21.0	–17.4
G.	**Current account (C + D + E + F)**	–21.9	–39.0	–28.8	–50.9	–29.2	5.9	37.6	45.8
H.	Direct investments	–54.7	–75.6	–4.2	–2.6	19.4	–2.7	23.1	–3.1
I.	Trade in shares	–29.8	–18.7	–3.0	10.3	32.2	33.3	–50.4	–23.2
J.	Loans/bonds in foreign currency	113.9	114.6	–26.7	13.9	–36.2	–21.2	5.0	–20.0
	Government and the Central bank	–13.0	–14.0	–18.9	155.0	69.1	25.9	54.8	9.0
	Other sectors	127.0	128.6	–7.8	–141.1	–105.3	–47.1	–49.9	–29.0
K.	Loans/bonds in SEK	36.8	64.5	64.7	27.2	36.5	1.2	39.7	–7.3
	Government and the Central bank	11.0	23.3	42.3	20.4	5.1	3.4	52.2	8.5
	Other sectors	25.9	41.2	22.3	6.7	31.5	–2.2	–12.5	–15.7
L.	Financial derivatives	0.0	0.0	0.0	0.0	0.0	0.0	–6.0	10.0
M.	Other capital transactions	–2.7	26.6	–36.9	10.0	29.7	35.2	–43.9	–31.0
N.	Official reserves	–9.3	–45.1	4.4	14.0	–75.5	–18.1	9.9	41.9
O.	Net errors and omissions	–32.3	–27.3	30.5	–21.9	23.1	–33.7	–14.7	–13.1
P.	**Financial account (H + I + J + K + L + M + N + O)**	21.9	39.0	28.8	50.9	29.2	–5.9	–37.6	–45.8

Source: Sveriges Riksbank.

Table F. **Labour**

	Labour force survey[1]					National labour market board					Participants in Labour Market Programmes[2, 3]		
	Labour force			Employed	Unemployed	Total	Insured unemployed			Jobs vacant[4]	Subsided	Training	Protected employment[5]
	Total	Men	Women				Total	Mining, manufacturing	Building construction				
	1 000 persons						Per cent of insured			Thousand	1 000 persons		
1987	4 410	2 292	2 117	4 316	94	78.0	2.3	2.9	3.8	46.1	22	37	76
1988	4 452	2 312	2 141	4 375	77	61.1	1.7	2.1	2.7	52.1	19	47	79
1989	4 508	2 346	2 163	4 442	66	56.3	1.6	2.1	2.2	50.0	13	42	81
1990	4 560	2 372	2 188	4 485	75	66.4	2.2	2.6	2.5	41.5	10	39	81
1991	4 530	2 356	2 174	4 396	134	114.8	3.2	2.8	7.1	18.6	16	59	80
1992	4 442	2 309	2 134	4 209	233	214.5	5.9	9.7	17.6	10.6	29	86	77
1993	4 320	2 245	2 076	3 964	356	326.1	8.7	14.1	24.5	8.6	23	53	76
1994	4 268	2 219	2 049	3 928	340	331.7	8.8	12.9	25.3	11.8	42	60	80
1995	4 320	2 251	2 069	3 986	334	328.7	8.7	11.3	24.0	15.0	35	55	85
1996	4 310	2 250	2 060	3 963	347	343.6	9.0	11.1	23.7	14.0	21	46	80

1. The employment concept covers all gainful employment, independently of the number of hours worked. Figures refer to persons aged between 16 and 64.
2. Averages of monthly observations.
3. The figure have been recalculated to conform with a new definition introduced in 1987.
4. From April 1989 only the vacancies with more than 10 days of duration are shown.
5. Employed in active work, sheltered workshops and semi-sheltered work.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics, Statistical Abstract of Sweden;* OECD, *Main Economic Indicators, Labour Force Statistics.*

Table G. **Production by sector and employment**

Mining and manufacturing

	Production, 1990 = 100									Employment[1] 1990 = 100
	Mining, manufacturing and electricity total	Mining and quarrying	Manufacturing							
			Total	Food, beverages, tobacco	Textiles, clothing, footwear	Wood and wood products	Chemicals, etc.	Basic metals	Metal products	
	ISIC 2-4	21-29	31-39	31	32	33	35	37	38	Hours worked
1987	94	108	94	96	115	92	98	101	89	103
1988	95	103	95	97	106	93	100	109	91	103
1989	99	97	99	100	100	98	100	110	97	102
1990	100	100	100	100	100	100	100	100	100	100
1991	95	104	95	97	84	93	102	92	93	90
1992	93	99	93	98	78	82	110	96	90	81
1993	93	94	93	100	70	86	119	105	86	76
1994	104	98	104	105	76	94	115	124	105	81
1995	113	107	114	108	85	104	117	125	125	87
1996	116	107	116	113	83	101	124	128	129	88

1. Monthly hours worked in mining and manufacturing.

Source: OECD, *Indicators of Industrial Activity; Main Economics Indicators.*

Table H. **Foreign trade, total and by area**[1]

$ million, monthly rates

	Total imports cif	Total exports fob	Imports by area						Exports by area					
			OECD			Non OECD			OECD			Non OECD		
			Total	OECD Europe		CEEC	OPEC	Other	Total	OECD Europe		CEEC	OPEC	Other
				EU[2]	Other					EU[2]	Other			
1987	3 392	3 710	3 068.7	2 251.4	313.5	143.0	41.2	139.6	3 291.5	2 179.9	528.5	64.7	77.0	276.9
1988	3 809	4 142	3 450.9	2 482.1	353.1	138.2	39.0	180.4	3 687.6	2 506.9	540.3	64.5	99.1	291.2
1989	4 090	4 300	3 690.9	2 606.0	410.5	148.2	45.2	205.7	3 833.3	2 668.8	506.9	79.6	87.8	299.3
1990	4 569	4 792	4 138.1	2 904.9	506.7	141.9	56.2	233.0	4 288.4	2 994.0	580.5	88.4	106.9	308.3
1991	4 164	4 601	3 744.5	2 639.1	450.3	109.7	74.6	235.7	4 077.5	2 863.7	567.4	85.9	151.2	286.5
1992	4 169	4 676	3 699.7	2 629.4	423.9	132.0	77.5	260.2	4 148.0	2 922.8	566.8	100.0	117.8	310.6
1993	3 567	4 155	3 153.5	2 235.3	354.2	68.3	70.4	275.1	3 587.0	2 460.4	515.1	61.9	105.1	400.6
1994	4 329	5 112	3 785.9	2 715.7	418.8	117.9	72.3	353.4	4 415.3	3 032.0	634.7	89.1	93.9	513.7
1995	5 138	6 454	4 652.4	3 554.6	585.8	121.4	62.5	301.5	5 567.0	3 780.6	816.8	129.1	104.7	653.4
1996	5 348	6 920	4 844.7	3 697.1	629.9	126.2	64.4	312.2	5 877.4	3 888.7	931.4	147.1	133.1	762.0

1. Break in data from 1994.
2. From 1993 EU-15.

Source: OECD, *Foreign Trade Statistics, Series A.*

Table I. **Prices and wages**

	Consumer prices, 1980 = 100				Net price index[2] 1980 = 100		Producer prices manufactured products 1990 = 100	Hourly earnings, Kronor[3] Manufacturing		
	Total		Food	Rent, fuel and electricity						
	Index	Annual growth rate[1]			Index	Annual growth rate[1]		Total	Males	Females
1987	167.0	4.2	191.0	159.0	159.8	2.9	84	56.80	57.71	51.02
1988	177.0	5.8	202.0	170.0	170.6	6.7	89	61.33	62.30	55.62
1989	188.0	6.4	213.0	183.0	181.2	6.4	96	67.40	68.42	61.18
1990	208.0	10.4	229.0	211.0	193.5	6.6	100	73.70	75.48	67.01
1991	227.2	9.7	239.1	249.9	204.2	5.5	102	76.40	82.21	69.58
1992	232.3	2.6	226.8	268.6	212.9	4.3	101	80.67	82.73	73.40
1993	243.2	4.7	228.4	283.9	222.1	4.3	106	82.32	84.21	75.11
1994	248.5	2.4	232.3	289.0	225.1	1.7	111	85.32	87.20	78.02
1995	254.8	2.8	235.5	296.4	228.6	1.5	122	89.17	91.09	81.26
1996	256.0	0.8	219.2	301.9	228.5	–0.2	119	96.00	98.16	87.57

1. Based on 12-month growth rates which are adjusted for the annual rebasing of the Consumer Price Index in December.
2. Consumer prices excluding indirect taxes.
3. Including direct wages for normal working hours, basic pay and supplements for overtime and shiftwork as well as supplements for staggered and inconvenient working hours.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics.*

Table J. **Money and credit**

	Interest rates							Central Bank		All banks	
				Bond yields				Holding of			
	Official discount rate[1]	Day-to-day money market rate	Treasury Bill 3 months	Government 5-year	Government 10-year[2]	Mortgage bonds	Industrial bonds	Foreign exchange reserves incl. gold	Domestic assets	Holding of domestic securities	*of which:* Government bonds
	Per cent							SKr million			
								End of period			
1987	7.50	9.20	9.39	11.44	11.68	11.99	12.21	50 288	95 697	149 344	44 491
1988	8.18	10.08	10.10	11.19	11.35	11.45	11.76	53 741	92 910	134 149	32 742
1989	9.25	11.57	11.50	11.30	11.18	11.64	12.16	60 704	58 461	154 533	41 149
1990	11.12	13.38	13.63	13.55	13.16	14.23	15.05	103 786	37 780	166 053	40 211
1991	9.39	11.76	11.56	10.90	10.69	11.84	12.47	99 751	131 721	152 755	25 682
1992	8.85	16.76	12.91	10.41	10.02	11.83	12.71	163 509	145 390	110 228	18 935
1993	6.78	9.05	8.33	7.65	8.51	8.95	8.97	179 651	98 397	191 984	42 005
1994	5.37	7.85	7.41	9.09	9.72	9.87	9.68	181 189	86 937	247 728	46 212
1995	7.13	8.55	8.75	9.87	10.24	10.27	10.51	171 320	71 025	246 186	54 066
1996	4.88	6.29	5.79	7.25	8.03	7.81	7.23	140 241	59 045	275 258	52 858

1. Annual average.
2. Except 1994: 9-year government bonds.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics;* Sveriges Riksbank, *Statistical Yearbook*; OECD, *Financial Statistics Monthly*.

Table J. **Money and credit** *(cont.)*

	Liquidity			Lending from financial institutions[2]			Bond, debentures, etc.[3]				Share issues (excluding bonus issues)
	Money supply (M3)	*of which:* Bank deposits	Extended M3[1]	Total	*of which:* Banks	*of which:* Mortgage institutes	*of which:* Total	*of which:* State	*of which:* Housing sector	*of which:* Business enterprises	
	End of year, SKr million			End of year, SKr billion			End of year, SKr million				SKr million
1987	500 500	..	562 780	1 193	480	539	56 640	–672	45 339	2 252	13 495
1988	519 650	..	610 471	1 454	630	639	86 512	–19 034	72 711	4 041	10 418
1989	571 681	503 867	664 315	1 763	791	768	77 343	–36 747	89 290	–6 303	14 498
1990	636 462	567 187	766 955	2 037	922	911	46 389	–32 910	71 733	–11 611	18 186
1991	661 787	591 955	854 613	2 110	909	1 026	83 545	13 121	74 383	3 240	21 261
1992	682 846	611 200	878 900	2 136	924	1 041	185 636	103 856	81 163	594	37 226
1993	710 062	636 846	820 696	2 007	749	1 088	296 261	196 036	77 349	4 588	14 980
1994	712 330	638 819	846 312	1 930	705	1 074	40 564	102 355	–39 171	–9 681	20 199
1995	731 814	655 223	847 224	1 944	701	1 088	126 933	212 294	–52 517	–16 790	51 426
1996	817 279	731 715	940 090	1 995	727	1 122	29 779	35 883	–5 710	–8 007	53 289

1. M3 + national savings accounts + treasury bills.
2. Amounts utilised.
3. Net issued.

Source: Statistics Sweden, *Monthly Digest of Swedish Statistics;* Sveriges Riksbank, Statistical Yearbook.

Table K. **Production structure and performance indicators**

A. Production structure

	Share of value added at factor cost[1]						Employment share (per cent of total)					
	1963	1970	1980	1990	1993	1996	1963	1970	1980	1990	1993	1996[2]
Tradeables												
Agriculture	8.0	5.7	3.8	3.7	3.5	2.9	16.3	10.8	7.8	5.6	5.9	5.3
Mining and quarrying	1.6	1.3	0.7	0.4	0.4	0.4	0.7	0.6	0.5	0.3	0.3	0.4
Manufacturing	35.4	34.9	29.8	28.1	27.1	29.8	34.4	34.2	35.0	30.8	28.1	28.7
of which:												
Food, beverages and tobacco	5.7	3.8	3.5	2.9	3.0	..	3.2	3.0	3.0	2.8	2.8	..
Textile, wearing apparels and leather industries	3.1	2.1	1.1	0.6	0.5	..	4.6	3.4	1.7	1.0	0.7	..
Wood and wood products, including furniture	2.5	2.9	2.4	2.2	1.7	..	3.0	3.1	2.9	2.3	1.9	..
Paper and paper products, printing and publishing	4.0	4.8	4.8	4.3	4.5	..	4.3	4.1	4.8	4.3	4.1	..
Chemicals and chemicals petroleum, coal, rubber and plastic products	2.6	2.9	3.3	3.3	3.9	..	1.9	2.3	2.6	2.3	2.2	..
Non-metallic mineral products except products of petroleum and coal	1.6	1.5	1.1	0.9	0.6	..	1.6	1.4	1.0	0.9	0.7	..
Basic metal industries	2.1	2.8	1.5	1.3	1.5	..	2.3	2.2	2.2	1.4	1.2	..
Fabricated metal products, machinery and equipment	13.4	13.7	12.1	12.5	11.3	..	13.1	13.8	15.6	14.6	12.9	..
Other manufacturing industries	0.5	0.4	0.1	0.2	0.2	..	0.4	0.8	1.2	1.3	1.5	..
Non-tradeables												
Electricity, gas and water	3.0	2.5	3.0	3.8	3.9	3.7	0.9	0.9	1.1	1.1	1.2	1.1
Construction	11.7	11.6	10.4	9.9	8.9	7.4	10.7	12.2	10.2	9.9	8.8	8.0
Wholesale and retail trade, restaurants and hotels	14.3	15.2	14.5	14.6	14.7	15.3	17.8	19.1	19.8	21.1	21.7	22.2
Transport, storage and communication	8.1	8.7	7.2	8.5	8.4	8.4	8.2	8.6	9.4	9.9	10.4	10.0
Finance, insurance, real estate and business services	13.3	15.3	25.2	26.0	27.8	29.1	3.9	5.7	8.1	12.5	13.4	15.7
Community, social and personal services	4.6	4.7	5.5	5.0	5.3	3.0	7.0	7.8	8.1	8.8	10.3	8.6
Memorandum items:												
Business GDP	86.1	81.5	74.3	76.2	76.1	78.8	82.9	77.5	68.2	67.4	66.2	67.7
Public sector	12.9	17.6	24.3	22.6	22.6	19.9	14.3	20.6	30.7	31.6	32.6	31.2
Other producers	1.0	0.9	1.4	1.2	1.3	1.3	2.8	1.9	1.2	1.1	1.2	1.2

1. At 1990 prices until 1993. At 1991 prices for 1996.
2. Break in the series.

Table K. **Production structure and performance indicators** *(cont.)*

B. Sector performance – Constant prices[1]

	Productivity growth (Sector GDP/sector employment)				Investment As a share of total business investment					
	1963-69	1970-79	1980-89	1991-96[2]	1963	1970	1980	1990	1993	1996
Tradeables										
Agriculture	7.9	2.8	5.5	3.0	4.2	3.6	4.9	3.2	3.2	3.3
Mining and quarrying	10.6	–0.3	2.4	2.7	1.3	0.9	1.0	0.6	0.8	0.9
Manufacturing	6.3	2.1	3.1	7.5	24.4	21.0	19.6	18.3	18.3	29.8
of which:										
Food, beverages and tobacco	3.6	1.2	1.6	..	..	2.3	2.2	1.3	2.3	..
Textile, wearing apparels and leather industries	7.5	4.5	1.8	..	..	0.6	0.6	0.3	0.2	..
Wood and wood products, including furniture	6.2	1.9	4.2	..	..	1.2	1.6	1.5	1.0	..
Paper and paper products, printing and publishing	5.6	2.3	2.5	..	..	4.5	3.7	4.1	3.6	..
Chemicals and chemicals petroleum, coal, rubber and plastic products	7.1	2.4	3.3	..	..	2.4	2.3	2.2	2.6	..
Non-metallic mineral products except products of petroleum and coal	7.0	2.4	2.8	..	..	0.9	0.7	0.6	0.3	..
Basic metal industries	7.4	1.9	5.7	..	..	2.7	2.2	1.0	0.8	..
Fabricated metal products, machinery and equipment	6.5	1.6	3.4	..	..	6.3	6.2	7.2	7.6	..
Other manufacturing industries	8.5	2.1	10.1	..	..	0.1	0.1	0.1	0.1	..
Non-tradeables										
Electricity, gas and water	6.9	5.1	5.2	1.7	9.5	9.9	9.6	5.2	6.5	5.5
Construction	1.5	3.6	2.4	2.7	3.4	2.1	2.3	3.1	1.9	1.8
Wholesale and retail trade, restaurants and hotels	2.8	1.7	1.9	3.7	7.5	5.8	6.5	9.2	8.7	8.5
Transport, storage and communication	4.2	3.2	2.8	3.3	12.5	10.3	11.3	12.4	14.6	16.5
Finance, insurance, real estate business services	–2.4	0.3	–1.1	2.1	35.9	43.5	42.9	45.3	41.1	30.5
Community, social and personal services	0.2	3.0	1.0	0.5	1.4	1.3	1.9	2.6	5.0	3.2
					As a share of total investment					
Business sector	4.5	2.5	2.1	4.1	83.5	80.1	83.8	88.7	80.2	86.0
Public sector	–0.9	–1.2	0.5	1.0	16.5	19.9	16.2	11.3	19.8	14.0
Other producers	1.2	2.8	2.7	2.2	..	..	..	..	..	..

1. At 1990 prices until 1993. At 1991 prices for 1996.
2. At 1991 prices.

Table K. **Production structure and performance indicators** *(cont.)*

C. Other indicators

	Size distribution of manufacturing enterprises in 1994				
	Under 10 employed	10-49 employed	50-99 employed	100-499 employed	500 employed and plus
Number of enterprises (per cent of total)	87.6	9.7	1.3	1.1	0.4
Number of employed (per cent of total)	7.5	11.6	5.6	14.0	61.4
	1989	1991	1993	1995	1996
Direct investment, per cent of business investment					
In Sweden	0.7	25.0	26.0	57.9	18.4
Abroad	38.4	27.8	9.2	45.0	20.1
Total R&D expenditure as % of total GDP	2.9	2.9	3.3	3.1	
Government funded R&D as % of total	38.1	34.0	31.4	..	..

Source: Statistics Sweden, *National Accounts;* OECD, *National Accounts* and *Main Science and Technology Indicators.*

Table L. **Labour market indicators**

	1986[1]	1990	1993	1994	1995	1996
A. Evolution of unemployment						
Unemployment rate (per cent of labour force)						
Total	2.5	1.7	8.2	8.0	7.7	8.1
Men	2.5	1.7	9.7	9.1	8.4	8.5
Women	2.5	1.6	6.6	6.7	6.9	7.5
Youth (age group 16-24 years)	5.3	3.7	18.5	16.7	15.4	15.6
Insured unemployed registered at the employment exchanges[2]	2.5	2.2	8.7	8.8	8.7	9.0
Unemployment average weeks[3]	16.6	14.7	23.5	27.6	26.6	28.1
Labour market policy						
Active measures (thousand persons)	92	54	185	224	189	196
Measures for handicapped, etc. (thousand persons)	74	81	76	80	85	80
B. Structural characteristics						
Participation rates						
Total	83.0	84.5	79.1	77.6	78.2	77.8
Men	85.4	86.6	80.9	79.5	80.2	80.0
Women	80.6	82.3	77.2	75.7	76.1	75.6
Employment, per cent of population between 16-64 years	80.9	83.1	72.6	71.5	72.2	71.6
Self-employed, per cent of total employed	9.8	9.2	10.8	11.1	11.2	11.0
Part-time employed, per cent of all employed	23.5	23.3	24.9	24.9	24.3	23.6
of which: Women (per cent of part-time employed)	86.6	83.4	81.3	80.1	80.1	79.5
Numbers of hours worked per week per person at work	36.2	36.8	36.4	36.9	36.9	36.9
Non-filled vacancies, per cent of employment	0.9	1.0	0.2	0.3	0.4	0.4
Overtime for employees, per cent of all employees	..	11.5	8.3	9.8	9.4	9.6

1. These figures have been recalculated to conform with a new definition introduced in 1987.
2. In per cent of the members of the recognized unemployment benefit funds.
3. Average number of weeks, which persons still registered as unemployed, have been looking for job.

Source: Statistics Sweden, *Statistical Abstract of Sweden*; OECD, *Labour Force Statistics.*

Table M. **The public sector**

	1970	1980	1990	1993	1994	1995	1996
Budget indicators :							
General government							
(per cent of GDP)							
Current receipts	46.9	56.1	63.3	58.8	57.9	57.6	61.6
Non-interest expenditure	43.4	60.5	59.0	70.0	66.3	62.8	60.9
Primary budget balance	3.8	–4.4	4.3	–11.3	–8.3	–5.2	0.8
Net interest expenses	–0.6	–0.4	0.1	1.0	2.0	2.6	2.9
General government budget balance	4.1	–4.0	4.2	–12.3	–10.3	–7.8	–2.2
Structure of expenditure and receipts							
(per cent of GDP)							
Expenditure							
Income transfers	12.2	20.1	21.5	27.9	27.5	26.2	25.2
Interest payments	1.9	4.1	5.1	6.3	6.9	7.1	7.2
Subsidies	1.7	4.3	4.7	5.9	5.3	5.1	4.7
Consumption	21.6	29.3	27.4	28.1	27.2	25.8	26.2
Education	5.5	6.0	5.3	5.4	5.3	5.1	..
Health	5.0	7.3	6.5	5.1	4.8	5.0	..
Social welfare	2.4	4.8	5.2	6.1	6.0	6.2	..
Housing	0.5	0.6	0.5	0.5	0.5	0.5	..
Economic services	1.2	1.4	1.8	2.1	2.2	2.2	..
Other	7.7	9.2	8.0	8.8	8.4	8.5	..
Gross investment	6.6	4.2	2.4	1.1	3.0	2.9	2.0
Taxes and charges							
Property income	3.8	5.6	6.5	7.2	6.3	6.2	5.9
Taxes	32.9	35.0	40.6	36.4	36.1	35.9	38.9
Indirect taxes	12.6	13.5	17.2	15.6	14.9	14.4	16.5
Direct taxes	20.3	21.5	23.4	20.8	21.2	21.6	22.3
Social security contributions	9.3	14.1	15.1	15.6	15.5	13.9	12.6
	1988	1990	1993	1994	1995	1996	1997
Tax rates (per cent)							
Personal income tax							
State income tax							
Top marginal rate	45.0	35.0	20.0	20.0	25.0	25.0	25.0
Lowest marginal rate	5.0	3.0	0.0	0.0	0.0	0.0	0.0
Average local income tax	30.56	31.16	31.04	31.05	31.50	31.65	31.66
Capital income tax[1]	n.a.	n.a.	30.0	30.0	30.0	30.0	30.0
Social security contributions	37.07	38.97	31.95	33.31	36.81	38.01	38.87
of which: Employees	–	–	0.95	1.95	3.95	4.95	5.95
Employers	37.07	38.97	31.00	31.36	32.86	33.06	32.92
Value-added tax (basic rate)	23.46	25.0	25.0	25.0	25.0	25.0	25.0
Corporate tax	52.0	52.0	30.0	28.0	28.0	28.0	28.0
Wealth tax	1.5-3.0	1.5-3.1	1.5	1.5	1.5	1.5	1.5
Real estate tax	0.47	0.47	1.5	1.5	1.5	1.7	1.7

1. Prior to 1991 capital income was added to earned income for tax purposes.

Source: Ministry of Finance, *The Swedish Budget*, various issues; The National Institute of Economic Research, *The Swedish Economy*, various issues; OECD.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1995	18 054	8 047	10 137	29 606	10 331	5 228	5 108	58 141	81 662	10 459
Inhabitants per sq. km	Number	1995	2	96	332	3	131	121	15	106	229	79
Net average annual increase over previous 10 years	%	1995	1.4	0.6	0.3	1.3	0	0.2	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1994	7 943	3 737	3 692 (93)	13 292	4 932	2 508	2 015	21 744	35 894	3 790
of which: Agriculture	% of TCE	1994	5.1	7.2	2.6 (92)	4.1	7	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.5	33.2	27.7 (92)	22.6	42.9	26.8	26.8	27.8 (93)	37.6	23.6
Services	% of TCE	1994	71.4	59.6	69.7 (92)	73.3	50.1	68.1	64.9	67.3 (93)	59.1	55.5
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1995	360.3	233.3	269.2	560	45.7	173.3	125	1 537.6	2 412.5	77.8 (94)
Per capita	US$	1995	19 957	28 997	26 556	18 915	4 420	33 144	24 467	26 445	29 542	7 458 (94)
At current prices using current PPP's[3]	Bill. US$	1995	349.4	167.2	210.8	622.6	..	112.6	90.9	1 159.3	1 673.8	127.3
Per capita	US$	1995	19 354	20 773	20 792	21 031	..	21 529	17 787	19 939	20 497	12 174
Average annual volume growth over previous 5 years	%	1995	3.3	2	1.2	1.5	..	2	–0.7	1.1	1.7	0.8 (94)
Gross fixed capital formation (GFCF)	% of GDP	1995	20.1	24.7	17.6	17.5	32.2	16	15.1	18	21.7	16.9 (94)
of which: Machinery and equipment	% of GDP	1995	10.5 (94)	9 (94)	7.4 (94)	6.5	..	7.2 (94)	5.9 (94)	8.1	7.6	7.9 (94)
Residential construction	% of GDP	1995	5.6 (94)	6.4 (94)	4.5 (94)	4.9	..	3 (94)	3.6 (94)	4.5	7.6	3.4 (94)
Average annual volume growth over previous 5 years	%	1995	3	3	–1.1	0.3	..	–0.5	–10.4	–1.3	2.5	1.2 (94)
Gross saving ratio[4]	% of GDP	1995	16.9	24.9	22.6	17.1	..	17.8	19.8	19.7	21.3	15.7 (94)
General government												
Current expenditure on goods and services	% of GDP	1995	17.2	18.9	14.8	19.6	..	25.1	21.8	19.3	19.5	18.5 (94)
Current disbursements[5]	% of GDP	1994	36.2	47.8	54.1	46.7	..	61.1	57.7	50.9	46.1	52.7
Current receipts	% of GDP	1994	34.2	47.3	50.6	42.4	..	59.1	53	46.4	46.4	44.2
Net official development assistance	% of GNP	1994	0.33	0.33	0.32	0.42	..	0.99	0.3	0.64	0.33	..
Indicators of living standards												
Private consumption per capita using current PPP's[3]	US$	1995	12 090	11 477	12 960	12 551	10 259	11 531	9 643	11 996	11 707	9 071
Passenger cars, per 1 000 inhabitants	Number	1993	438	418	402	455 (92)	164	312	367	419	533[8]	187
Telephones, per 1 000 inhabitants	Number	1992	482	451	437	592	190	589	544	536	457[8]	457
Television sets, per 1 000 inhabitants	Number	1992	482	480	453	640	..	537	505	408	558	201
Doctors, per 1 000 inhabitants	Number	1994	2.2 (91)	2.4	3.7	2.2	3.2	2.8 (93)	2.7	2.9	3.2 (92)	3.9 (93)
Infant mortality per 1 000 live births	Number	1994	5.9	6.3	7.6	6.8 (93)	..	5.4 (93)	4.6	6.1	5.8 (93)	7.9
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1995	2	5	2.8	2.7	..	3.3	4.2	3	4.8	13.4
Consumer prices	%	1995	2.5	3.2	2.4	2.2	20.3	2	2.3	2.2	3.5	13.9
Foreign trade												
Exports of goods, fob*	Mill. US$	1995	53 092	57 200	170 230[7]	192 502	21 654	49 045	39 995	286 762	523 000	11 761
As % of GDP	%	1995	14.7	24.5	63.2	34.4	47.4	28.3	32	18.7	21.7	12.2 (94)
Average annual increase over previous 5 years	%	1995	6	6.9	7.6	8.6	..	7	8.6	5.8	5	8
Imports of goods, cif*	Mill. US$	1995	57 406	65 293	155 449[7]	164 443	26 523	43 728	28 928	267 059	463 472	27 718
As % of GDP	%	1995	15.9	28	57.7	29.4	58	25.2	23.1	17.4	19.2	28.3 (94)
Average annual increase over previous 5 years	%	1995	8.1	5.9	5.3	7.1	..	6.7	1.4	3.5	6.1	6.9
Total official reserves[6]	Mill. SDRs	1995	8 003	12 600	10 883[7]	10 124	9 312	7 411	6 753	18 065	57 185	9 943
As ratio of average monthly imports of goods	Ratio	1995	1.7	2.3	0.8	0.7	..	2	2.8	0.8	1.5	4.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1995	10 229	267	3 580	57 283	125 250	44 851	413	91 120	15 457	3 580
Inhabitants per sq. km	Number	1995	111	3	51	190	332	444	159	46	379	13
Net average annual increase over previous 10 years	%	1995	–0.3	1	0.1	0	0.4	0.9	1.2	2.1	0.7	0.9
Employment												
Total civilian employment (TCE)[2]	Thousands	1994	3 643	138	1 207	20 022	64 530	19 831	207	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9	9.4	12	7.7	5.8	13.6	2.9	25.8	4	10.4
Industry	% of TCE	1995	34	26.1	27.6	32.1	34	33.2	30.7 (90)	22.2	23	24.9
Services	% of TCE	1995	57.1	65.2	60.5	60.2	60.2	53.2	66.1 (90)	52.1	73	64.6
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1995	43.7	7	64.3	1 087.2	5 114	455.5	10.6 (92)	246.1	395.5	59.7
Per capita	US$	1995	4 273	23 366	17 965	18 984	40 726	10 155	26 866 (92)	2 597	25 597	16 689
At current prices using current PPP's[3]	Bill. US$	1995	..	5.9	61.7	1 114.7	2 736.8	..	12.8	699.7	305.6	60.3
Per capita	US$	1995	..	21 938	17 228	19 465	21 795	..	31 303	7 383	19 782	16 851
Average annual volume growth over previous 5 years	%	1995	..	0.9	5.7	1.1	1.3	..	4.1 (92)	3.0 (93)	2.1	2.9
Gross fixed capital formation (GFCF)	% of GDP	1995	19.3	15.2	15.1	17	28.5	36.6	27.7 (92)	16.6	19.4	20.5
of which: Machinery and equipment	% of GDP	1993	8	4.7	5.5	8.6	9.6 (94)	13.2	..	9.4 (93)	9.1	10
Residential construction	% of GDP	1993	9.8	4	5	4.8	5.7 (94)	7.8	..	4.9 (93)	5.2	5.5
Average annual volume growth over previous 5 years	%	1995	..	–4.1	0.9	–1.7	–0.1	..	6.5 (92)	7.7 (94)	1.2	4.5
Gross saving ratio[4]	% of GDP	1995	..	16.4	19.5	20.5	30.8	35.8	60.2 (92)	15.1 (94)	24.6	18.4
General government												
Current expenditure on goods and services	% of GDP	1995	24.9	20.8	14.7	16.3	9.7	10.4	17.1 (92)	10.6[7]	14.3	14.3
Current disbursements[5]	% of GDP	1994	..	34.4	40.4 (93)	51	27	15.3	..	..	52.8	..
Current receipts	% of GDP	1994	..	35.4	38.9 (93)	45	32.2	24.2	..	..	51.6	..
Net official development assistance	% of GNP	1994	..	..	0.2	0.27	0.28	..	0.4	..	0.75	0.22
Indicators of living standards												
Private consumption per capita using current PPP's[3]	US$	1995	11 197	13 208	9 467	11 952	13 102	12 287	16 827	5 368	11 854	10 396
Passenger cars, per 1 000 inhabitants	Number	1993	204	435	251	516 (92)	326	95	506	88	372	439
Telephones, per 1 000 inhabitants	Number	1993	146	544	328	418	468	378	541	88	499	460
Television sets, per 1 000 inhabitants	Number	1992	414	319	304	421	614	211	267	149	488	443
Doctors, per 1 000 inhabitants	Number	1994	..	3 (93)	2	1.7 (92)	1.8	..	2.2 (93)	1	2.5 (90)	2.1
Infant mortality per 1 000 live births	Number	1994	11.5	4.8 (93)	5.9	7.3 (93)	4.2	..	8.5 (92)	17	5.6	7.3 (93)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1995	..	..	4.3	5	2	..	..	1.9	2.9	1.6
Consumer prices	%	1995	25.4	3.5	2.5	5	1.4	6.2	2.8	17.6	2.7	2.1
Foreign trade												
Exports of goods, fob*	Mill. US$	1995	12 540	1 802	44 708	233 868	441 512	125 058	..	79 542	197 087	13 805
As % of GDP	%	1995	28.7	25.6	69.5	21.5	8.6	27.5	..	32.3	49.8	23.1
Average annual increase over previous 5 years	%	1995	5.2	2.5	13.4	6.6	9	14	..	24.3	8.5	7.9
Imports of goods, cif*	Mill. US$	1995	15 073	1 754	33 024	206 246	335 392	135 119	..	72 453	177 912	13 990
As % of GDP	%	1995	34.5	24.9	51.3	19	6.6	29.7	..	29.4	45	23.4
Average annual increase over previous 5 years	%	1995	11.7	1.2	9.8	2.5	7.5	14.1	..	18.3	7.1	8.1
Total official reserves[6]	Mill. SDRs	1995	8 108	207	5 806	23 482	123 277	21 983	..	11 333	22 680	2 967
As ratio of average monthly imports of goods	Ratio	1995	..	1.4	2.1	1.4	4.4	..	..	1.9	1.5	2.5

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1995	4 360	38 588	9 921	39 210	8 827	7 081	61 644	58 613	263 058
Inhabitants per sq. km	Number	1995	13	119	107	78	20	171	79	239	28
Net average annual increase over previous 10 years	%	1995	0.5	0.4	–0.1	0.2	0.6	0.8	2.1	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	2 003	14 658	4 372	11 760	3 926	3 772	19 664	25 579	123 060
of which: Agriculture	% of TCE	1994	5.3	23.8	11.5	9.8	3.4	4	44.8	2.1	2.9
Industry	% of TCE	1994	23.4	31.9	32.8	30.1	25	28.8	22.2	27.7	24
Services	% of TCE	1994	71.3	44.1	55.7	60.2	71.6	67.3	33	70.2	73.1
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1995	103.4 (93)	118	99.8	559.6	230.6	306.1	169.3	1 101.8	6 954.8
Per capita	US$	1995	23 984 (93)	3 057	10 060	14 272	26 096	43 233	2 747	18 799	26 438
At current prices using current PPP's[3]	Bill. US$	1995	98.8	..	123.5	557.8	165	175.7	350.8	1 041.9	6 954.8
Per capita	US$	1995	22 672	..	12 457	14 226	18 673	24 809	5 691	17 776	26 438
Average annual volume growth over previous 5 years	%	1995	2.1 (93)	..	1.4	1.3	0.4	0	3.2	1.2	2.3
Gross fixed capital formation (GFCF)	% of GDP	1995	22 (93)	17.1	23.7	20.6	14.5	22.7	23.3	15.1	17.6
of which: Machinery and equipment	% of GDP	1995	..	..	11.7 (93)	5.8 (94)	7.8	8.3	10.8	7.3 (94)	7.7 (93)
Residential construction	% of GDP	1995	..	..	5.2 (93)	4.1 (94)	1.6	14.4[7]	9.3 (94)	3.1 (94)	4.0 (93)
Average annual volume growth over previous 5 years	%	1995	–3.3 (93)	..	1.8	–0.9	–5.3	–0.2	4	–1.6	5.8
Gross saving ratio[4]	% of GDP	1995	21.9 (93)	..	21.6	21.5	16.6	30.1	19.7	13.8	15.9
General government											
Current expenditure on goods and services	% of GDP	1995	22.1 (93)	..	18.1	16.6	25.8	14	10.8	21.4	16.2
Current disbursements[5]	% of GDP	1994	..	..	42.5 (93)	42.6	66.4	36.8.	..	42.3	35.8 (93)
Current receipts	% of GDP	1994	..	..	39.8 (93)	39.1	57.4	36.6	..	37.3	31.7 (93)
Net official development assistance	% of GNP	1994	1.05	..	0.36	0.27	0.92	0.38	..	0.31	0.15
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1995	11 194	..	8 150	8 812	9 778	14 594	4 021	11 319	17 834
Passenger cars, per 1 000 inhabitants	Number	1993	375	175	332	343	404	440	43	375 (92)	556
Telephones, per 1 000 inhabitants	Number	1993	542	115	311	364	678	611	184	494	574
Television sets, per 1 000 inhabitants	Number	1992	424	295	188	402	469	407	176	435	815
Doctors, per 1 000 inhabitants	Number	1994	3.3 (93)	..	2.9	4.1 (93)	3	3.1	1.1	1.5 (93)	2.5 (93)
Infant mortality per 1 000 live births	Number	1994	5.1 (93)	..	7.9	7.6 (93)	4.8 (93)	5.6 (93)	46.8	6.2	8.5 (92)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1995	3.5	..	..	6.4	4.6	..	..	5.7	2.7
Consumer prices	%	1995	2.4	43	7.2	5.2	4.4	3.2	78.6	3.4	3.1
Foreign trade											
Exports of goods, fob*	Mill. US$	1995	41 836	22 892	23 356	91 615	79 595	81 499	21 853	242 692	584 742
As % of GDP	%	1995	30.9 (93)	19.4	23.4	16.4	34.5	26.6	12.9	22	8.4
Average annual increase over previous 5 years	%	1995	4.4	..	7.4	10.6	6.7	5	10.8	5.7	8.2
Imports of goods, cif*	Mill. US$	1995	32 804	29 050	33 539	114 835	64 469	80 193	36 060	265 696	743 445
As % of GDP	%	1995	23.3 (93)	24.6	33.6	20.5	28	26.2	21.3	24.1	10.7
Average annual increase over previous 5 years	%	1995	3.8	..	6.2	5.6	3.3	2.8	9.8	3.5	8.5
Total official reserves[6]	Mill. SDRs	1995	15 148	9 939	10 663	23 199	16 180	24 496	8 370	28 265	50 307
As ratio of average monthly imports of goods	Ratio	1995	5.5	..	3.8	2.4	3	3.7	2.8	1.3	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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